In [135…
```python
import pandas as pd
from pathlib import Path
import os
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde, skew, kurtosis, norm
from scipy.stats import pearsonr
import statsmodels.api as sm
import seaborn as sns
from statsmodels.nonparametric.smoothers_lowess import lowess
from difflib import get_close_matches
```

In [138…
```python
def find_column(df, name_candidates):
    cols = list(df.columns)

    for cand in name_candidates:
        if cand in cols:
            return cand

    lower_map = {c.lower(): c for c in cols}
    for cand in name_candidates:
        if cand.lower() in lower_map:
            return lower_map[cand.lower()]

    for cand in name_candidates:
        m = get_close_matches(cand, cols, n=1, cutoff=0.75)
        if m:
            return m[0]
    return None

def ensure_sorted_safe(df, id_col_hint='country', time_col_hint='year'):
    id_found = find_column(df, [id_col_hint, 'country', 'id', 'iso', 'country_name'])
    time_found = find_column(df, [time_col_hint, 'year', 'time', 'date', 'period'])
    if id_found is None or time_found is None:
        raise KeyError(f"Column not found. id_hint={id_col_hint} -> {id_found}, time_hint={time_col_hint} -> {time_found}. Available columns: {list(df.columns)}")
    # make it a numeric
    if pd.api.types.is_string_dtype(df[time_found]):
        sample = df[time_found].dropna().astype(str).head(10).tolist()
        if all(s.isdigit() and len(s) == 4 for s in sample):
            df[time_found] = pd.to_numeric(df[time_found], errors='coerce')
    sorted_df = df.sort_values([id_found, time_found], na_position='last').reset_index(drop=True)
    print(f"Sorted by detected columns: id='{id_found}', time='{time_found}'")
    return sorted_df

# Run diagnostic when cell executed
print('--- ensure_sorted_safe diagnostic ---')
if 'df' not in globals():
    print("No DataFrame named 'df' found in this kernel. Define 'df' and re-run this cell.")
else:
    print('DataFrame df found. Columns:')
    print(list(df.columns))
    print(Column dtypes:)
    print(df.dtypes)
    # pick hints from existing variables if present
    id_hint = globals().get('id_col', 'country')
    time_hint = globals().get('time_col', 'year')
    print(f'Using id hint: {id_hint}, time hint: {time_hint}')
    try:
        df0 = ensure_sorted_safe(df.copy(), id_hint, time_hint)
        print('
Sorted sample (first 8 rows):')
        display(df0.head(8))
    except Exception as e:
        print('
ERROR while sorting:')
        print(e)
        print('
Sample rows to help debugging:')
        display(df.head(10))
```

```
  Cell In[138], line 48
    print('
          ^
SyntaxError: unterminated string literal (detected at line 48)
```

Question 1 --> Within Transforms, Pooled etc

In [139…
```python
# Question 1 --> create the between-transformed and one-way within-transformed versions of each variable in the dataset.
# getting teh path to my data set

final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final = pd.read_excel(final_data_path / "final_panel_dataset.xlsx")

# In the study as one of teh explanatory varibles they create a 5 year lagged GDP per capita variable.
# Therefore, prior to plotting the distributions, we create this variable in the final data set.

df_Final = df_Final.sort_values(["country", "year"])
df_Final = df_Final[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]
df_Final = df_Final.sort_values(["country", "year"])

# Logging GDP per capita first
df_Final['ln_gdppc'] = np.log(df_Final['gdppc'])
df_Final["ln_gdppc_lag5"] = df_Final.groupby("country")["ln_gdppc"].shift(5)
# Creating the 5 year lagged GDP per capita growth variable
df_Final["ln_gdp_growth_5y"] = df_Final["ln_gdppc"] - df_Final["ln_gdppc_lag5"]

# Doing of rthe alternative dependent variable: labour productivity growth
df_Final['ln_labour_productivity'] = np.log(df_Final['labour_productivity'])
df_Final["ln_labour_productivity_lag5"] = df_Final.groupby("country")["ln_labour_productivity"].shift(5)
df_Final["ln_labour_productivity_growth_5y"] = df_Final["ln_labour_productivity"] - df_Final["ln_labour_productivity_lag5"]

# taking teh logarithm of the other variables as well
df_Final['ln_cpi'] = np.log(df_Final['cpi'])
df_Final['ln_trade_openness'] = np.log(df_Final['trade_openness'])


variables = [ 'ln_gdppc','ln_labour_productivity', 'ln_cpi', 'ln_trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', "ln_gdp_growth_5y", "ln_labour_productivity_growth_5y" ]

for var in variables:   # Between transformation
    df_Final[f'{var}_between'] = df_Final.groupby('country')[var].transform('mean')

    # One-way Within transformation
    df_Final[f'{var}_within'] = df_Final[var] - df_Final.groupby('country')[var].transform('mean')

rows = []

for var in variables:
    pooled_variance = df_Final[var].var(ddof=1)
    var_between = df_Final[f'{var}_between'].var(ddof=1)
    var_within = df_Final[f'{var}_within'].var(ddof=1)
    if pooled_variance != 0 and not np.isnan(pooled_variance):
        share_between = var_between / pooled_variance
        share_within = var_within / pooled_variance
    else:
        share_between = np.nan
        share_within = np.nan
    rows.append({"variable": var,"var_total_pooled": pooled_variance,"var_between": var_between,"var_within": var_within,"share_between": share_between,"share_within": share_within})

# vardecomp tabel
var_decomp = pd.DataFrame(rows).set_index("variable")


pd.set_option("display.float_format", "{:,.4f}".format)
print("Variance decomposition")
print(var_decomp)

df_Final.describe()
```

```
Variance decomposition
                                  var_total_pooled  var_between  var_within  \
variable
ln_gdppc                                    0.8785       0.0980      0.7805
ln_labour_productivity                      1.1241       0.1079      1.0162
ln_cpi                                     18.9989       3.4977     15.5012
ln_trade_openness                          20.0214      19.6485      0.3729
liquid_liabilities                          0.0664       0.0125      0.0539
mcap_gdp                                    0.1390       0.0447      0.0944
bank_loans_gdp                              0.1293       0.0602      0.0691
ln_gdp_growth_5y                            0.0172       0.0002      0.0170
ln_labour_productivity_growth_5y            0.0167       0.0004      0.0163

                                  share_between  share_within
variable
ln_gdppc                                 0.1115        0.8885
ln_labour_productivity                   0.0960        0.9040
ln_cpi                                   0.1841        0.8159
ln_trade_openness                        0.9814        0.0186
liquid_liabilities                       0.1877        0.8123
mcap_gdp                                 0.3212        0.6789
bank_loans_gdp                           0.4654        0.5346
ln_gdp_growth_5y                         0.0140        0.9860
ln_labour_productivity_growth_5y         0.0221        0.9779
```

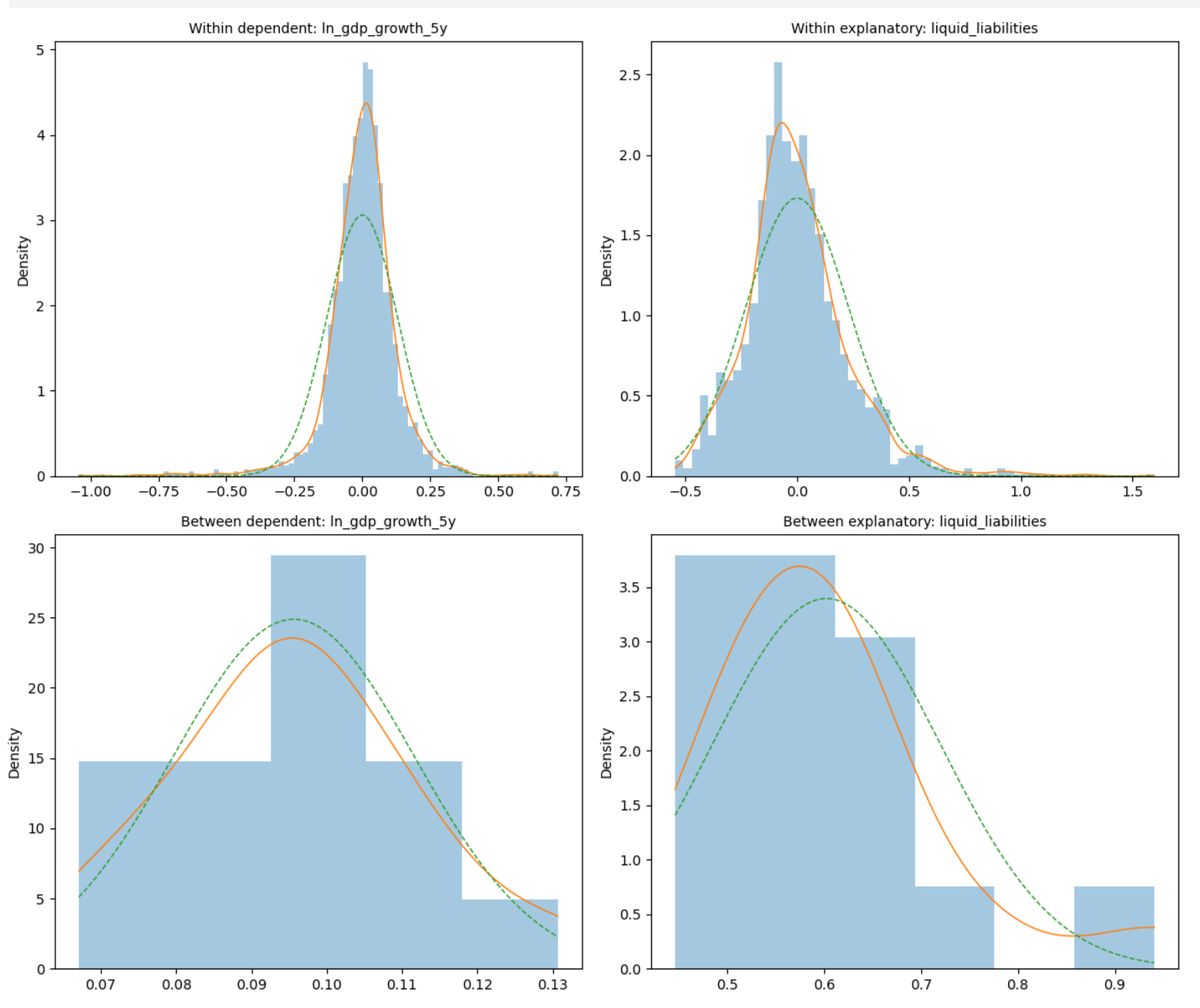
Out[139…

	year	gdppc	labour_productivity	cpi	trade_openness	liquid_liabilities	mcap_gdp	bank_loans_gdp	ln_gdppc	ln_gdppc_lag5	...	liquid_liabilities_between	liquid_liabilities_within	mcap_gdp_betw
count	2,224.0000	2,224.0000	2,096.0000	2,224.0000	2,224.0000	2,224.0000	1,488.0000	1,668.0000	2,224.0000	2,144.0000	...	2,224.0000	2,224.0000	1,668.0
mean	1,951.0000	16,107.7353	24.2423	44.4844	0.4052	0.5938	0.4301	0.6261	9.2763	9.2239	...	0.5938	-0.0000	0.4
std	40.1338	14,479.9899	21.7278	57.8501	0.6467	0.2577	0.3729	0.3595	0.9373	0.9123	...	0.1116	0.2322	0.2
min	1,882.0000	1,581.0000	0.9705	0.0000	0.0000	0.0579	0.0043	0.0507	7.3658	7.3658	...	0.4385	-0.5263	0.1
25%	1,916.0000	4,997.2500	6.2426	2.6660	0.0005	0.4316	0.1678	0.3417	8.5166	8.4918	...	0.5121	-0.1393	0.2
50%	1,951.0000	9,841.0760	14.5815	11.9223	0.1717	0.5484	0.3218	0.5532	9.1943	9.1301	...	0.5888	-0.0253	0.3
75%	1,986.0000	24,807.5099	40.4610	85.5400	0.4826	0.7076	0.5780	0.8386	10.1189	10.0575	...	0.6261	0.1109	0.6
max	2,020.0000	85,115.3231	97.5205	226.8252	2.9965	2.2599	2.7971	2.0357	11.3518	11.3231	...	0.9261	1.5988	0.7

8 rows × 34 columns

Question 2 --> Distributions Plots, etc --> Model is:

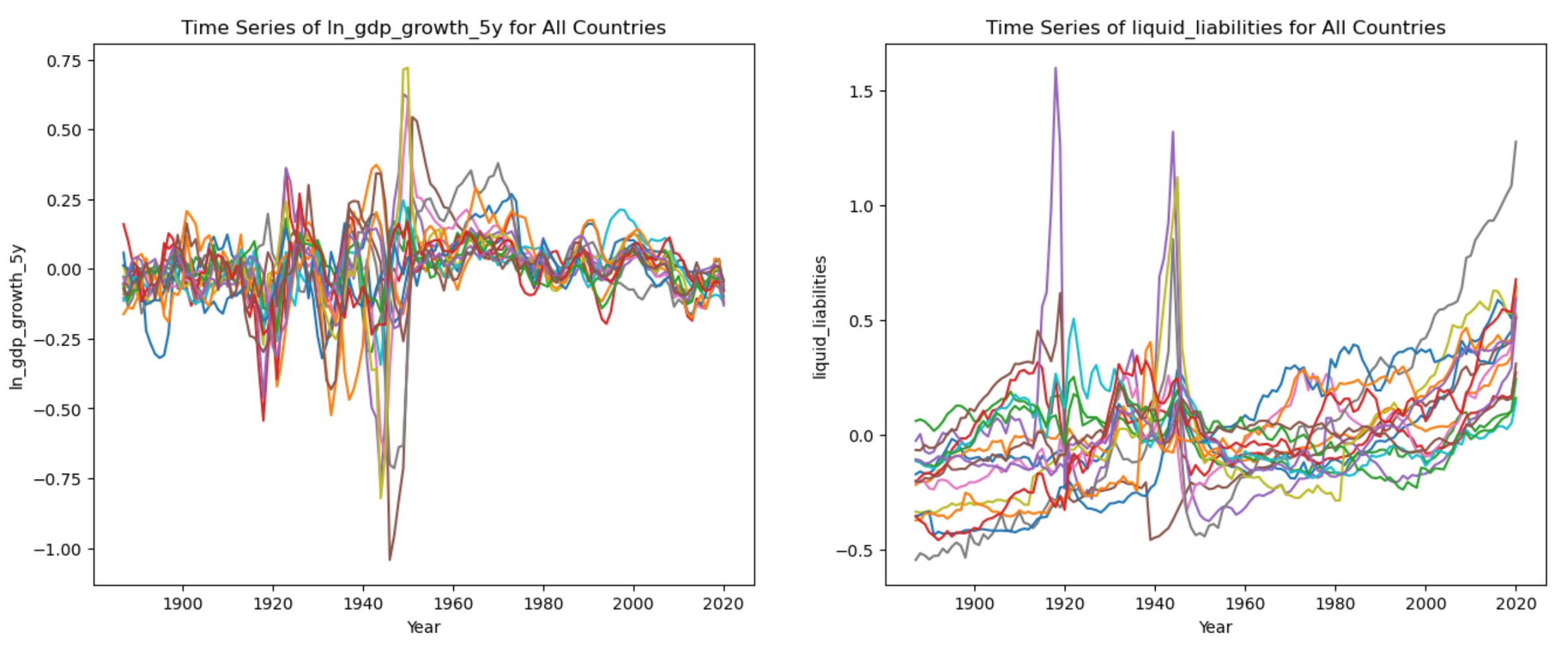
In [140…
```python
# Defining my Dependent and Explanatory Variables
Dependent_Variables_Growth = ['ln_gdp_growth_5y','ln_labour_productivity_growth_5y' ]
Explanatory_Variables = ['ln_gdppc_lag5', 'ln_labour_productivity_lag5', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Control_Variables = ['ln_cpi', 'ln_trade_openness']

# defining my individuals, time, dependent and explanatory variables

individuals = df_Final['country']
time = df_Final['year']
dep_var_col = 'ln_gdp_growth_5y'
exp_var_col = 'liquid_liabilities'
cols_to_keep = ["country", "year", dep_var_col, exp_var_col, "ln_cpi", "ln_trade_openness"]
df_Final_defined = df_Final[cols_to_keep].dropna()

df_Final_defined["dep_var_col_mean_i"] = df_Final_defined.groupby(individuals)[dep_var_col].transform("mean")
df_Final_defined["exp_var_col_mean_i"] = df_Final_defined.groupby(individuals)[exp_var_col].transform("mean")

df_Final_defined["dep_var_col_within"] = df_Final_defined[dep_var_col] - df_Final_defined['dep_var_col_mean_i']
df_Final_defined["exp_var_col_within"] = df_Final_defined[exp_var_col] - df_Final_defined['exp_var_col_mean_i']

between_df_Final_defined = (df_Final_defined.groupby("country").agg(dep_var_col_between=(dep_var_col, 'mean'),exp_var_col_between=(exp_var_col, 'mean')).reset_index())

def plot_dist_subplot(ax, values, title, bins='auto'):
    values = pd.to_numeric(values, errors='coerce')
    values = values[np.isfinite(values)]
    if len(values) == 0:
        ax.set_title(f"No valid data for {title}", fontsize=10)
        return
    mu = values.mean()
    sigma = values.std(ddof=1)
    # Histogram
    ax.hist(values, bins=bins, density=True, alpha=0.4)
    # KDE
    kde = gaussian_kde(values)
    xmin, xmax = values.min(), values.max()
    x_grid = np.linspace(xmin, xmax, 500)
    ax.plot(x_grid, kde(x_grid), linewidth=1)
    # Normal pdf
    if sigma > 0:
        normal_pdf = (
            1 / (np.sqrt(2 * np.pi) * sigma)
            * np.exp(-0.5 * ((x_grid - mu) / sigma) ** 2)
        )
        ax.plot(x_grid, normal_pdf, linestyle="--", linewidth=1)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel("")
    ax.set_ylabel("Density")
fig, axs = plt.subplots(2, 2, figsize=(12, 10))
plot_dist_subplot(axs[0, 0],df_Final_defined["dep_var_col_within"],f"Within dependent: {dep_var_col}")
plot_dist_subplot(axs[0, 1],df_Final_defined["exp_var_col_within"],f"Within explanatory: {exp_var_col}")
plot_dist_subplot(axs[1, 0],between_df_Final_defined["dep_var_col_between"],f"Between dependent: {dep_var_col}")
plot_dist_subplot(axs[1, 1],between_df_Final_defined["exp_var_col_between"],f"Between explanatory: {exp_var_col}")
plt.tight_layout()
plt.show()

def compute_stats(values, name):
    values = pd.to_numeric(values, errors='coerce')
    values = values[np.isfinite(values)]
    mu = values.mean()
    sd = values.std(ddof=1)
    sk = skew(values)
    kt = kurtosis(values, fisher=False)

    # cool high leverage figure calc make odds ones stand out
    if sd > 0:
        z = (values - mu) / sd
        high_lev = np.sum(np.abs(z) > 3)
    else:
        high_lev = 0

    return {
        "variable": name,
        "n": len(values),
        "mean": mu,
        "sd": sd,
        "skewness": sk,
        "kurtosis": kt,
        "high_leverage(|z|>3)": high_lev,
    }

stats_list = []
stats_list.append(
    compute_stats(df_Final_defined["dep_var_col_within"],
                  f"{dep_var_col}_within")
)
stats_list.append(
    compute_stats(df_Final_defined["exp_var_col_within"],
                  f"{exp_var_col}_within")
)
stats_list.append(
    compute_stats(between_df_Final_defined["dep_var_col_between"],
                  f"{dep_var_col}_between")
)
stats_list.append(
    compute_stats(between_df_Final_defined["exp_var_col_between"],
                  f"{exp_var_col}_between")
)

stats_df = pd.DataFrame(stats_list)
print("\nSummary statistics for within / between variables:")
print(stats_df.to_string(index=False))
```



```
Summary statistics for within / between variables:
                 variable    n   mean     sd  skewness  kurtosis  high_leverage(|z|>3)
   ln_gdp_growth_5y_within 2144 0.0000 0.1304   -1.1084   12.7469                    37
 liquid_liabilities_within 2144 0.0000 0.2305    1.1249    6.8723                    23
  ln_gdp_growth_5y_between   16 0.0956 0.0160    0.2492    2.9390                     0
liquid_liabilities_between   16 0.6026 0.1175    1.4965    5.4128                     0
```

In [141…
```python
df_timeseries_plot = df_Final_defined.sort_values(["country", "year"])
countries = sorted(df_timeseries_plot["country"].unique())
fig, axes = plt.subplots(1, 2, figsize=(16, 6), sharex=False)
for c in countries:
    df_c = df_timeseries_plot[df_timeseries_plot["country"] == c]
    axes[0].plot(df_c["year"], df_c['dep_var_col_within'], label=c)
axes[0].set_title(f"Time Series of {dep_var_col} for All Countries")
axes[0].set_xlabel("Year")
axes[0].set_ylabel(dep_var_col)
for c in countries:
    df_c = df_timeseries_plot[df_timeseries_plot["country"] == c]
    axes[1].plot(df_c["year"], df_c['exp_var_col_within'], label=c)

axes[1].set_title(f"Time Series of {exp_var_col} for All Countries")
axes[1].set_xlabel("Year")
axes[1].set_ylabel(exp_var_col)
```

Out[141…
```
Text(0, 0.5, 'liquid_liabilities')
```



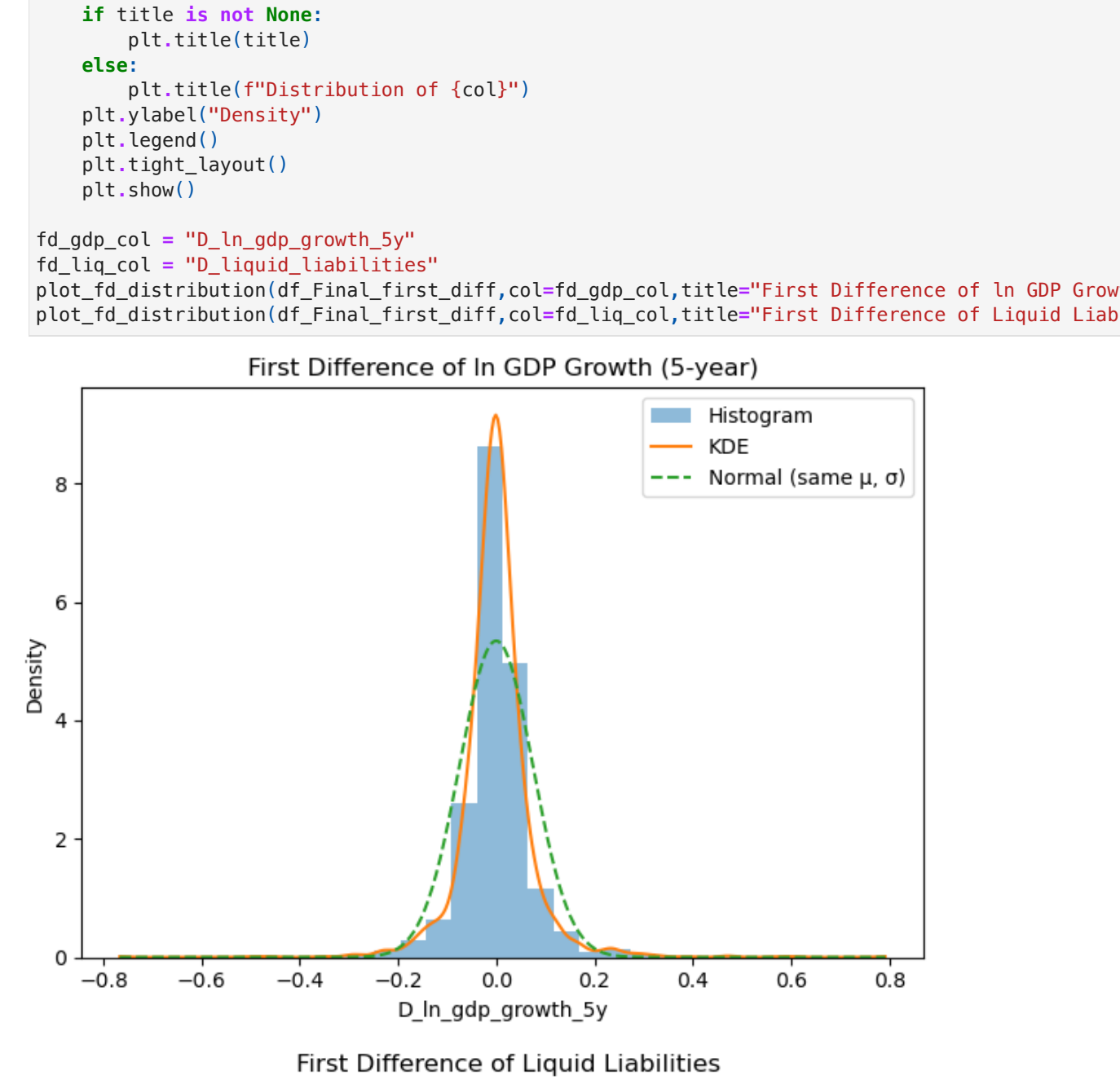
Question 3 --> 1st difference,

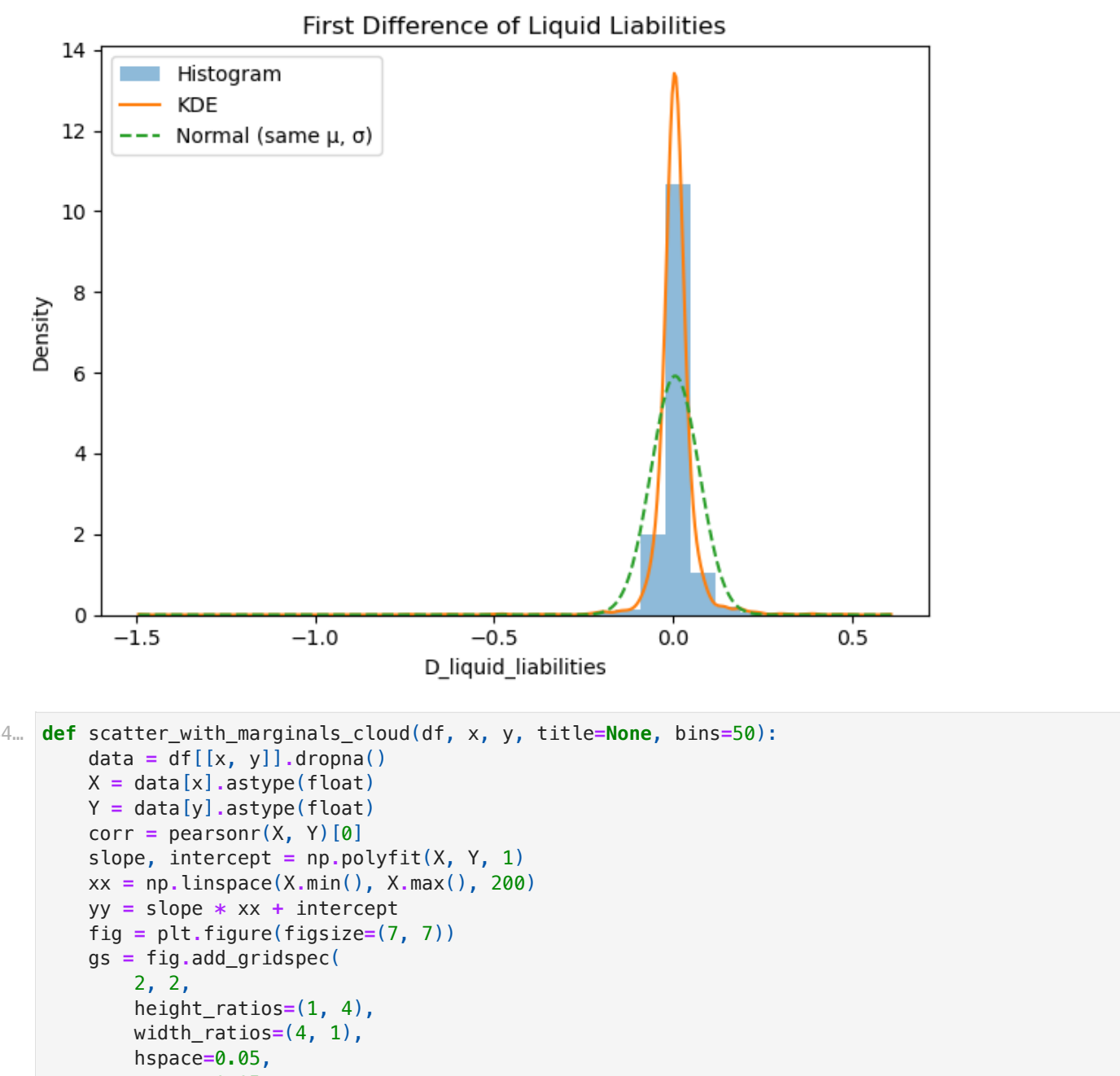
In [142…
```python
# Create first differences for the selected columns.
# Define which columns we want first differences for (use the dependent and the explanatory of interest)
x_col = ["ln_gdp_growth_5y", "liquid_liabilities"]

# Build df_Final_first_diff by grouping by country and taking first differences
df_Final_first_diff = df_Final.sort_values(["country", "year"]).copy()
for col in x_col:
    df_Final_first_diff[f"D_{col}"] = df_Final_first_diff.groupby("country")[col].diff()

# just tryna keep the first diff colums
first_diff = [f"D_{c}" for c in x_col]
fd_cols = first_diff

def check_first_diffs_boundaries(df, id_col, time_col, fd_cols):
    df = df.sort_values([id_col, time_col]).copy()
    first_id = df[id_col].iloc[0]
    T = (df[id_col] == first_id).sum()
    n = min(3 * T + 1, len(df))
    sub = df.iloc[:n].copy()
    changes = []
    ids = sub[id_col].to_numpy()
    for i in range(1, len(sub)):
        if ids[i] != ids[i - 1]:
            changes.append(i)
            if len(changes) == 3:
                break


    for idx in changes:
        row = sub.iloc[idx]
        print("\n change at stacked row", idx)
        print(row[[id_col, time_col]])


        for col in fd_cols:
            val = row.get(col, np.nan)
            if not pd.isna(val):

                print("\nProblem in column:", col)
                print(sub.iloc[max(idx-2, 0): idx+3][[id_col, time_col] + fd_cols])
                raise AssertionError(
                    f"First difference in column '{col}' at change of individual "
                    f"should be NaN, but is {val}."
                )


    print("\n✓ Check passed: at the first 3 changes of individuals,")
    print("   all specified first-difference columns are NaN at the boundary.")
    return True

check_first_diffs_boundaries(
    df_Final_first_diff,
    id_col="country",
    time_col="year",
    fd_cols=fd_cols
)
```

```
 change at stacked row 139
country    Canada
year         1882
Name: 139, dtype: object

 change at stacked row 278
country    Denmark
year          1882
Name: 278, dtype: object

 change at stacked row 417
country    Finland
year          1882
Name: 417, dtype: object

✓ Check passed: at the first 3 changes of individuals,
   all specified first-difference columns are NaN at the boundary.
```

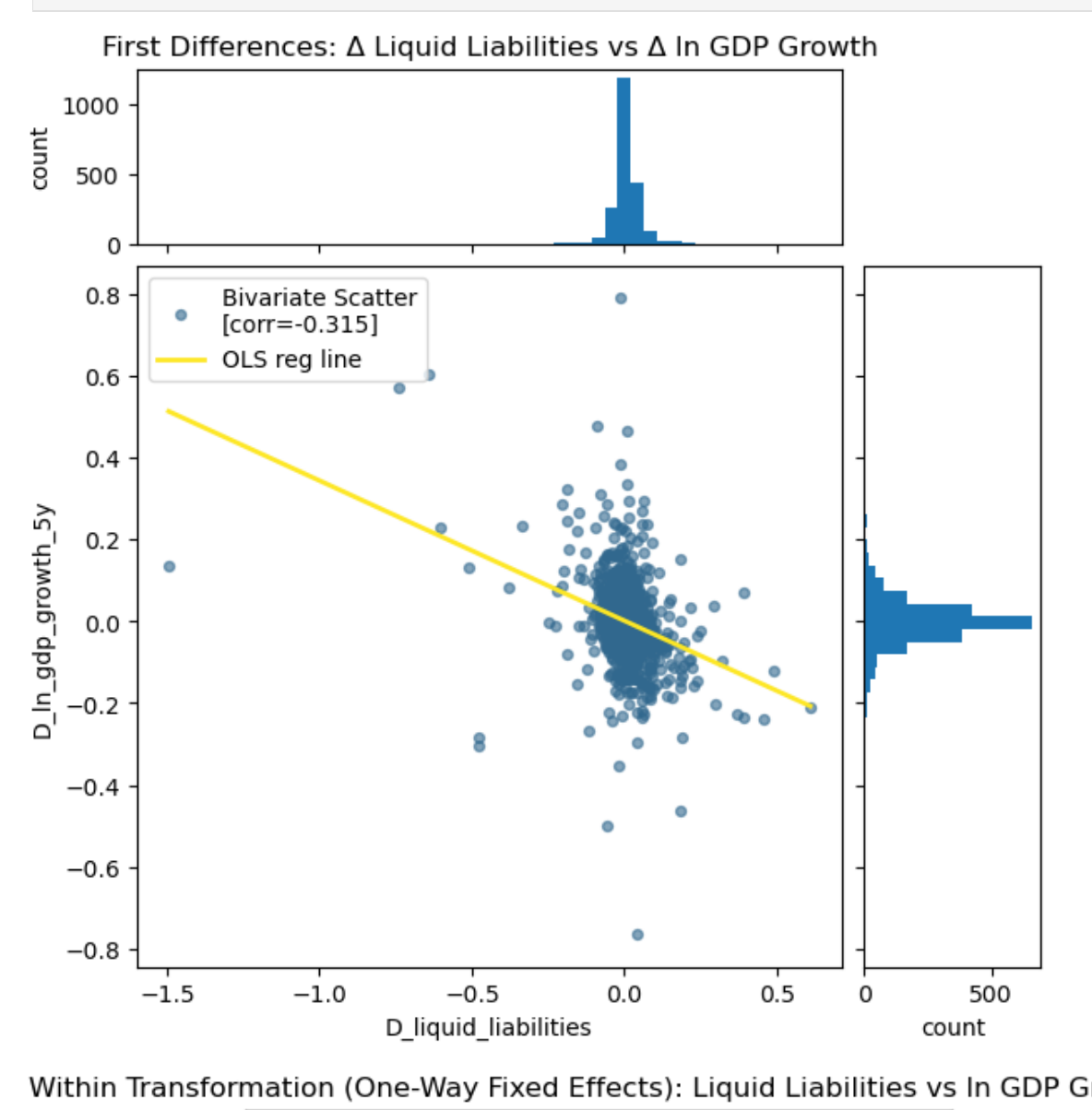
Out[142…
```
True
```

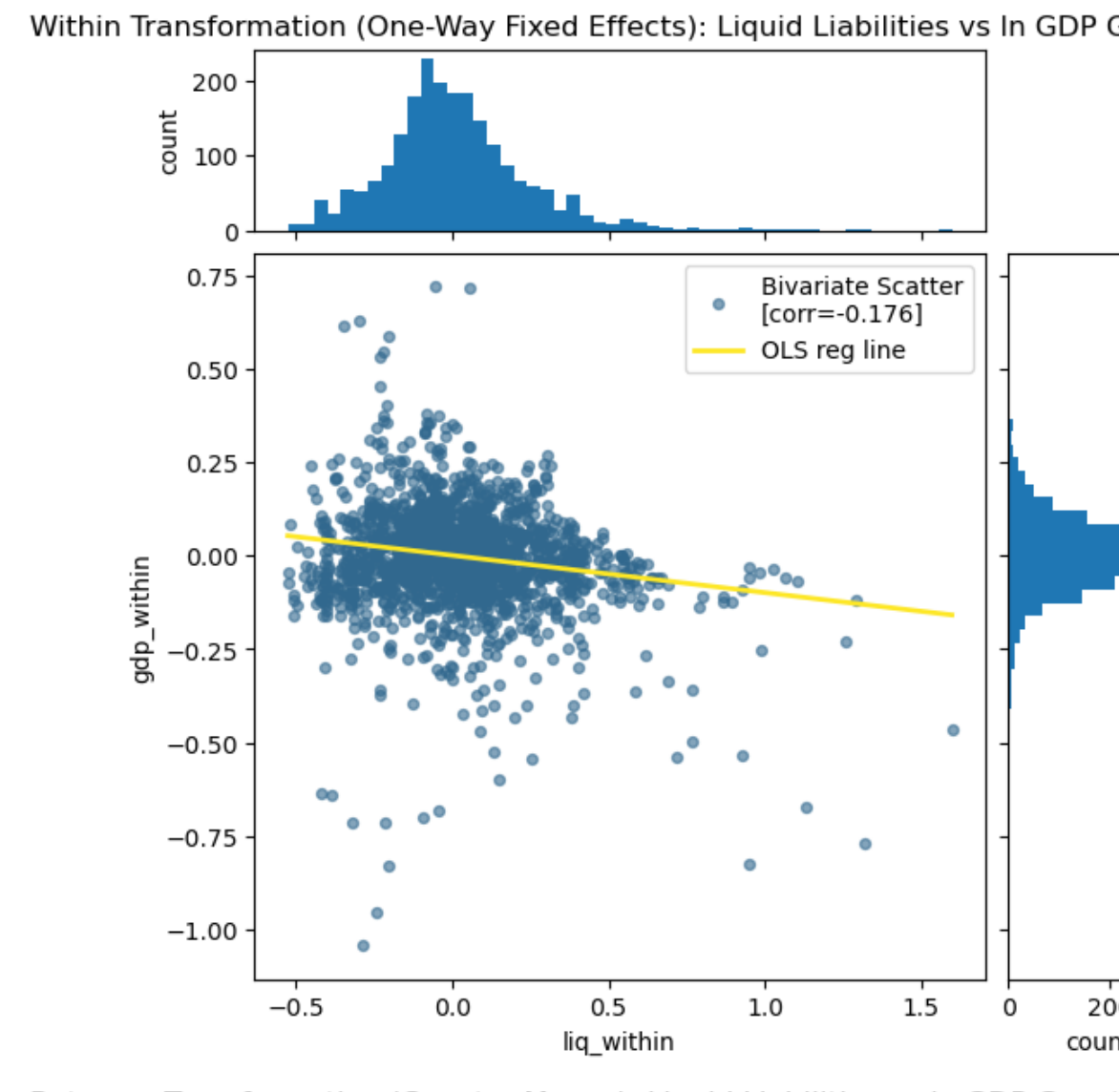
In [143…
```python
def plot_fd_distribution(df, col, title=None, bins=30):
    data = df[col].dropna().to_numpy()
    if data.size == 0:
        print(f"No non-missing data in column {col}.")
        return
    mu = data.mean()
    sigma = data.std(ddof=1)
    x_min, x_max = data.min(), data.max()
    x_vals = np.linspace(x_min, x_max, 400)
    kde = gaussian_kde(data)
    kde_vals = kde(x_vals)
    normal_vals = norm.pdf(x_vals, loc=mu, scale=sigma)
    plt.figure()
    plt.hist(data, bins=bins, density=True, alpha=0.5, label="Histogram")
    plt.plot(x_vals, kde_vals, label="KDE")
    plt.plot(x_vals, normal_vals, linestyle="--", label="Normal (same μ, σ)")
    plt.xlabel(col)
    if title is not None:
        plt.title(title)
    else:
        plt.title(f"Distribution of {col}")
    plt.ylabel("Density")
    plt.legend()
    plt.tight_layout()
    plt.show()

fd_gdp_col = "D_ln_gdp_growth_5y"
fd_liq_col = "D_liquid_liabilities"
plot_fd_distribution(df_Final_first_diff,col=fd_gdp_col,title="First Difference of ln GDP Growth (5-year)")
plot_fd_distribution(df_Final_first_diff,col=fd_liq_col,title="First Difference of Liquid Liabilities")
```

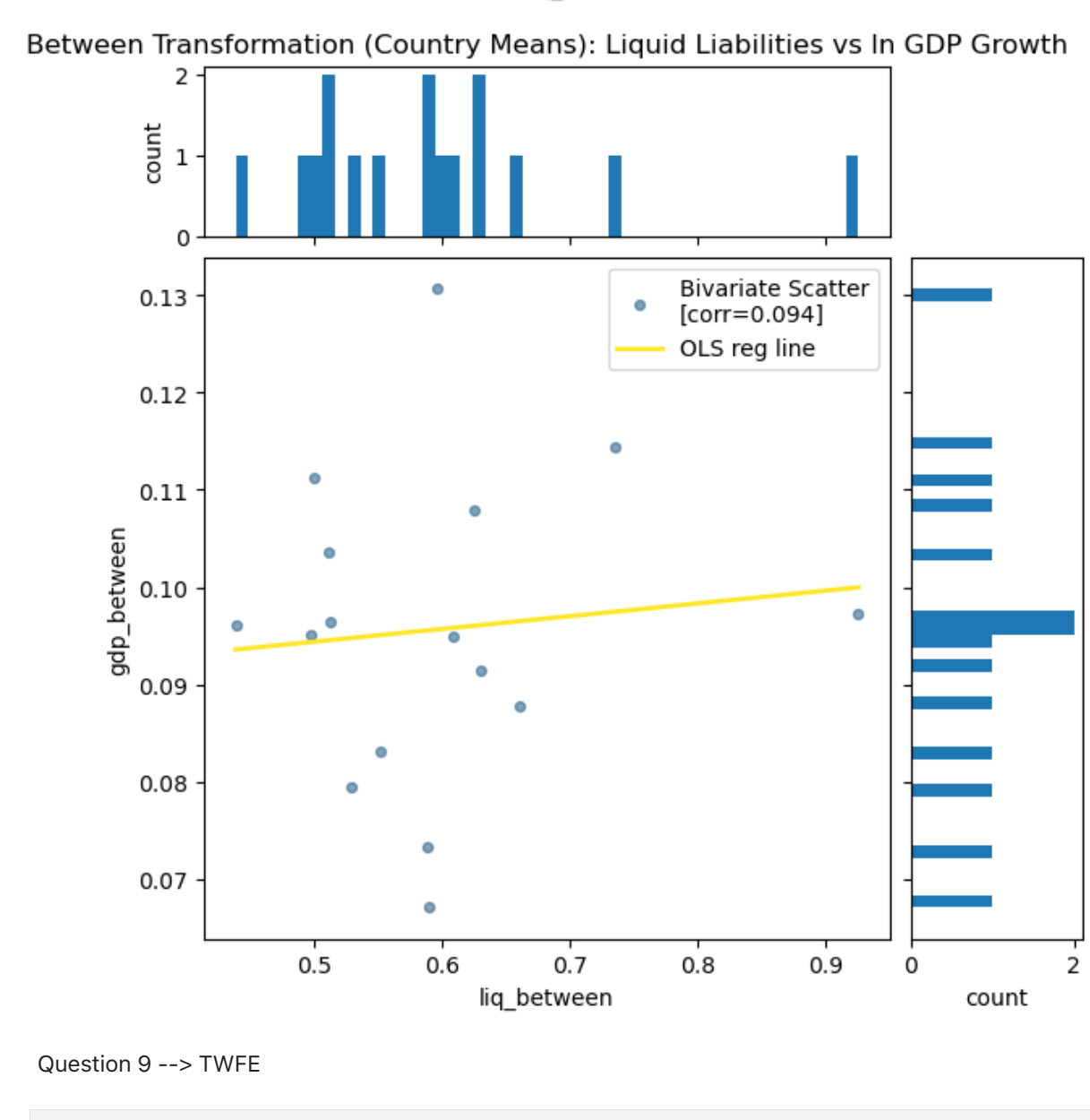




In [144…
```python
def scatter_with_marginals_cloud(df, x, y, title=None, bins=50):
    data = df[[x, y]].dropna()
    X = data[x].astype(float)
    Y = data[y].astype(float)
    corr = pearsonr(X, Y)[0]
    slope, intercept = np.polyfit(X, Y, 1)
    xx = np.linspace(X.min(), X.max(), 200)
    yy = slope * xx + intercept
    fig = plt.figure(figsize=(7, 7))
    gs = fig.add_gridspec(
        2, 2,
        height_ratios=(1, 4),
        width_ratios=(4, 1),
        hspace=0.05,
        wspace=0.05
    )
    ax_top   = fig.add_subplot(gs[0, 0])
    ax_main  = fig.add_subplot(gs[1, 0], sharex=ax_top)
    ax_right = fig.add_subplot(gs[1, 1], sharey=ax_main)
    ax_main.scatter(
        X, Y,
        s=18, alpha=0.6,
        color="#31688E",
        label=f"Bivariate Scatter\n[corr={corr:.3f}]"
    )
    ax_main.plot(xx, yy, linewidth=2, color="#FDE725", label="OLS reg line")
    ax_main.set_xlabel(x)
    ax_main.set_ylabel(y)
    ax_main.legend()
    ax_top.hist(X, bins=bins)
    ax_top.tick_params(axis="x", labelbottom=False)
    ax_top.set_ylabel("count")
    ax_right.hist(Y, bins=bins, orientation="horizontal")
    ax_right.tick_params(axis="y", labelleft=False)
    ax_right.set_xlabel("count")
    if title is None:
        title = f"{x} vs {y} (corr={corr:.3f})"
    ax_top.set_title(title)

    plt.show()

x_fd = "D_liquid_liabilities"
y_fd = "D_ln_gdp_growth_5y"

scatter_with_marginals_cloud( df_Final_first_diff,x=x_fd,y=y_fd,title="First Differences: Δ Liquid Liabilities vs Δ ln GDP Growth")


df_within = df_Final_first_diff.copy()
df_within["liq_within"] = df_within.groupby("country")["liquid_liabilities"] \
.transform(lambda v: v - v.mean())
df_within["gdp_within"] = df_within.groupby("country")["ln_gdp_growth_5y"] \
    .transform(lambda v: v - v.mean())

scatter_with_marginals_cloud(df_within,x="liq_within",y="gdp_within",title="Within Transformation (One-Way Fixed Effects): Liquid Liabilities vs ln GDP Growth")
df_between = df_Final_first_diff.groupby("country").agg({"liquid_liabilities": "mean","ln_gdp_growth_5y": "mean"}).reset_index()
df_between.rename(columns={"liquid_liabilities": "liq_between","ln_gdp_growth_5y": "gdp_between"}, inplace=True)
scatter_with_marginals_cloud(df_between,x="liq_between",y="gdp_between",title="Between Transformation (Country Means): Liquid Liabilities vs ln GDP Growth")
```

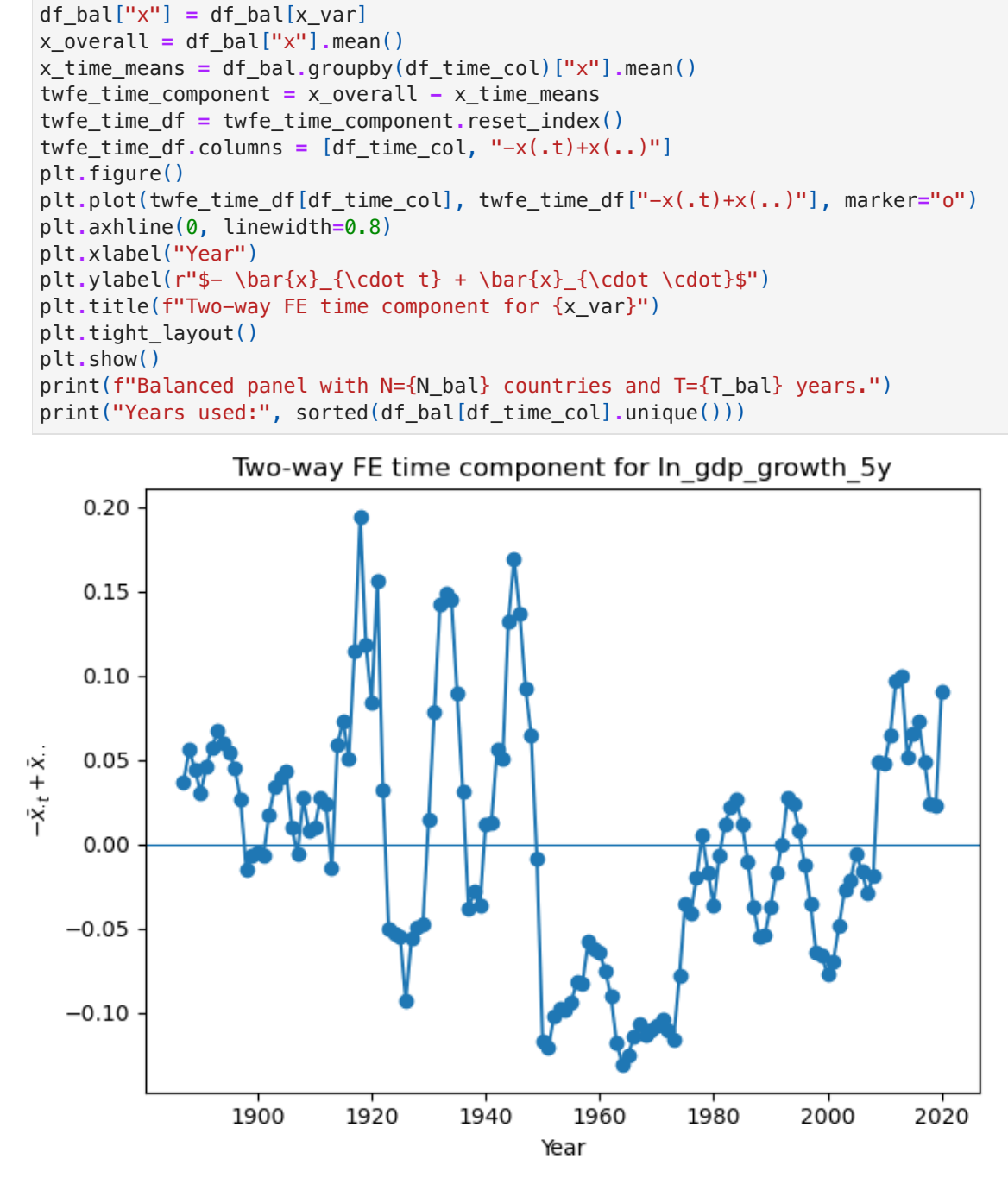






Question 9 --> TWFE

In [145…
```python
df_TWFE_Q = df_Final.copy()
df_id_col = "country"
df_time_col = "year"
T_per_country = df_TWFE_Q.groupby(df_id_col)[df_time_col].nunique()
T_max = T_per_country.max()
countries_longest = T_per_country[T_per_country == T_max].index
df_long = df_TWFE_Q[df_TWFE_Q[df_id_col].isin(countries_longest)].copy()
N = len(countries_longest)
year_counts = df_long.groupby(df_time_col)[df_id_col].nunique()
years_balanced = year_counts[year_counts == N].index
df_bal = df_long[df_long[df_time_col].isin(years_balanced)].copy()
df_bal = df_bal.sort_values([df_id_col, df_time_col])
N_bal = df_bal[df_id_col].nunique()
T_bal = df_bal[df_time_col].nunique()
x_var = "ln_gdp_growth_5y"
df_bal = df_bal.copy()
df_bal["x"] = df_bal[x_var]
x_overall = df_bal["x"].mean()
x_time_means = df_bal.groupby(df_time_col)["x"].mean()
twfe_time_component = x_overall - x_time_means
twfe_time_df = twfe_time_component.reset_index()
twfe_time_df.columns = [df_time_col, "-x(.t)+x(..)"]
plt.figure()
plt.plot(twfe_time_df[df_time_col], twfe_time_df["-x(.t)+x(..)"], marker="o")
plt.axhline(0, linewidth=0.8)
plt.xlabel("Year")
plt.ylabel(r"$- \bar{x}_{\cdot t} + \bar{x}_{\cdot\cdot}$")
plt.title(f"Two-way FE time component for {x_var}")
plt.tight_layout()
plt.show()
print(f"Balanced panel with N={N_bal} countries and T={T_bal} years.")
print("Years used:", sorted(df_bal[df_time_col].unique()))
```



```
Balanced panel with N=16 countries and T=139 years.
Years used: [np.int64(1882), np.int64(1883), np.int64(1884), np.int64(1885), np.int64(1886), np.int64(1887), np.int64(1888), np.int64(1889), np.int64(1890), np.int64(1891), np.int64(1892), np.int64(1
893), np.int64(1894), np.int64(1895), np.int64(1896), np.int64(1897), np.int64(1898), np.int64(1899), np.int64(1900), np.int64(1901), np.int64(1902), np.int64(1903), np.int64(1904), np.int64(1905), n
p.int64(1906), np.int64(1907), np.int64(1908), np.int64(1909), np.int64(1910), np.int64(1911), np.int64(1912), np.int64(1913), np.int64(1914), np.int64(1915), np.int64(1916), np.int64(1917), np.int6
4(1918), np.int64(1919), np.int64(1920), np.int64(1921), np.int64(1922), np.int64(1923), np.int64(1924), np.int64(1925), np.int64(1926), np.int64(1927), np.int64(1928), np.int64(1929), np.int64(193
0), np.int64(1931), np.int64(1932), np.int64(1933), np.int64(1934), np.int64(1935), np.int64(1936), np.int64(1937), np.int64(1938), np.int64(1939), np.int64(1940), np.int64(1941), np.int64(1942), np.
int64(1943), np.int64(1944), np.int64(1945), np.int64(1946), np.int64(1947), np.int64(1948), np.int64(1949), np.int64(1950), np.int64(1951), np.int64(1952), np.int64(1953), np.int64(1954), np.int64(1
955), np.int64(1956), np.int64(1957), np.int64(1958), np.int64(1959), np.int64(1960), np.int64(1961), np.int64(1962), np.int64(1963), np.int64(1964), np.int64(1965), np.int64(1966), np.int64(1967), n
p.int64(1968), np.int64(1969), np.int64(1970), np.int64(1971), np.int64(1972), np.int64(1973), np.int64(1974), np.int64(1975), np.int64(1976), np.int64(1977), np.int64(1978), np.int64(1979), np.int6
4(1980), np.int64(1981), np.int64(1982), np.int64(1983), np.int64(1984), np.int64(1985), np.int64(1986), np.int64(1987), np.int64(1988), np.int64(1989), np.int64(1990), np.int64(1991), np.int64(199
2), np.int64(1993), np.int64(1994), np.int64(1995), np.int64(1996), np.int64(1997), np.int64(1998), np.int64(1999), np.int64(2000), np.int64(2001), np.int64(2002), np.int64(2003), np.int64(2004), np.
int64(2005), np.int64(2006), np.int64(2007), np.int64(2008), np.int64(2009), np.int64(2010), np.int64(2011), np.int64(2012), np.int64(2013), np.int64(2014), np.int64(2015), np.int64(2016), np.int64(2
017), np.int64(2018), np.int64(2019), np.int64(2020)]
```

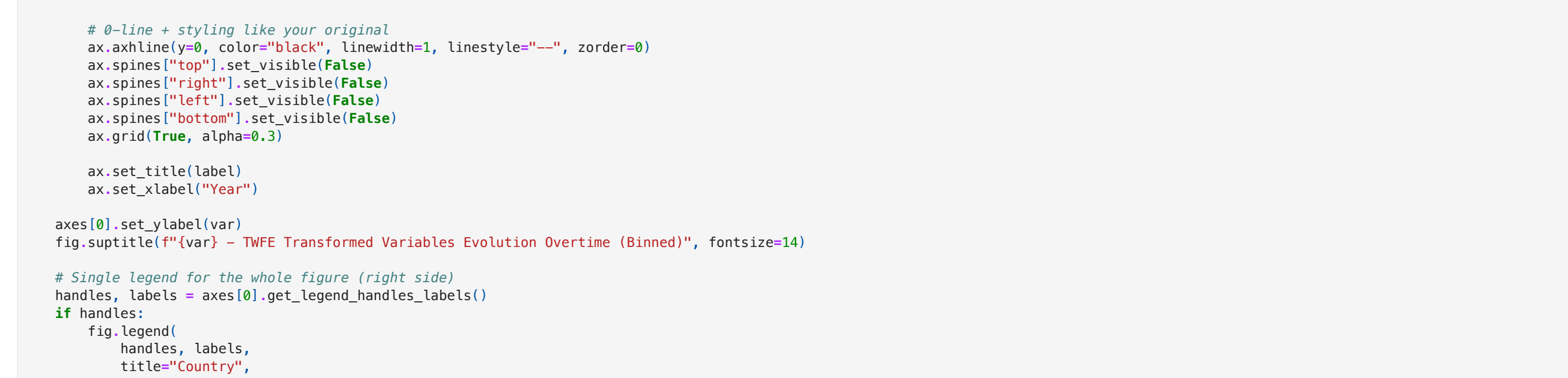
In [155…
```python
vars_for_twfe = ["ln_gdp_growth_5y", "liquid_liabilities", "ln_cpi", "ln_trade_openness", "mcap_gdp", "bank_loans_gdp", "ln_labour_productivity_growth_5y", 'ln_gdppc_lag5']

df_twfe = df_bal.copy()

for x_var in vars_for_twfe:
    x = df_twfe[x_var]
    x_i_dot = df_twfe.groupby(df_id_col)[x_var].transform("mean")       # x(i.)
    x_dot_t = df_twfe.groupby(df_time_col)[x_var].transform("mean")     # x(.t)
    x_dot_dot = df_twfe[x_var].mean()                                   # x(..)
    df_twfe[f"{x_var}_twfe"] = x - x_i_dot - x_dot_t + x_dot_dot
df_twfe[[df_id_col, df_time_col, "ln_gdp_growth_5y", "ln_gdp_growth_5y_twfe",
        "liquid_liabilities", "liquid_liabilities_twfe"]].head()
vars_for_twfe = ["ln_gdp_growth_5y", "liquid_liabilities"]
overall_means = df_bal[vars_for_twfe].mean()
time_means = df_bal.groupby(df_time_col)[vars_for_twfe].mean()
time_component = overall_means - time_means
table = time_component.reset_index()
table = table.rename(columns={df_time_col: "Time"})
new_cols = ["Time"] + [f"-{v}(.t)+{v}(..)" for v in vars_for_twfe]
table.columns = new_cols

print(table.to_string(index=False))
```

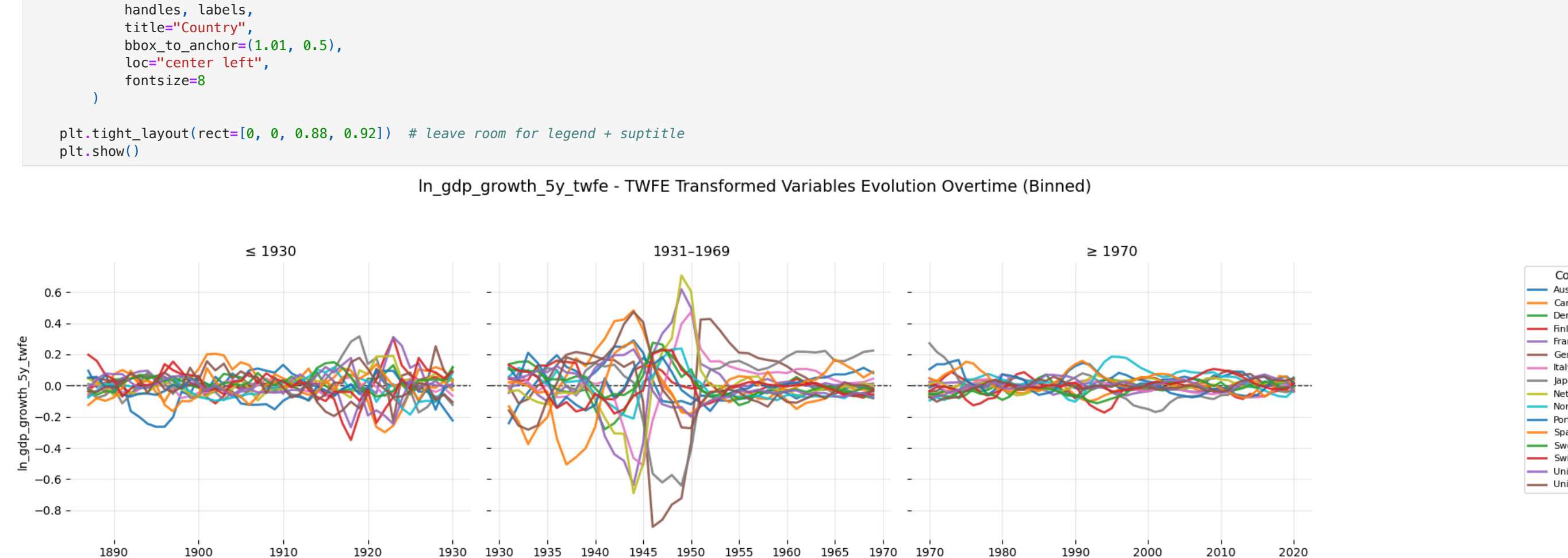
```
Time  -ln_gdp_growth_5y(.t)+ln_gdp_growth_5y(..)  -liquid_liabilities(.t)+liquid_liabilities(..)
1882                                         NaN                                          0.2593
1883                                         NaN                                          0.2499
1884                                         NaN                                          0.2346
1885                                         NaN                                          0.2225
1886                                         NaN                                          0.2125
1887                                      0.0372                                          0.1996
1888                                      0.0558                                          0.1947
```

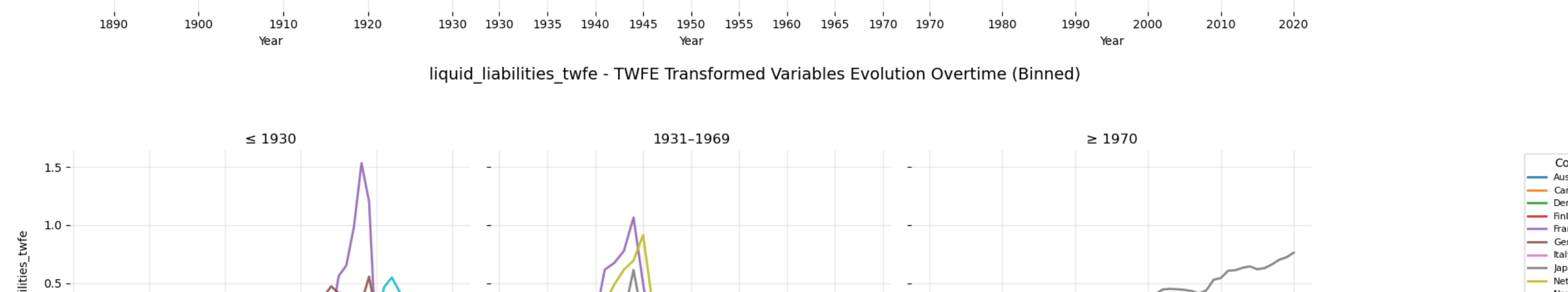
```
1889                    0.0446                    0.1992
1890                    0.0303                    0.2061
1891                    0.0464                    0.2124
1892                    0.0575                    0.1979
1893                    0.0670                    0.1837
1894                    0.0596                    0.1702
1895                    0.0542                    0.1588
1896                    0.0451                    0.1540
1897                    0.0263                    0.1496
1898                   -0.0147                    0.1525
1899                   -0.0071                    0.1408
1900                   -0.0050                    0.1409
1901                   -0.0068                    0.1266
1902                    0.0175                    0.1152
1903                    0.0341                    0.1167
1904                    0.0392                    0.0998
1905                    0.0435                    0.0864
1906                    0.0103                    0.0837
1907                   -0.0061                    0.0959
1908                    0.0272                    0.0651
1909                    0.0079                    0.0557
1910                    0.0099                    0.0517
1911                    0.0273                    0.0571
1912                    0.0236                    0.0682
1913                   -0.0143                    0.0711
1914                    0.0580                    0.0183
1915                    0.0733                    0.0040
1916                    0.0504                    0.0323
1917                    0.1147                   -0.0049
1918                    0.1941                   -0.0656
1919                    0.1181                   -0.0616
1920                    0.0840                    0.0968
1921                    0.1557                    0.0499
1922                    0.0321                    0.0394
1923                   -0.0504                    0.0609
1924                   -0.0530                    0.0899
1925                   -0.0545                    0.1025
1926                   -0.0928                    0.0850
1927                   -0.0554                    0.0662
1928                   -0.0492                    0.0660
1929                   -0.0470                    0.0638
1930                    0.0144                    0.0205
1931                    0.0786                   -0.0354
1932                    0.1422                   -0.0766
1933                    0.1490                   -0.0776
1934                    0.1453                   -0.0592
1935                    0.0895                   -0.0491
1936                    0.0315                   -0.0364
1937                   -0.0379                   -0.0155
1938                   -0.0283                   -0.0440
1939                   -0.0365                   -0.0338
1940                    0.0121                   -0.0429
1941                    0.0125                   -0.0726
1942                    0.0559                   -0.0946
1943                    0.0508                   -0.1500
1944                    0.1321                   -0.2553
1945                    0.1693                   -0.2183
1946                    0.1364                   -0.0824
1947                    0.0927                   -0.0121
1948                    0.0647                    0.0260
1949                   -0.0088                    0.0407
1950                   -0.1164                    0.0675
1951                   -0.1207                    0.1071
1952                   -0.1021                    0.1062
1953                   -0.0971                    0.0974
1954                   -0.0978                    0.0933
1955                   -0.0938                    0.0938
1956                   -0.0815                    0.1086
1957                   -0.0022                    0.1104
1958                   -0.0577                    0.0942
1959                   -0.0618                    0.0857
1960                   -0.0644                    0.0837
1961                   -0.0748                    0.0813
1962                   -0.0902                    0.0748
1963                   -0.1178                    0.0641
1964                   -0.1307                    0.0639
1965                   -0.1250                    0.0618
1966                   -0.1142                    0.0602
1967                   -0.1060                    0.0519
1968                   -0.1128                    0.0402
1969                   -0.1098                    0.0440
1970                   -0.1077                    0.0555
1971                   -0.1036                    0.0380
1972                   -0.1097                    0.0160
1973                   -0.1154                    0.0121
1974                   -0.0783                    0.0331
1975                   -0.0354                    0.0258
1976                   -0.0405                    0.0237
1977                   -0.0198                    0.0286
1978                    0.0051                    0.0037
1979                   -0.0163                   -0.0115
1980                   -0.0361                    0.0090
1981                   -0.0069                    0.0072
1982                    0.0122                   -0.0134
1983                    0.0219                   -0.0168
1984                    0.0270                   -0.0173
1985                    0.0117                   -0.0237
1986                   -0.0101                   -0.0292
1987                   -0.0372                   -0.0377
1988                   -0.0548                   -0.0425
1989                   -0.0539                   -0.0467
1990                   -0.0368                   -0.0483
1991                   -0.0172                   -0.0504
1992                   -0.0001                   -0.0549
1993                    0.0279                   -0.0674
1994                    0.0235                   -0.0580
1995                    0.0080                   -0.0587
1996                   -0.0121                   -0.0621
1997                   -0.0504                   -0.0584
1998                   -0.0643                   -0.0596
1999                   -0.0659                   -0.0704
2000                   -0.0773                   -0.0569
2001                   -0.0695                   -0.0762
2002                   -0.0483                   -0.0914
2003                   -0.0270                   -0.1107
2004                   -0.0216                   -0.1167
2005                   -0.0053                   -0.1378
2006                   -0.0157                   -0.1549
2007                   -0.0292                   -0.1864
2008                   -0.0189                   -0.2170
2009                    0.0486                   -0.2579
2010                    0.0476                   -0.2527
2011                    0.0649                   -0.2581
2012                    0.0970                   -0.2832
2013                    0.0996                   -0.2941
2014                    0.0520                   -0.3054
2015                    0.0653                   -0.3298
2016                    0.0725                   -0.3545
2017                    0.0491                   -0.3639
2018                    0.0239                   -0.3644
2019                    0.0233                   -0.3782
2020                    0.0909                   -0.5318
```

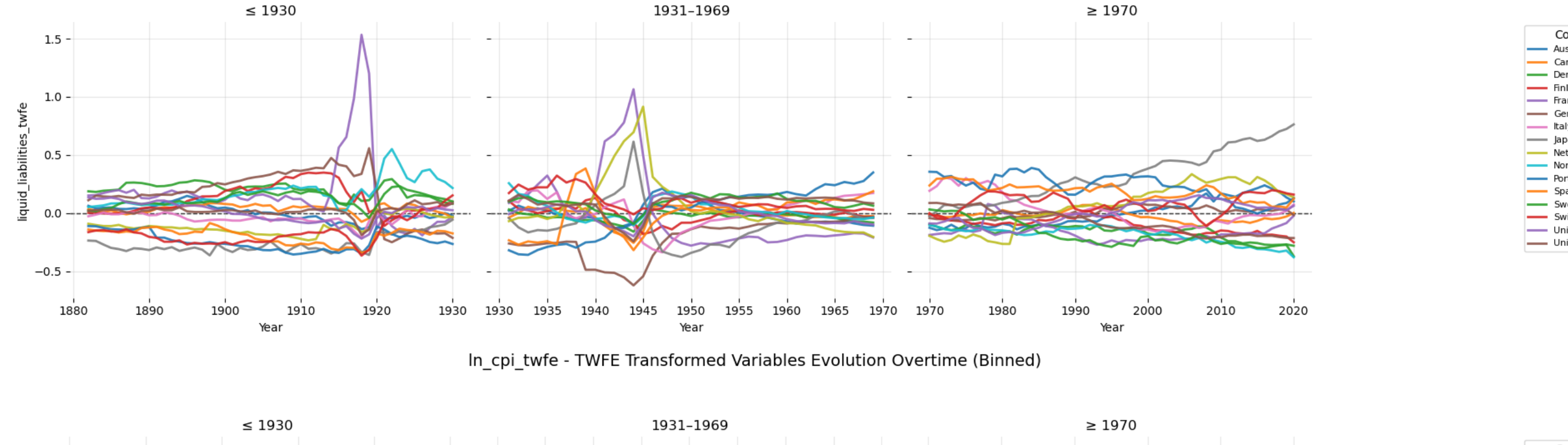
In [156…

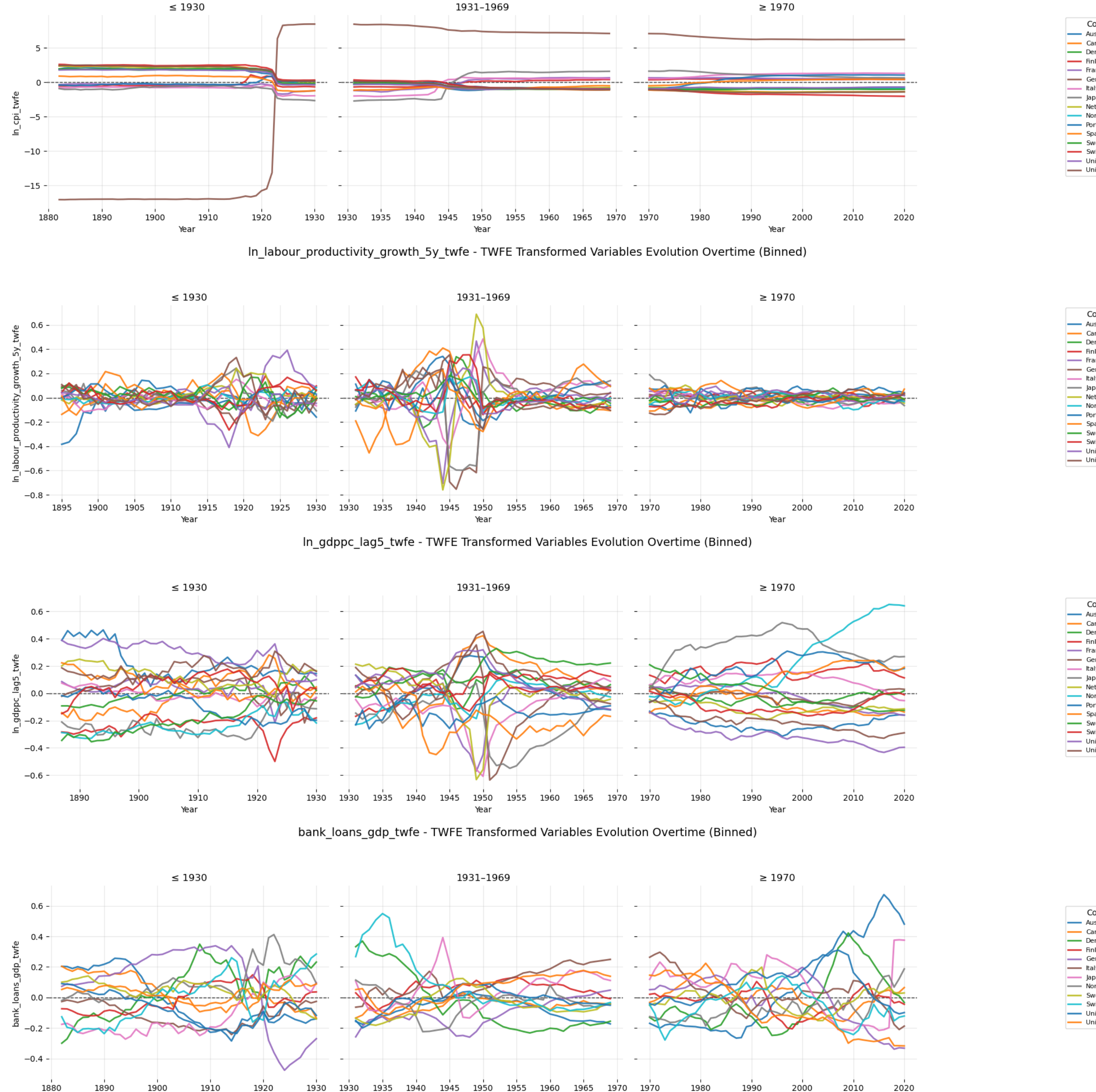
```
id_col = "country"
time_col = "year"

# ALL countries (or replace with your subset list)
target_countries = sorted(df_twfe[id_col].unique())

vars_16 = [
    "ln_gdp_growth_5y",
    "liquid_liabilities",
    "ln_cpi",
    "ln_labour_productivity_growth_5y",
    "ln_gdppc_lag5",
    "bank_loans_gdp",
    "mcap_gdp",
    "ln_trade_openness"
]
target_variables = [v + "_twfe" for v in vars_16]

# Optional: custom colors (leave empty to let matplotlib choose automatically)
country_colors = {}  # e.g. {"Italy": "#576b6b", ...}

# Sort once
df_twfe_plot = df_twfe.sort_values([id_col, time_col]).copy()

# Time bins (label, min_year, max_year)
time_bins = [
    ("≤ 1930", None, 1930),
    ("1931–1969", 1931, 1969),
    ("≥ 1970", 1970, None),
]

def slice_by_period(d, min_year, max_year):
    if min_year is not None:
        d = d[d[time_col] >= min_year]
    if max_year is not None:
        d = d[d[time_col] <= max_year]
    return d

for var in target_variables:
    fig, axes = plt.subplots(1, 3, figsize=(18, 5), sharey=True)

    for ax, (label, min_y, max_y) in zip(axes, time_bins):
        for country in target_countries:
            d = df_twfe_plot[df_twfe_plot[id_col] == country][[time_col, var]].dropna()
            d = slice_by_period(d, min_y, max_y)

            if d.empty:
                continue

            ax.plot(
                d[time_col],
                d[var],
                linewidth=2,
                alpha=0.9,
                label=country,
                color=country_colors.get(country, None)  # None => default cycle
            )

        # 0-line + styling like your original
        ax.axhline(y=0, color="black", linewidth=1, linestyle="--", zorder=0)
        ax.spines["top"].set_visible(False)
        ax.spines["right"].set_visible(False)
        ax.spines["left"].set_visible(False)
        ax.spines["bottom"].set_visible(False)
        ax.grid(True, alpha=0.3)

        ax.set_title(label)
        ax.set_xlabel("Year")

    axes[0].set_ylabel(var)
    fig.suptitle(f"{var} - TWFE Transformed Variables Evolution Overtime (Binned)", fontsize=14)

    # Single legend for the whole figure (right side)
    handles, labels = axes[0].get_legend_handles_labels()
    if handles:
        fig.legend(
            handles, labels,
            title="Country",
            bbox_to_anchor=(1.01, 0.5),
            loc="center left",
            fontsize=8
        )

    plt.tight_layout(rect=[0, 0, 0.88, 0.92])  # leave room for legend + suptitle
    plt.show()
```

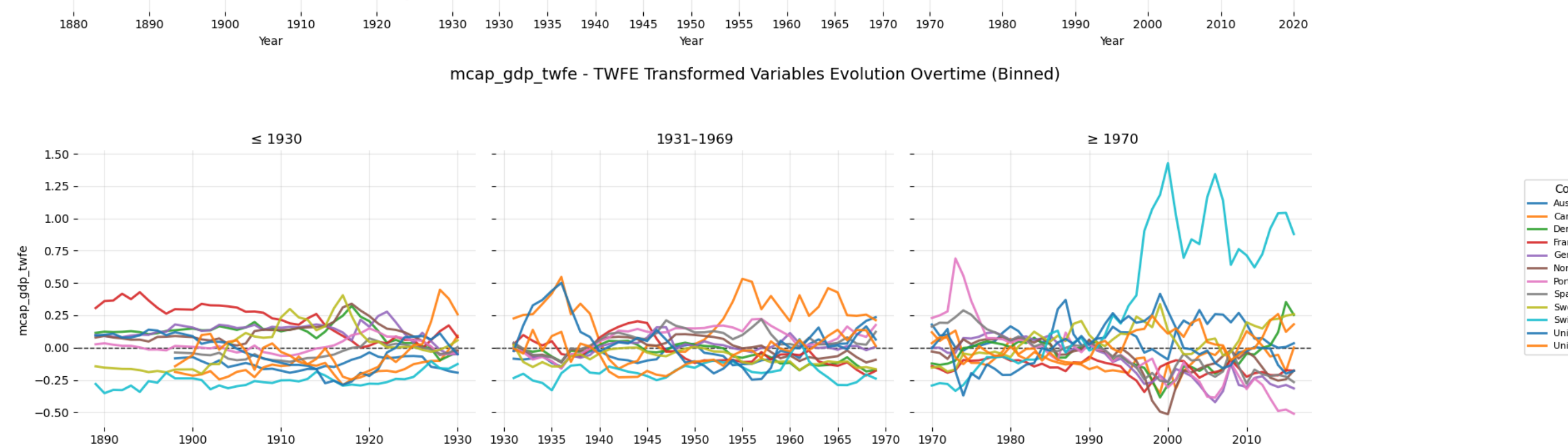


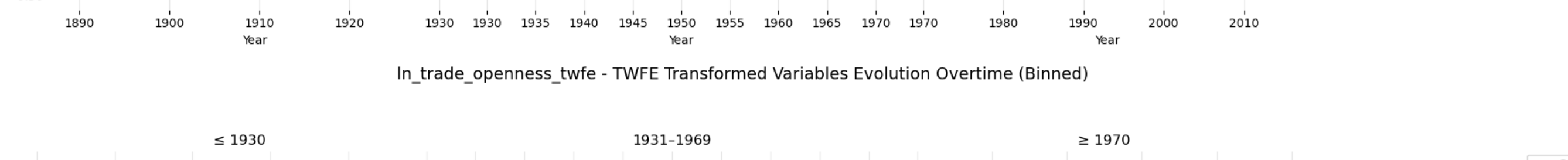


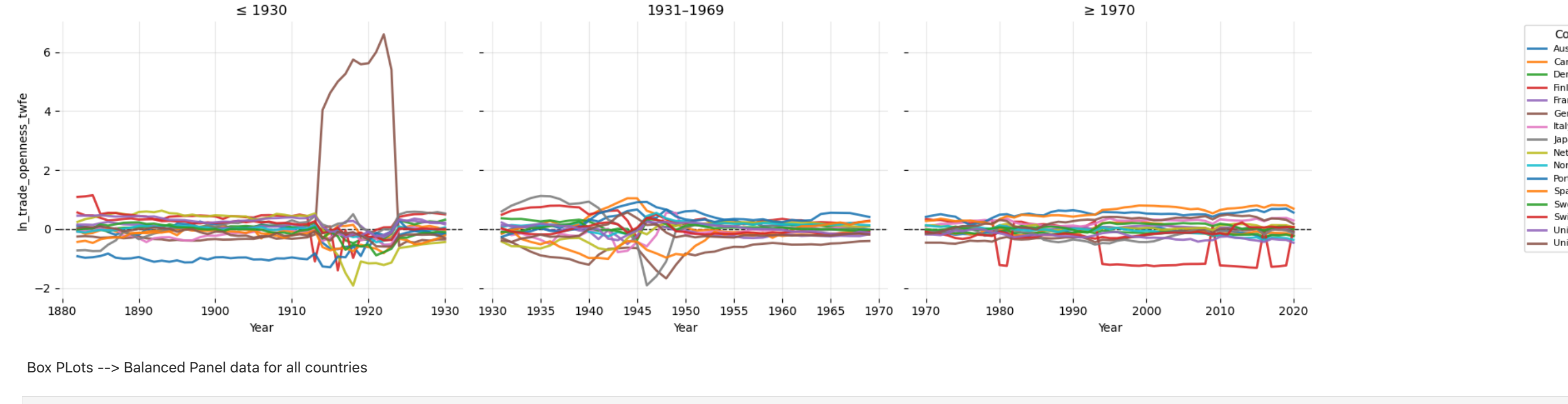












Box PLots --> Balanced Panel data for all countries

In [157…

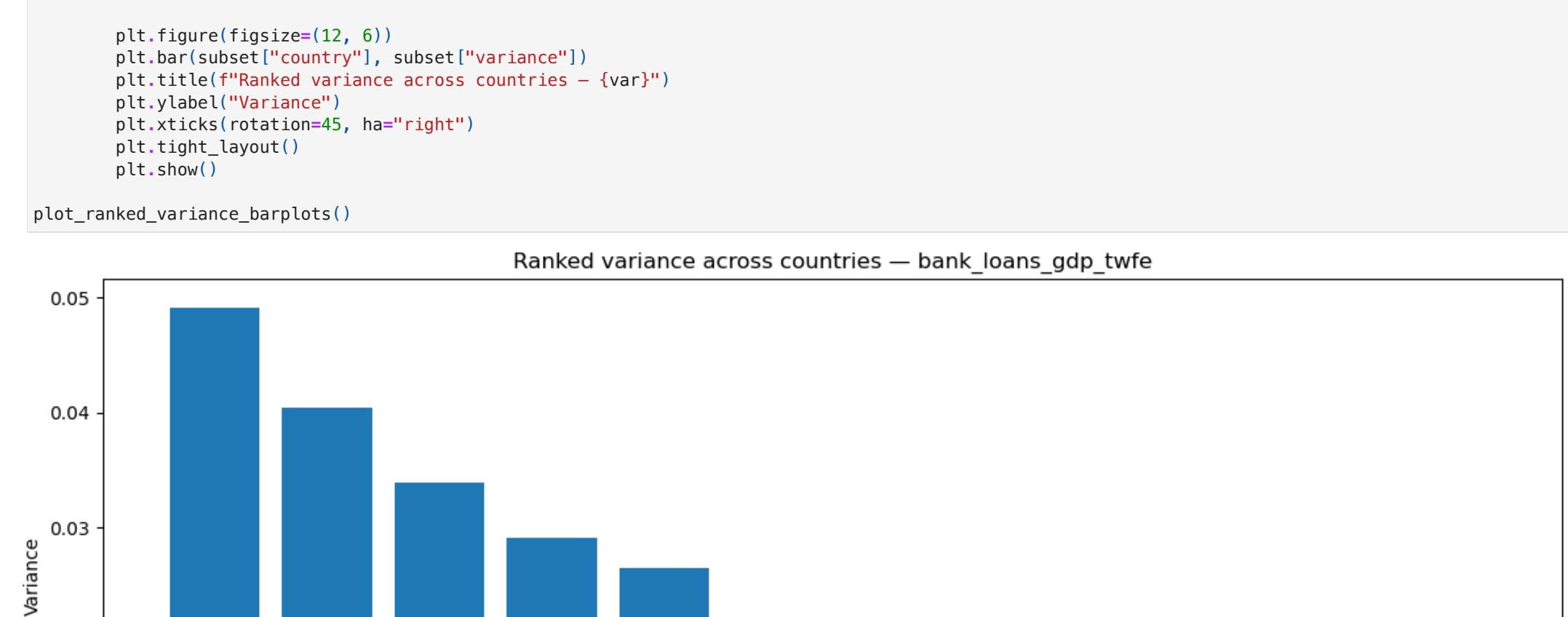
```
df_twfe.describe()
df_twfe_describe = df_twfe.copy()
df_twfe_describe = df_twfe_describe[['ln_gdp_growth_5y_twfe', 'liquid_liabilities_twfe']]
df_twfe_describe.describe()
df_twfe.groupby("country")[f"{v}_twfe"].mean()
```

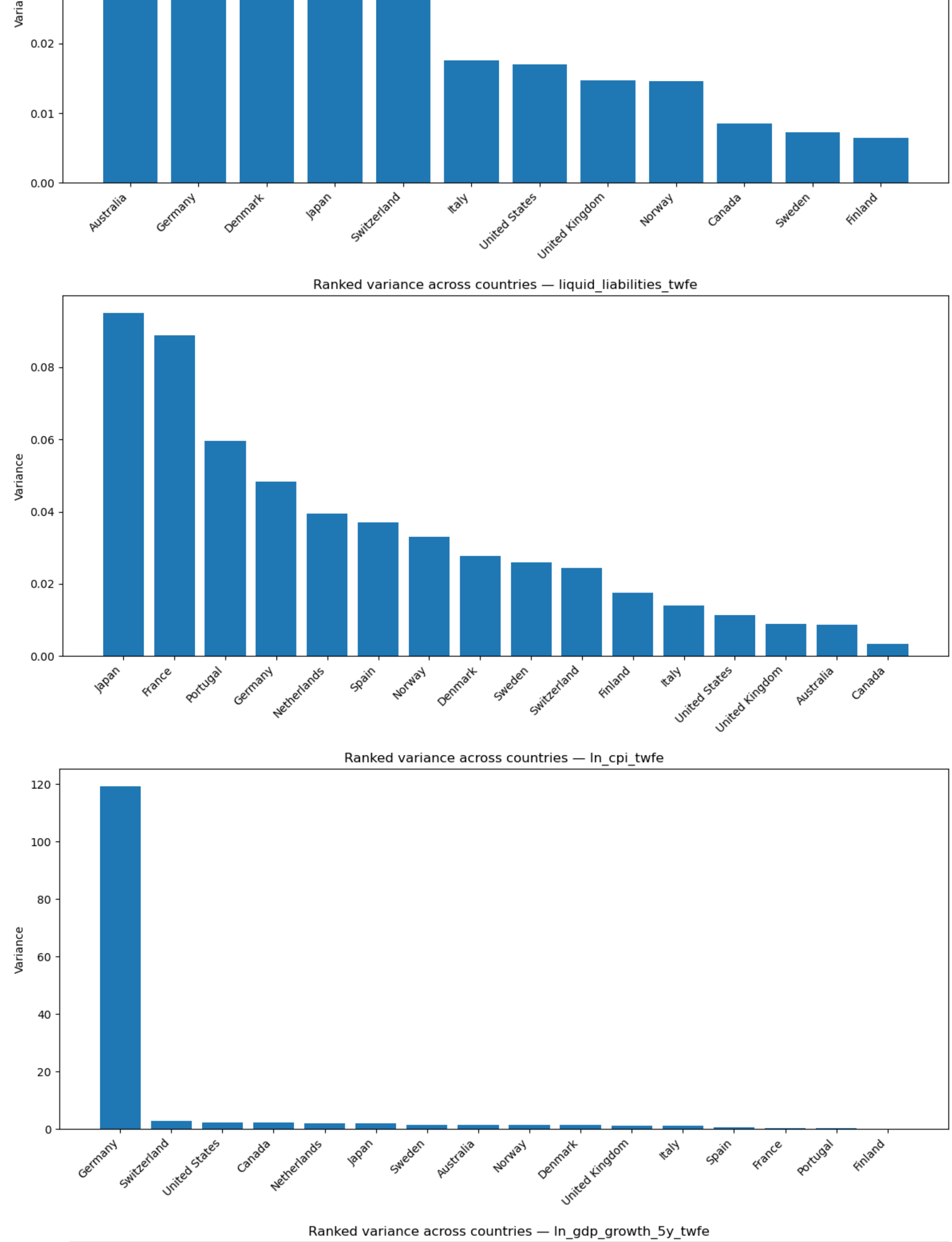
Out[157…

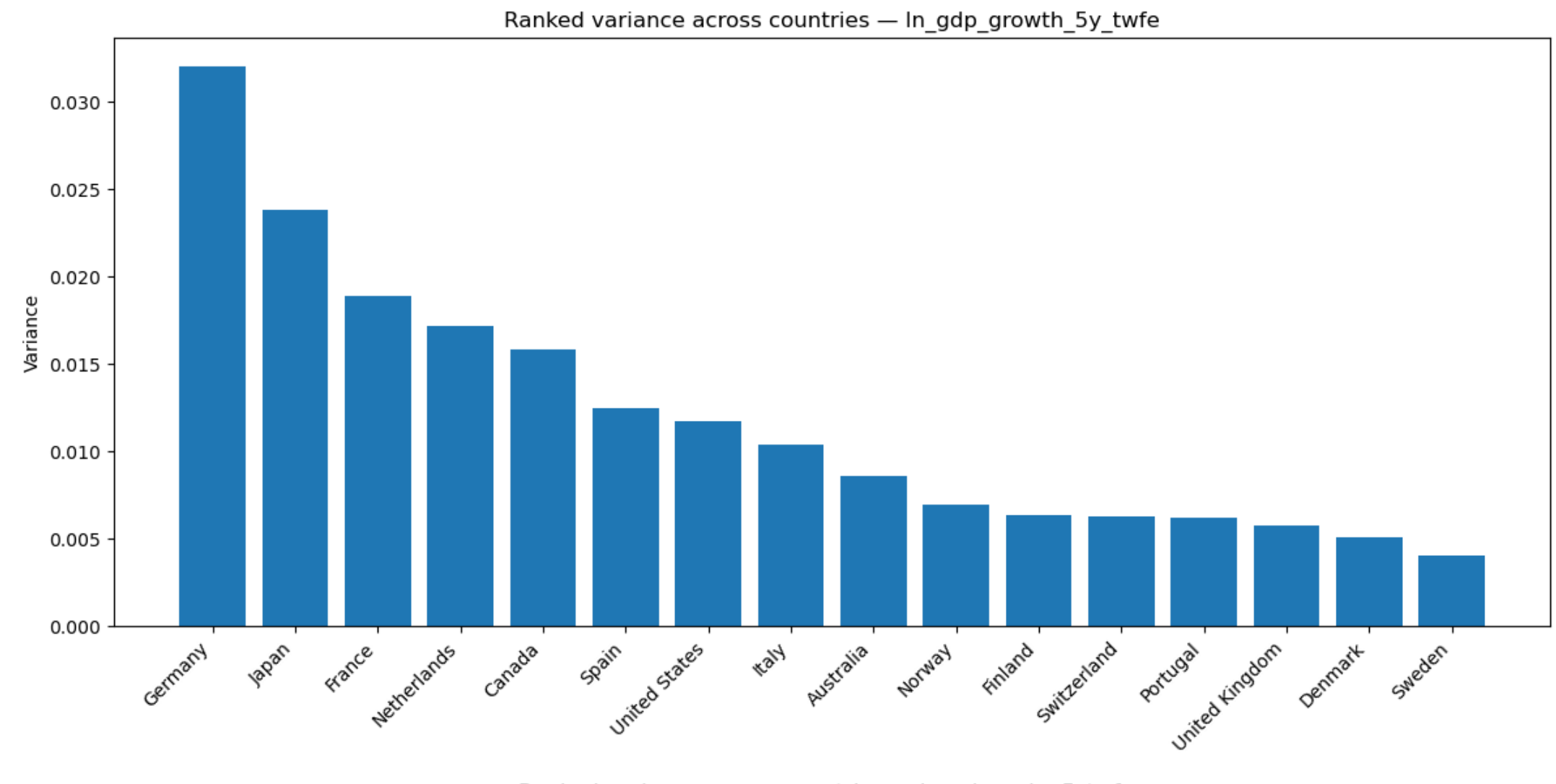
```
country
Australia         0.0000
Canada            0.0000
Denmark           0.0000
Finland           0.0000
France            0.0000
Germany           0.0000
Italy             0.0000
Japan             0.0000
Netherlands       0.0000
Norway            0.0000
Portugal          0.0000
Spain             0.0000
Sweden            0.0000
Switzerland       0.0000
United Kingdom    0.0000
United States     0.0000
Name: ln_labour_productivity_growth_5y_twfe, dtype: float64
```

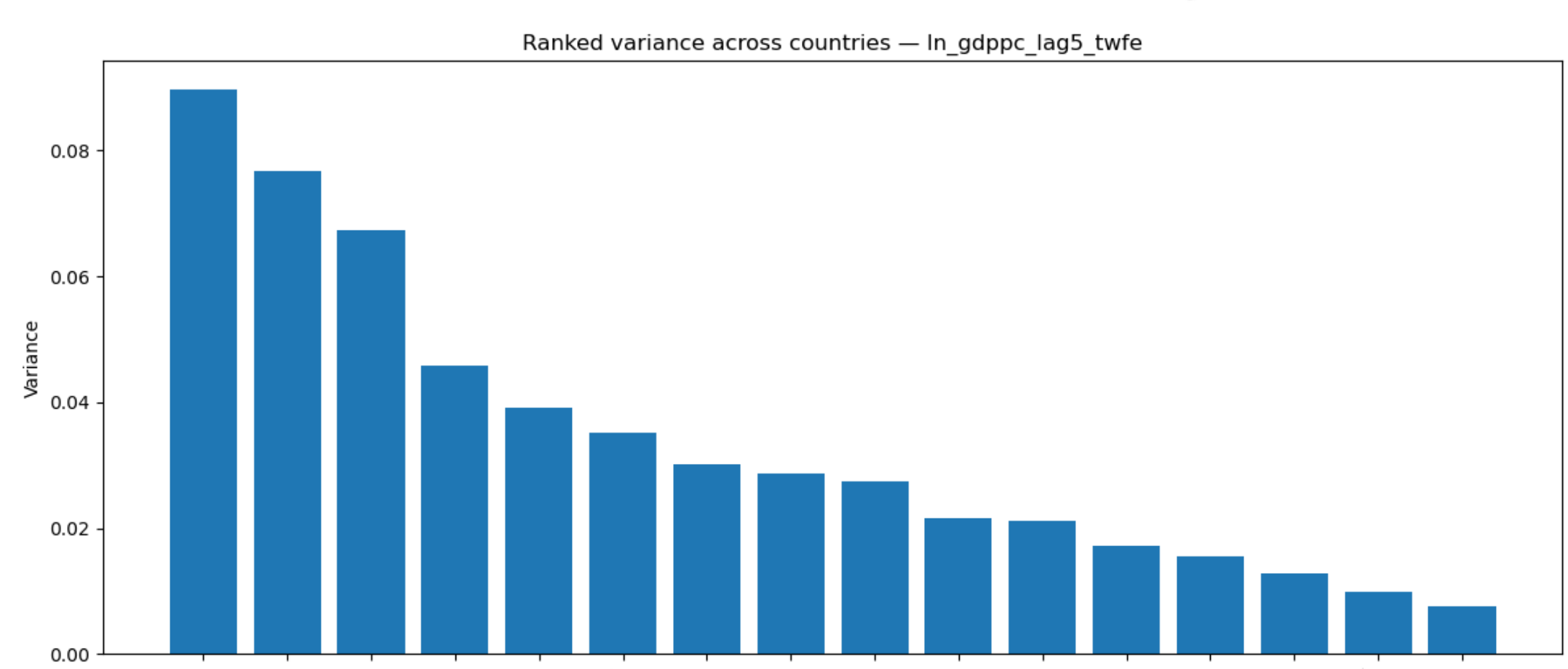
In [158…

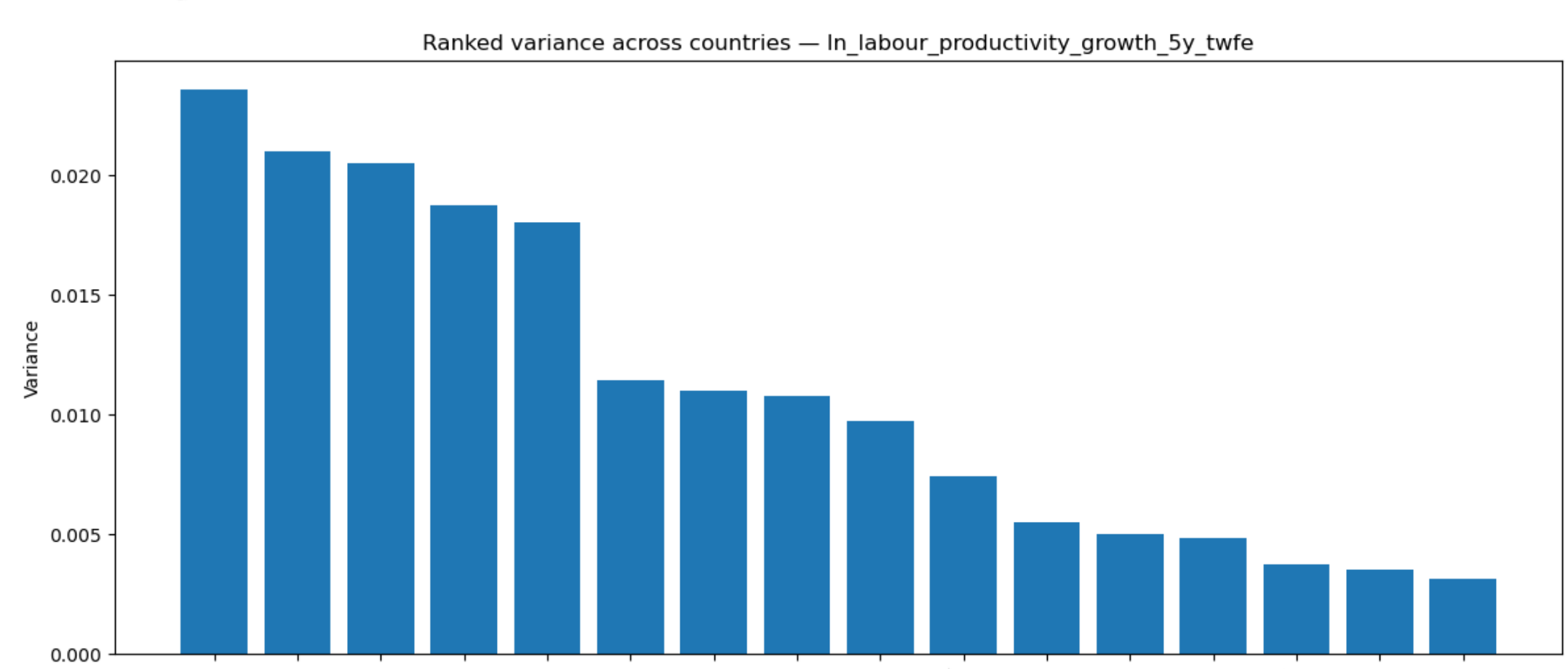
```
#Descriptive statistics by country
cols = ["ln_gdp_growth_5y_twfe","liquid_liabilities_twfe","ln_cpi_twfe","ln_labour_productivity_growth_5y_twfe","ln_gdppc_lag5_twfe","bank_loans_gdp_twfe","mcap_gdp_twfe","ln_trade_openness_twfe",]
country_dessy_stats = df_twfe.groupby('country')[cols].agg(['count', 'mean', 'std', 'var', 'min', 'max'])
countries_var_by_diff_var = country_dessy_stats.loc[:, pd.IndexSlice[:, 'var']].copy()
countries_var_by_diff_var.columns = countries_var_by_diff_var.columns.get_level_values(0)

# long- table
countries_var_by_diff_var_long = ( countries_var_by_diff_var .reset_index() .melt(id_vars='country', var_name='variable', value_name='variance'))

countries_var_by_diff_var_long = countries_var_by_diff_var_long.sort_values( ["variable","variance" ], ascending= [True,True])



def plot_ranked_variance_barplots():
    variables = countries_var_by_diff_var_long["variable"].unique()

    for var in variables:
        subset = countries_var_by_diff_var_long[
            countries_var_by_diff_var_long["variable"] == var
        ].sort_values("variance", ascending=False)

        plt.figure(figsize=(12, 6))
        plt.bar(subset["country"], subset["variance"])
        plt.title(f"Ranked variance across countries — {var}")
        plt.ylabel("Variance")
        plt.xticks(rotation=45, ha="right")
        plt.tight_layout()
        plt.show()

plot_ranked_variance_barplots()
```

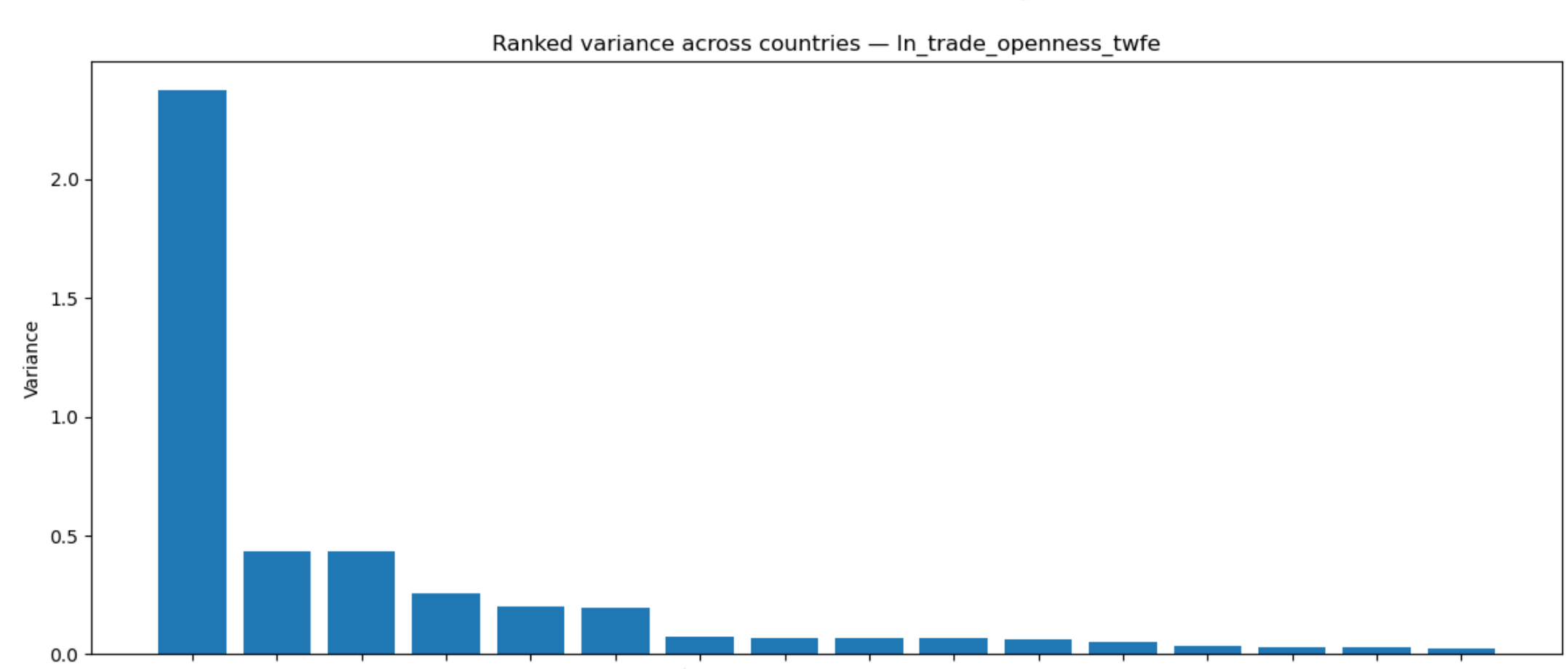


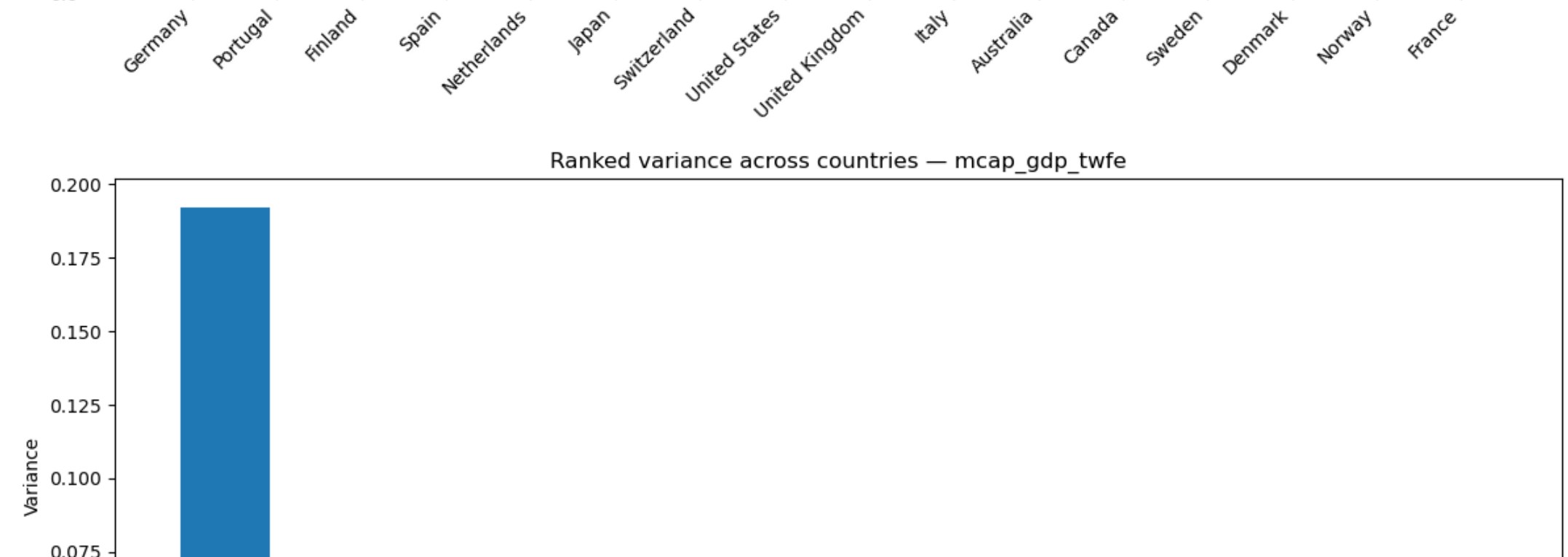


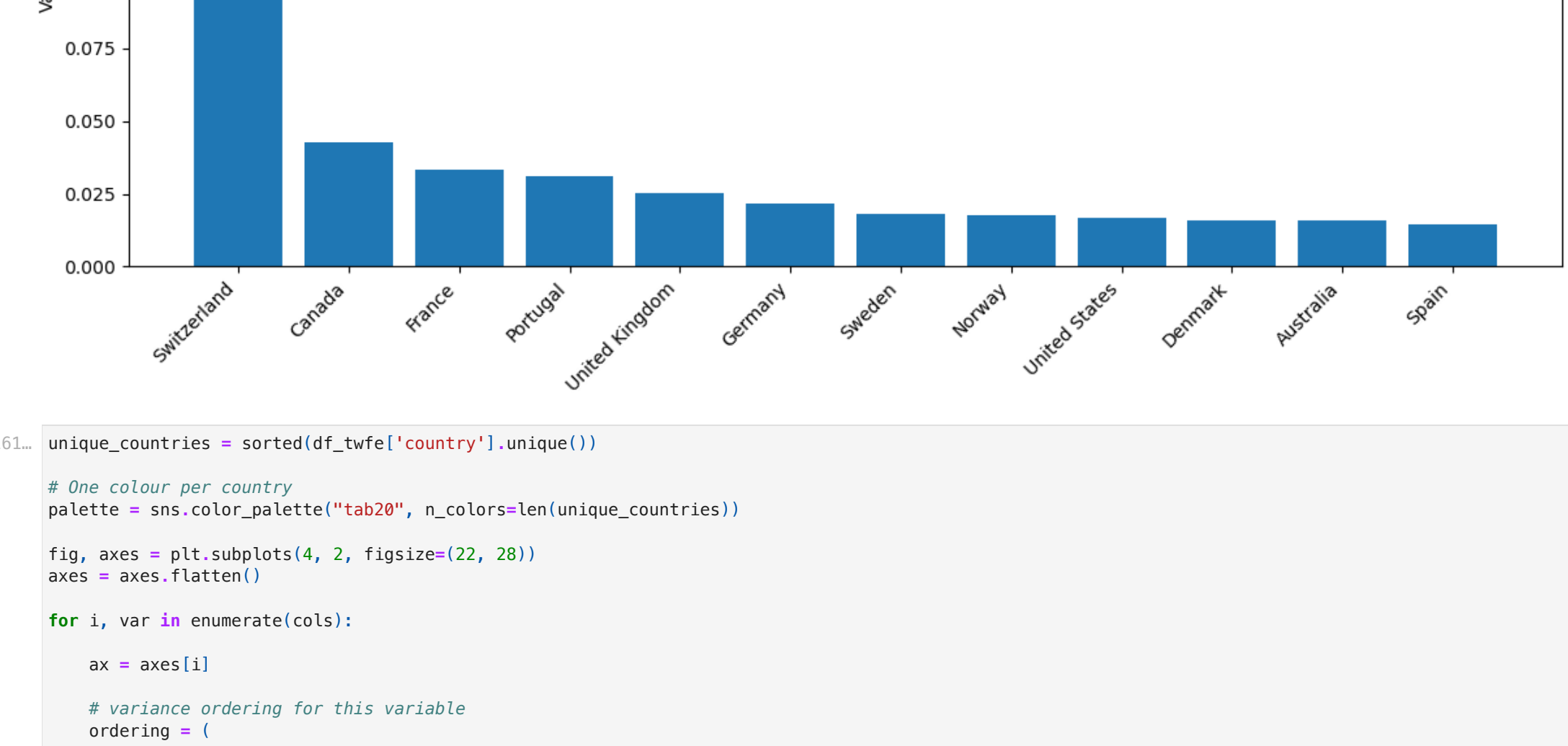








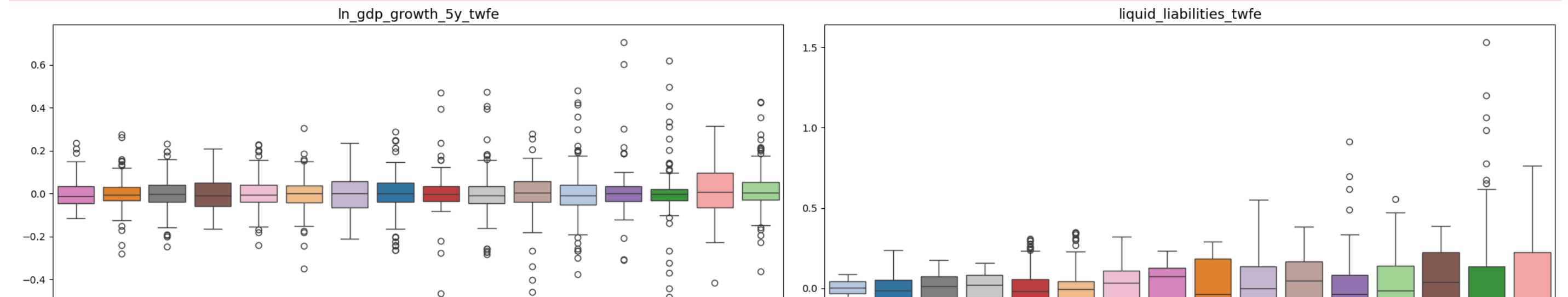


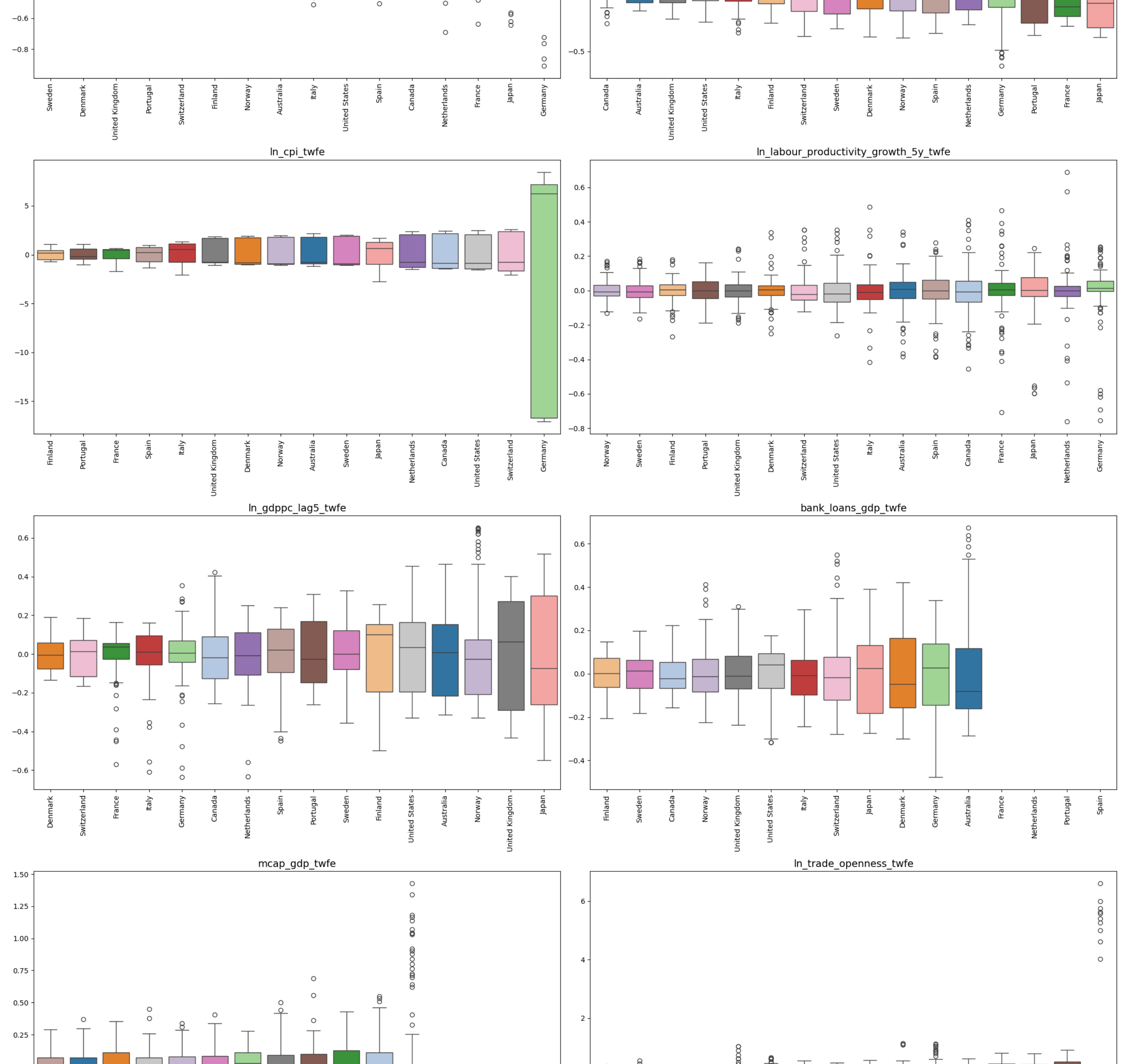


In [161…

```
unique_countries = sorted(df_twfe['country'].unique())

# One colour per country
palette = sns.color_palette("tab20", n_colors=len(unique_countries))

fig, axes = plt.subplots(4, 2, figsize=(22, 28))
axes = axes.flatten()

for i, var in enumerate(cols):

    ax = axes[i]

    # variance ordering for this variable
    ordering = (
        countries_var_by_diff_var_long
        .query("variable == @var")
        .sort_values("variance")["country"]
        .tolist()
    )

    # dataset for plotting
    df_plot = df_twfe[["country", var]].copy()
    df_plot["country"] = pd.Categorical(df_plot["country"], categories=ordering, ordered=True)

    # country → colour mapping dictionary
    colour_map = {country: palette[j] for j, country in enumerate(unique_countries)}

    sns.boxplot(
        data=df_plot,
        x="country",
        y=var,
        order=ordering,
        ax=ax,
        palette=[colour_map[c] for c in ordering]  # colours aligned with ordering
    )

    ax.set_title(f"{var}", fontsize=14)
    ax.set_xlabel("")
    ax.set_ylabel("")
    ax.tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()
```

```
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
```





In [162…

```python
overall_desc = df_twfe[cols].describe().T
overall_desc['median'] = df_twfe[cols].median()
overall_desc = overall_desc[['count','mean','std','min','25%','median','75%','max']]
overall_desc
```

Out[162…

	count	mean	std	min	25%	median	75%	max
ln_gdp_growth_5y_twfe	2,144.0000	0.0000	0.1091	-0.9063	-0.0415	-0.0023	0.0418	0.7056
liquid_liabilities_twfe	2,224.0000	0.0000	0.1836	-0.6222	-0.1271	-0.0005	0.1041	1.5332
ln_cpi_twfe	2,224.0000	0.0000	2.9673	-17.0637	-0.9511	-0.2890	1.0618	8.4539
ln_labour_productivity_growth_5y_twfe	2,016.0000	0.0000	0.1052	-0.7599	-0.0408	-0.0015	0.0417	0.6881
ln_gdppc_lag5_twfe	2,144.0000	-0.0000	0.1841	-0.6361	-0.1243	0.0075	0.1186	0.6517
bank_loans_gdp_twfe	1,668.0000	-0.0000	0.1482	-0.4767	-0.1039	-0.0070	0.0868	0.6733
mcap_gdp_twfe	1,488.0000	0.0000	0.1928	-0.5146	-0.1173	-0.0123	0.0857	1.4266
ln_trade_openness_twfe	2,224.0000	-0.0000	0.5240	-1.9155	-0.1881	0.0001	0.1906	6.5972

In [163…

```python
country_stats = (
    df_twfe.groupby('country')[cols]
    .agg(['count','mean','std','var','min','max'])
)
country_stats
```

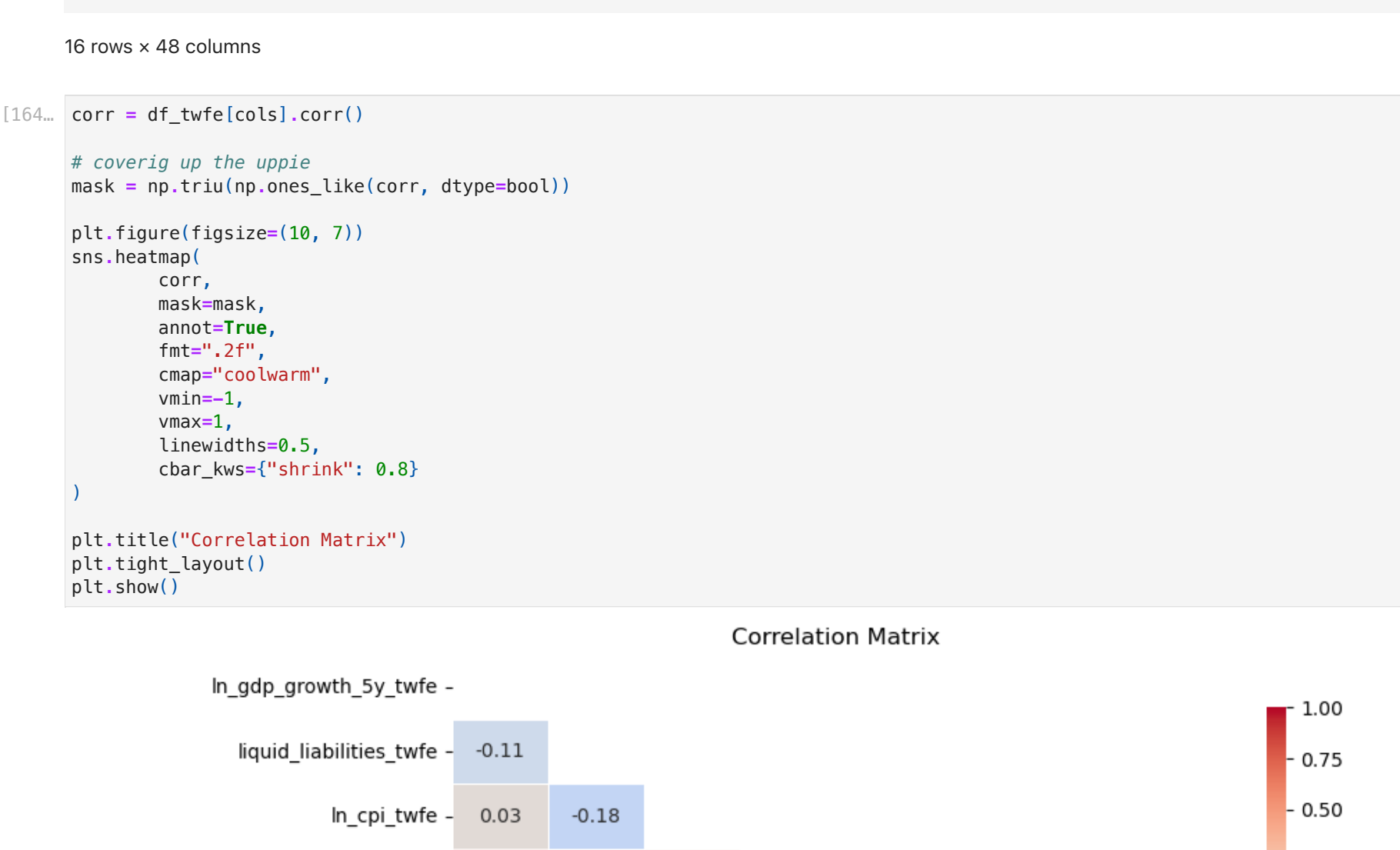
Out[163…

	ln_gdp_growth_5y_twfe						liquid_liabilities_twfe				...	mcap_gdp_twfe				ln_trade_openness_twfe					
	count	mean	std	var	min	max	count	mean	std	var	...	std	var	min	max	count	mean	std	var	min	max
country																					
Australia	134	0.0000	0.0929	0.0086	-0.2649	0.2893	139	0.0000	0.0934	0.0087	...	0.1252	0.0157	-0.1936	0.3696	139	-0.0000	0.2499	0.0624	-0.6827	0.6628
Canada	134	0.0000	0.1258	0.0158	-0.3753	0.4814	139	0.0000	0.0588	0.0035	...	0.2065	0.0426	-0.3517	0.5471	139	-0.0000	0.2338	0.0547	-0.3552	1.0431
Denmark	134	0.0000	0.0715	0.0051	-0.2804	0.2747	139	0.0000	0.1667	0.0278	...	0.1260	0.0159	-0.3848	0.3528	139	-0.0000	0.1782	0.0318	-0.4799	0.3642
Finland	134	-0.0000	0.0796	0.0063	-0.3494	0.3059	139	0.0000	0.1324	0.0175	...	NaN	NaN	NaN	NaN	139	-0.0000	0.6590	0.4343	-1.4003	0.7940
France	134	-0.0000	0.1375	0.0189	-0.6381	0.6174	139	0.0000	0.2980	0.0888	...	0.1829	0.0335	-0.3429	0.4288	139	-0.0000	0.1569	0.0246	-0.5060	0.3523
Germany	134	-0.0000	0.1790	0.0320	-0.9063	0.4267	139	0.0000	0.2196	0.0482	...	0.1465	0.0215	-0.4214	0.2782	139	-0.0000	1.5414	2.3760	-1.6752	6.5972
Italy	134	-0.0000	0.1018	0.0104	-0.5105	0.4711	139	0.0000	0.1181	0.0140	...	NaN	NaN	NaN	NaN	139	-0.0000	0.2592	0.0672	-0.7846	0.5501
Japan	134	-0.0000	0.1544	0.0238	-0.6420	0.3151	139	0.0000	0.3082	0.0950	...	NaN	NaN	NaN	NaN	139	-0.0000	0.4429	0.1961	-1.9085	1.1233
Netherlands	134	0.0000	0.1311	0.0172	-0.6897	0.7056	139	0.0000	0.1984	0.0394	...	NaN	NaN	NaN	NaN	139	-0.0000	0.4505	0.2029	-1.9155	0.6240
Norway	134	0.0000	0.0834	0.0070	-0.2120	0.2368	139	0.0000	0.1820	0.0331	...	0.1331	0.0177	-0.5146	0.3396	139	-0.0000	0.1668	0.0278	-0.4456	0.5469
Portugal	134	-0.0000	0.0790	0.0062	-0.1630	0.2096	139	0.0000	0.2442	0.0596	...	0.1756	0.0308	-0.5112	0.6898	139	-0.0000	0.6603	0.4360	-1.2967	0.9186
Spain	134	0.0000	0.1119	0.0125	-0.5064	0.2794	139	0.0000	0.1924	0.0370	...	0.1210	0.0146	-0.2784	0.2885	139	-0.0000	0.5059	0.2559	-1.0125	0.8168
Sweden	134	0.0000	0.0636	0.0040	-0.1138	0.2353	139	0.0000	0.1613	0.0260	...	0.1349	0.0182	-0.2019	0.4055	139	-0.0000	0.1953	0.0381	-0.8909	0.3304
Switzerland	134	0.0000	0.0796	0.0063	-0.2394	0.2273	139	0.0000	0.1560	0.0243	...	0.4384	0.1922	-0.3518	1.4266	139	-0.0000	0.2723	0.0741	-0.3853	1.1450
United Kingdom	134	0.0000	0.0761	0.0058	-0.2463	0.2326	139	0.0000	0.0941	0.0089	...	0.1595	0.0255	-0.3701	0.5012	139	-0.0000	0.2644	0.0699	-0.4668	0.4728
United States	134	0.0000	0.1085	0.0118	-0.2831	0.4725	139	0.0000	0.1066	0.0114	...	0.1289	0.0166	-0.2476	0.4476	139	0.0000	0.2672	0.0714	-0.5959	0.5633

16 rows × 48 columns

In [164…

```python
corr = df_twfe[cols].corr()

# coverig up the uppie
mask = np.triu(np.ones_like(corr, dtype=bool))

plt.figure(figsize=(10, 7))
sns.heatmap(
        corr,
        mask=mask,
        annot=True,
        fmt=".2f",
        cmap="coolwarm",
        vmin=-1,
        vmax=1,
        linewidths=0.5,
        cbar_kws={"shrink": 0.8}
)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

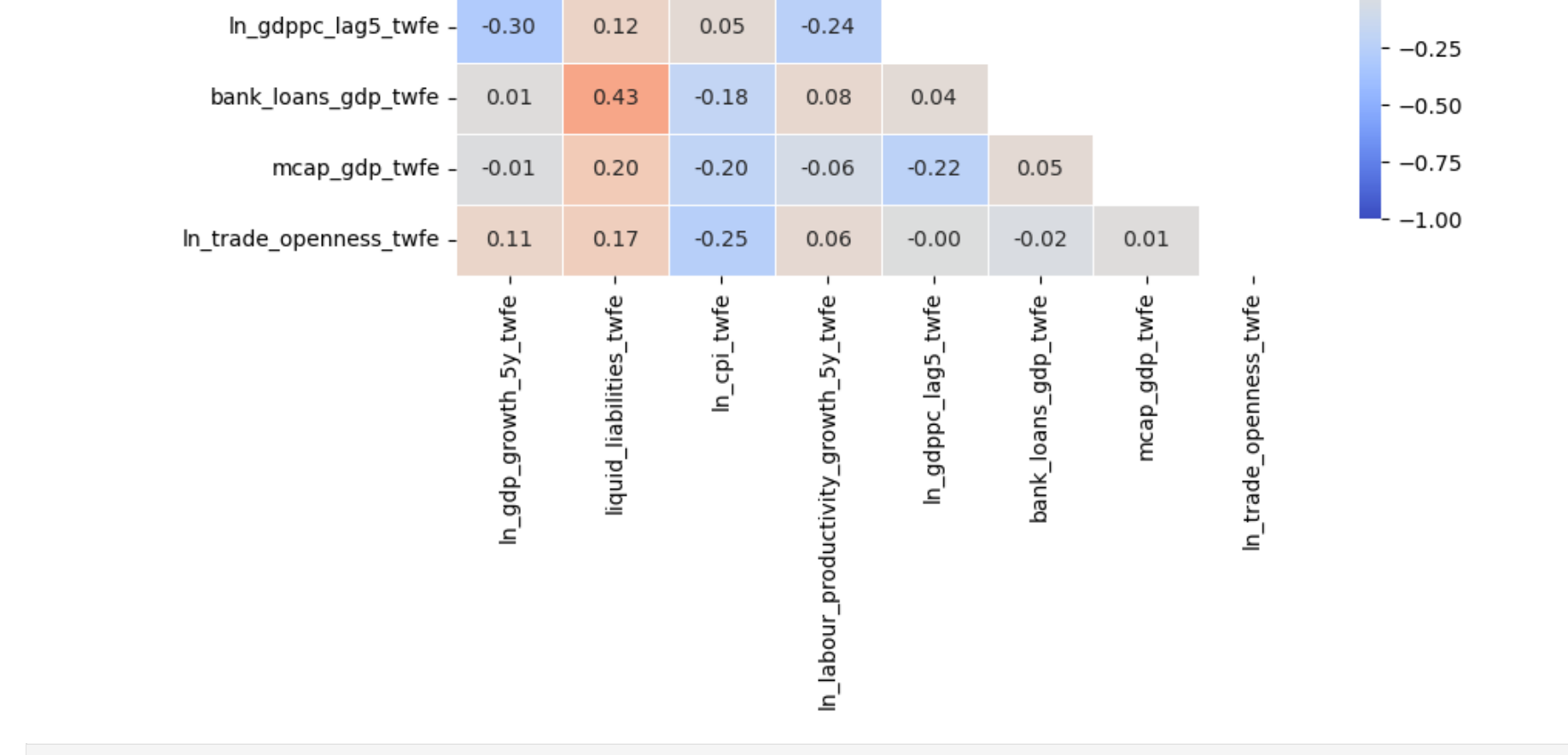


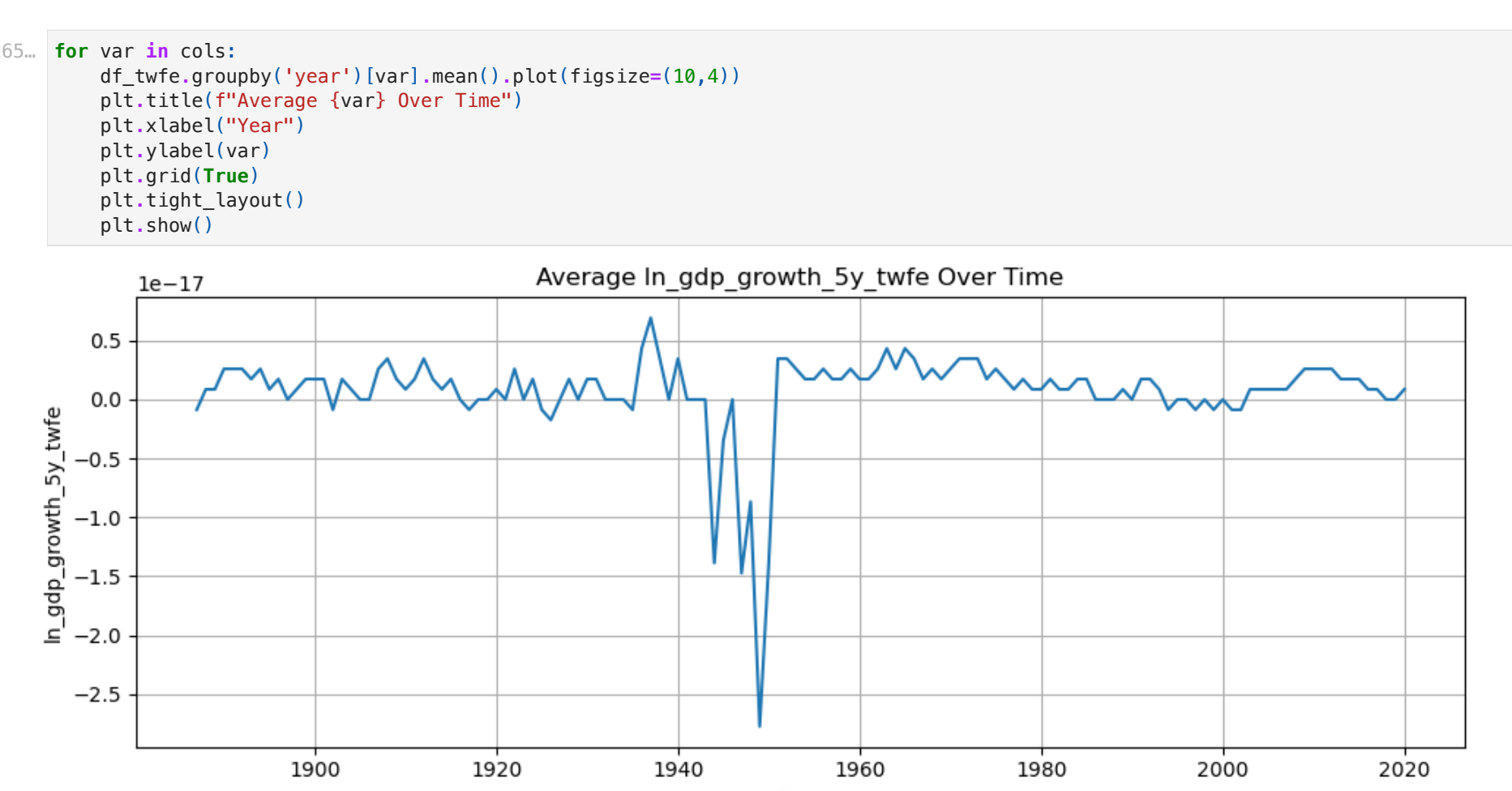
Correlation Matrix

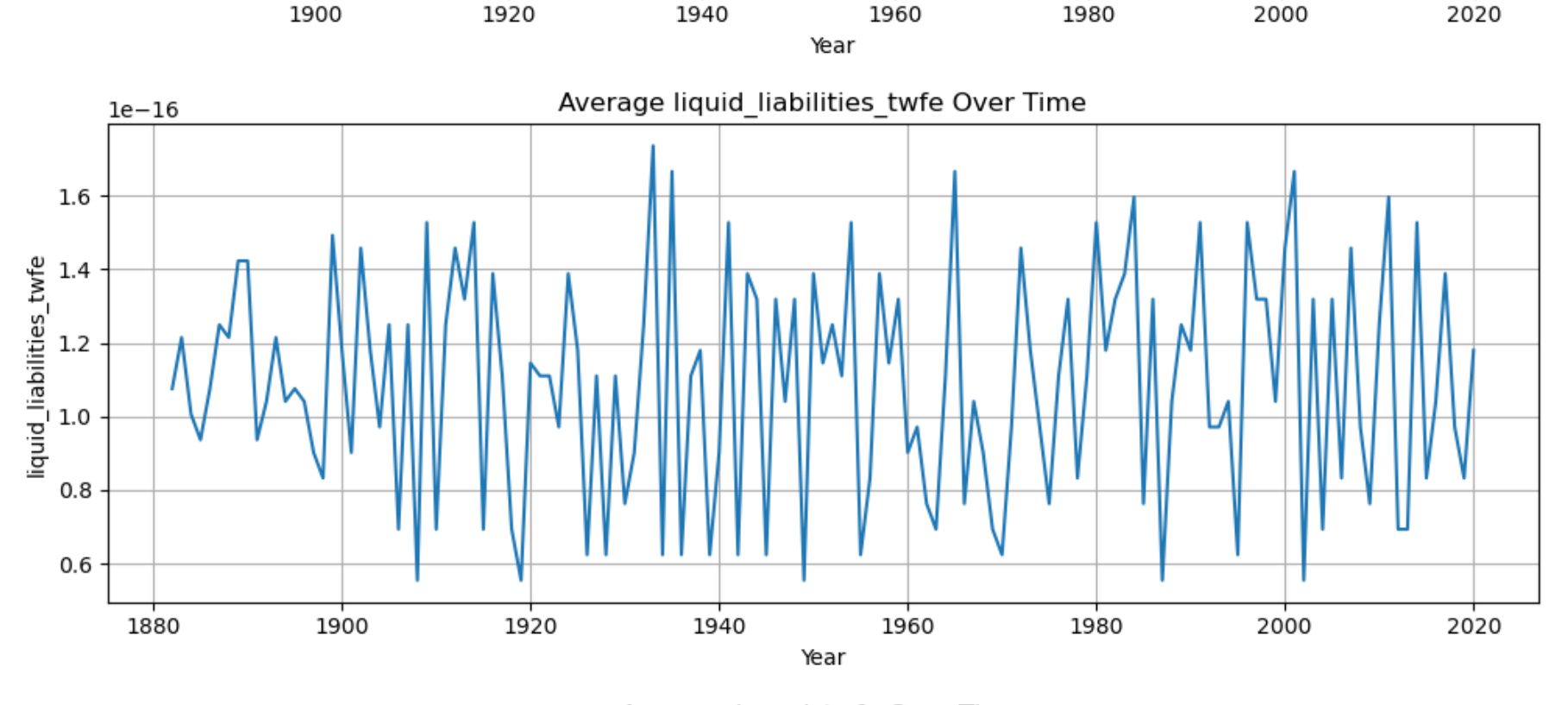
	ln_gdp_growth_5y_twfe	liquid_liabilities_twfe	ln_cpi_twfe	ln_labour_productivity_growth_5y_twfe	ln_gdppc_lag5_twfe	bank_loans_gdp_twfe	mcap_gdp_twfe
ln_gdp_growth_5y_twfe							
liquid_liabilities_twfe	-0.11						
ln_cpi_twfe	0.03	-0.18					
ln_labour_productivity_growth_5y_twfe	0.78	-0.12	0.05				
ln_gdppc_lag5_twfe	-0.30	0.12	0.05	-0.24			
bank_loans_gdp_twfe	0.01	0.43	-0.18	0.08	0.04		
mcap_gdp_twfe	-0.01	0.20	-0.20	-0.06	-0.22	0.05	
ln_trade_openness_twfe	0.11	0.17	-0.25	0.06	-0.00	-0.02	0.01

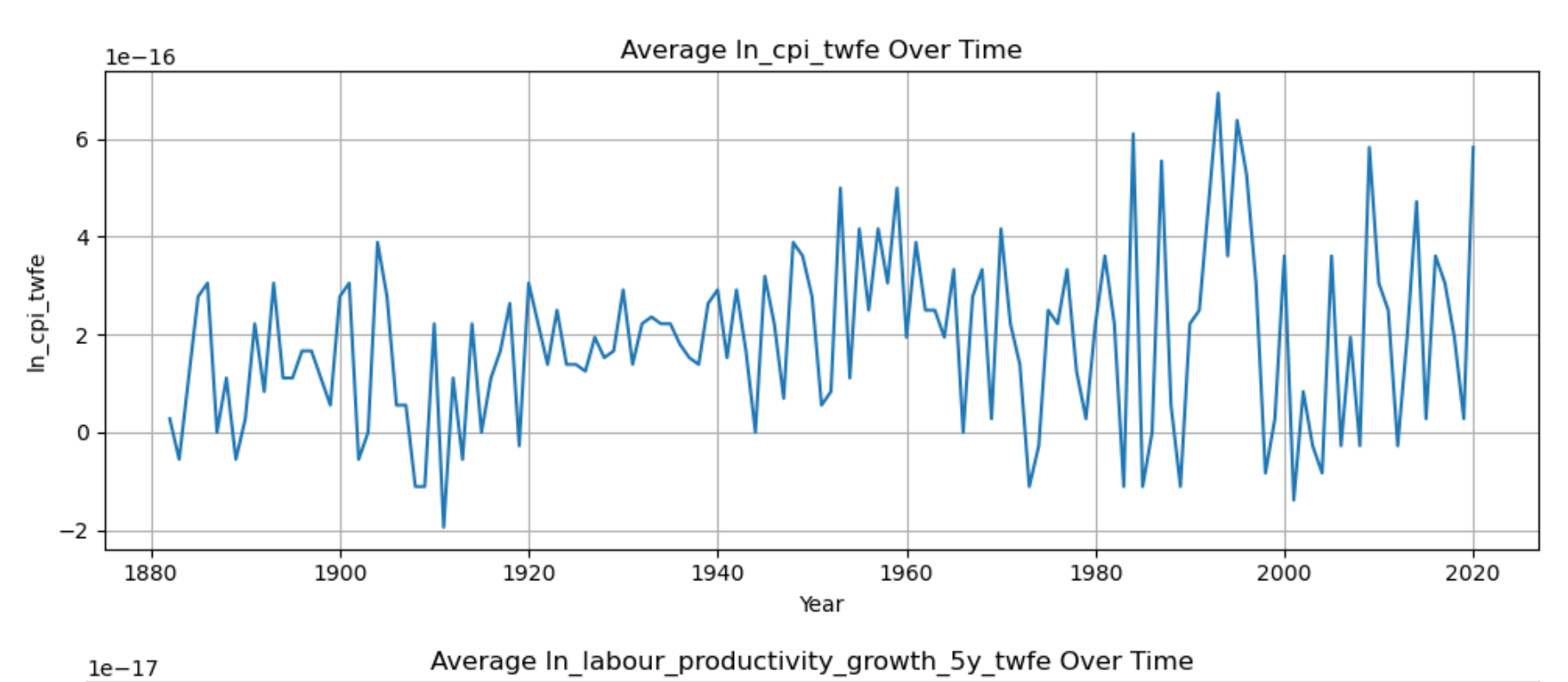
In [165…

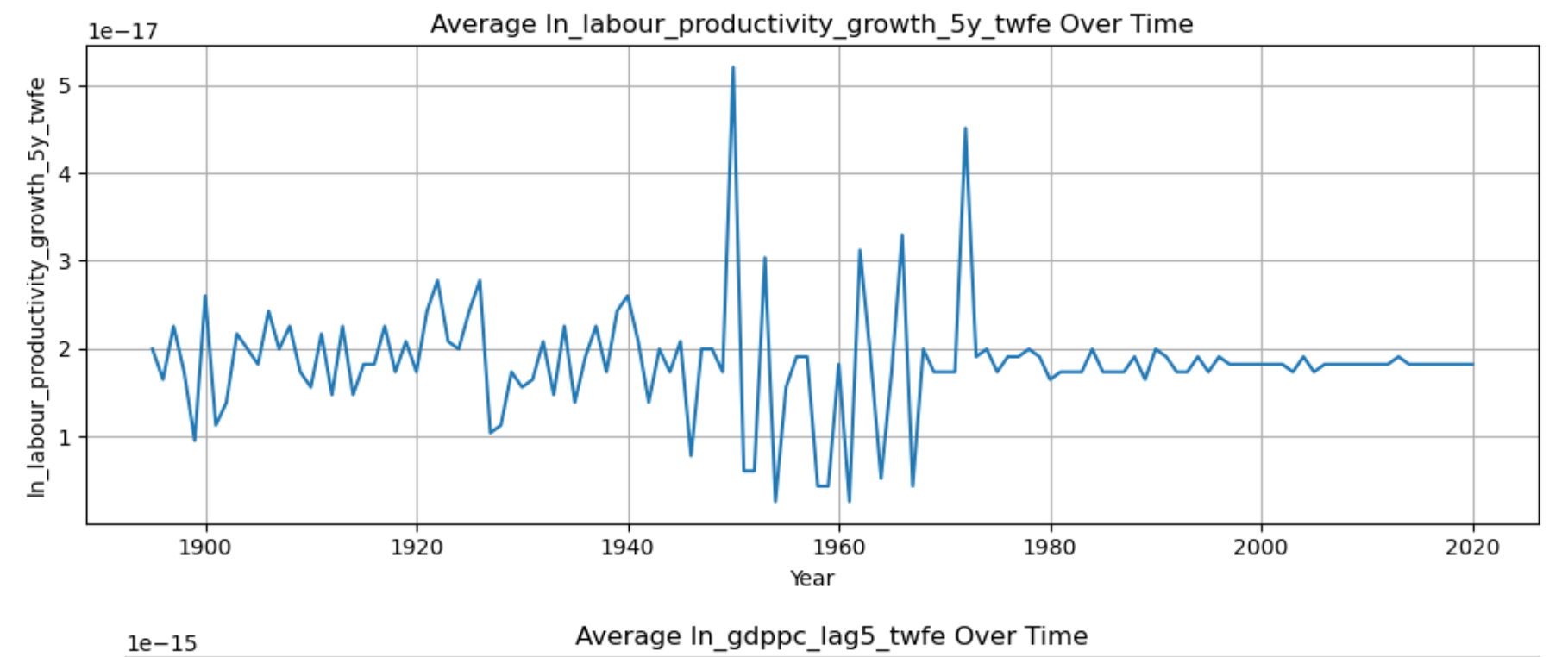
```python
for var in cols:
    df_twfe.groupby('year')[var].mean().plot(figsize=(10,4))
    plt.title(f"Average {var} Over Time")
    plt.xlabel("Year")
    plt.ylabel(var)
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```

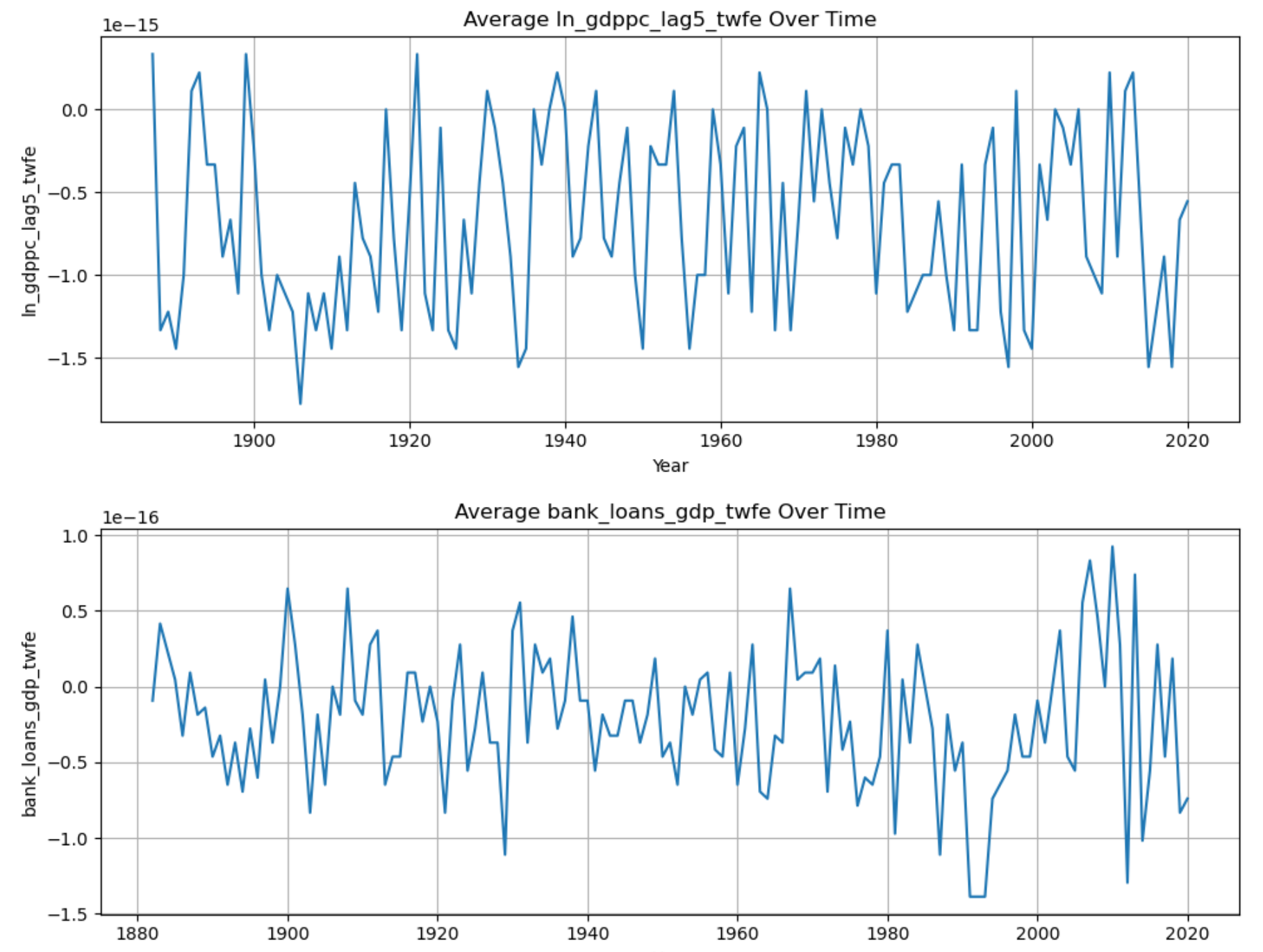


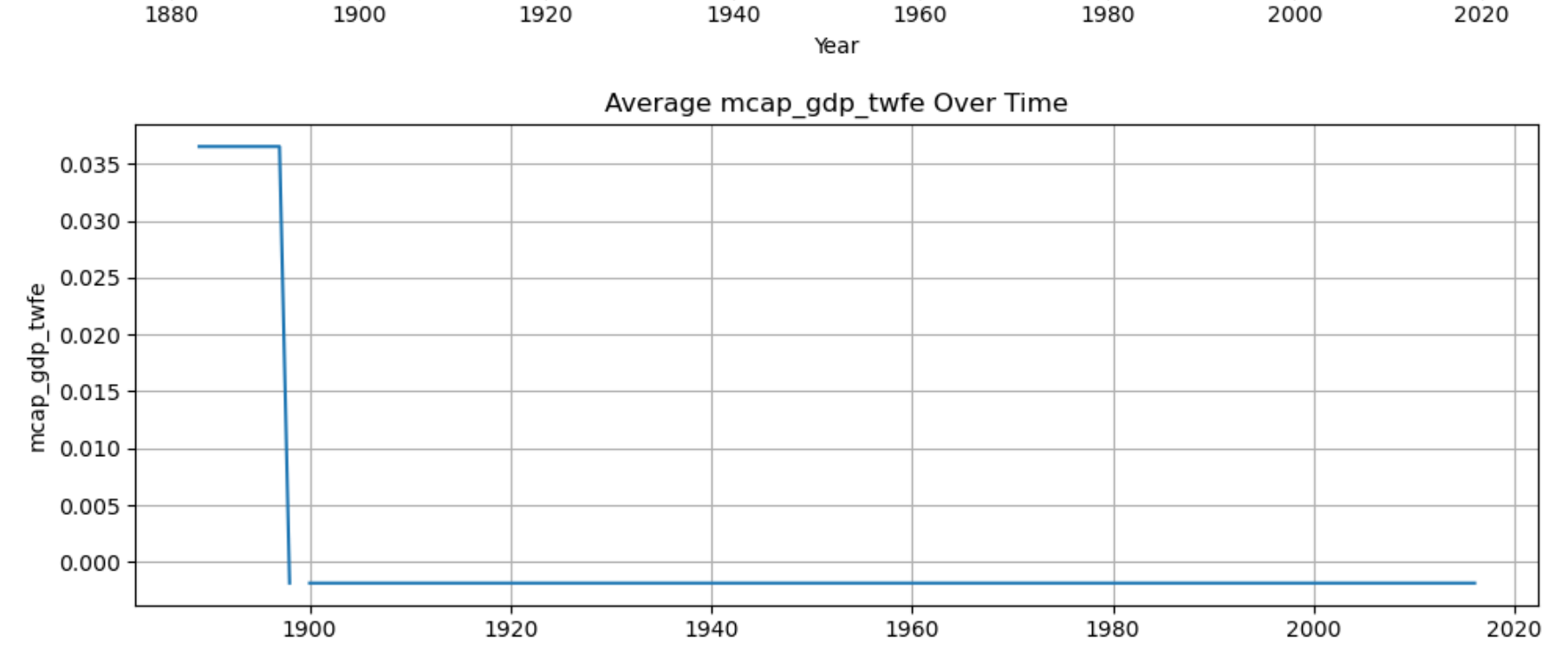


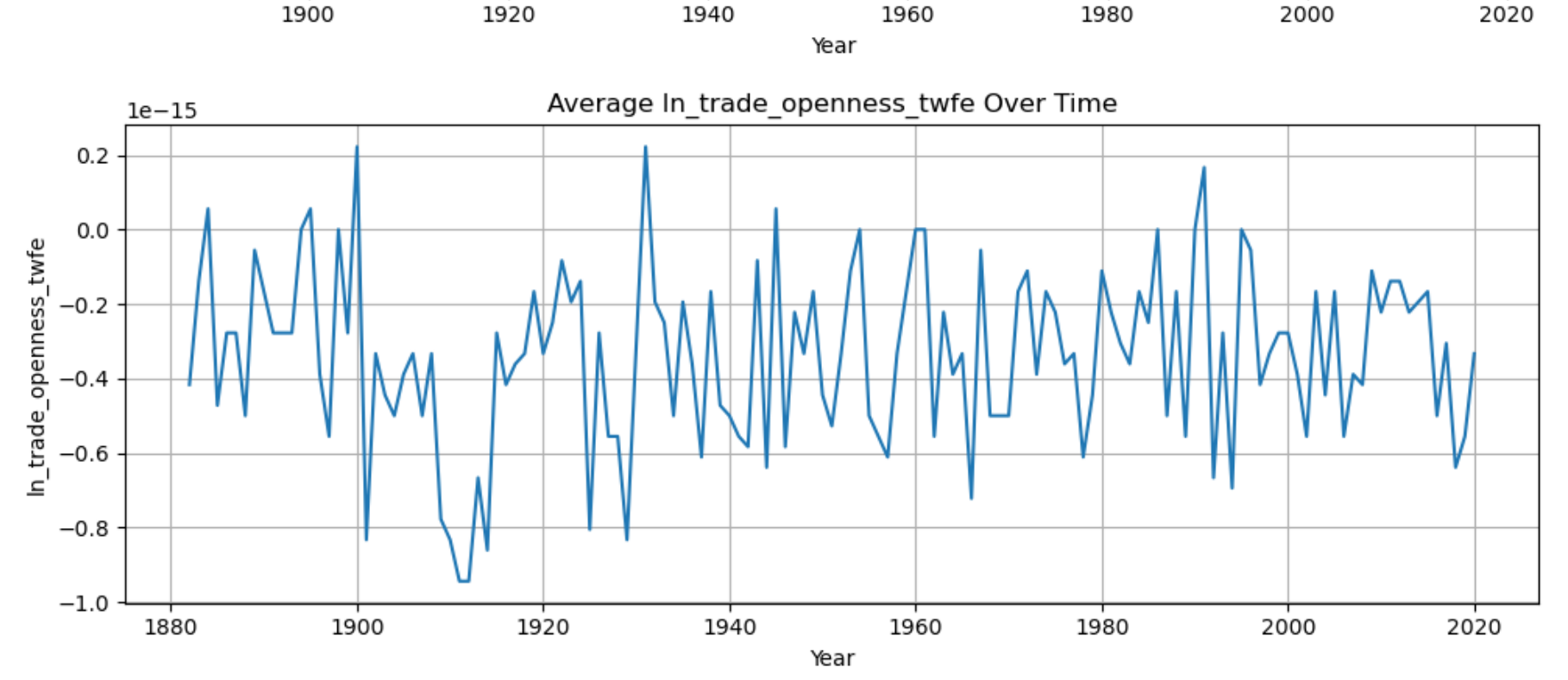












In [ ]:

```python
xtsum_table = {}

for var in cols:
    mean_overall = df_twfe[var].mean()

    mean_country = df_twfe.groupby('country')[var].mean()
    between_var = mean_country.var()

    df_twfe["wv"] = df_twfe[var] - df_twfe.groupby('country')[var].transform('mean')
    within_var = df_twfe["wv"].var()

    xtsum_table[var] = {
        "overall_mean": mean_overall,
        "overall_sd": df_twfe[var].std(),
        "between_sd": between_var**0.5,
        "within_sd": within_var**0.5
    }

xtsum_table
```

Out[ ]:

```
{'ln_gdp_growth_5y_twfe': {'overall_mean': np.float64(2.4791010869370043e-18),
  'overall_sd': 0.10912585709740846,
  'between_sd': 4.963734527954227e-18,
  'within_sd': 0.10912585709740846},
 'liquid_liabilities_twfe': {'overall_mean': np.float64(1.0837653716102585e-16),
  'overall_sd': 0.18363363000607902,
  'between_sd': 4.042370940802248e-17,
  'within_sd': 0.18363363000607902},
 'ln_cpi_twfe': {'overall_mean': np.float64(1.916931841079407e-16),
  'overall_sd': 2.9672821765078976,
  'between_sd': 2.1320899695677157e-16,
  'within_sd': 2.9672821765078976},
 'ln_labour_productivity_growth_5y_twfe': {'overall_mean': np.float64(1.9171447938870667e-17),
  'overall_sd': 0.1051628450342546,
  'between_sd': 8.817028499520694e-18,
  'within_sd': 0.1051628450342546},
 'ln_gdppc_lag5_twfe': {'overall_mean': np.float64(-6.744190612275205e-16),
  'overall_sd': 0.18407344415994994,
  'between_sd': 5.120898971657714e-16,
  'within_sd': 0.18407344415994992},
 'bank_loans_gdp_twfe': {'overall_mean': np.float64(-2.1432362945401702e-17),
  'overall_sd': 0.14815060196502217,
  'between_sd': 6.98769017106518e-17,
  'within_sd': 0.1481506019650222},
 'mcap_gdp_twfe': {'overall_mean': np.float64(1.0818705549102668e-17),
  'overall_sd': 0.19277883500406054,
  'between_sd': 0.004200566042937384,
  'within_sd': 0.1927379166514744},
 'ln_trade_openness_twfe': {'overall_mean': np.float64(-3.358624329891211e-16),
  'overall_sd': 0.5240223833901163,
  'between_sd': 6.492684364933169e-16,
  'within_sd': 0.5240223833901163}}
```

Question 12 --> Simple correlation coefficent

In [166…

```python
df_Twfe_r_coeffiecent_ordering = df_twfe

df_Twfe_r_coeffiecent_ordering = df_Twfe_r_coeffiecent_ordering[["country",  "ln_gdp_growth_5y_twfe", "liquid_liabilities_twfe",  "ln_labour_productivity_growth_5y_twfe", "ln_gdppc_lag5_twfe", "bank_

y = "ln_gdp_growth_5y_twfe"

X = ["liquid_liabilities_twfe","ln_gdppc_lag5_twfe","bank_loans_gdp_twfe", "mcap_gdp_twfe"]
corr_list = []
for x in X:
    tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
    tmp["variable"] = x
    corr_list.append(tmp)
corr_table = pd.concat(corr_list).reset_index(drop=True)
sd_table = ( df_Twfe_r_coeffiecent_ordering.groupby("country").agg(SD_Dependent = (y, "std"),SD_Explanatory  = (x, "std")).reset_index())
corr_table = corr_table.merge(sd_table, on="country", how="left")
corr_table["beta_simple"] = corr_table["corr"] * corr_table["SD_Dependent"] / corr_table["SD_Explanatory"]
corr_table = corr_table.sort_values("corr", ascending=False).reset_index(drop=True)

print(corr_table)
```

```
       country   corr                 variable  SD_Dependent  SD_Explanatory  \
0     Portugal 0.3518            mcap_gdp_twfe        0.0790          0.1756
1     Portugal 0.3228  liquid_liabilities_twfe        0.0790          0.1756
2      Denmark 0.2631  liquid_liabilities_twfe        0.0715          0.1260
3       Sweden 0.2481  liquid_liabilities_twfe        0.0636          0.1349
4        Japan 0.2173      bank_loans_gdp_twfe        0.1544             NaN
..         ...    ...                      ...           ...             ...
59       Spain    NaN      bank_loans_gdp_twfe        0.1119          0.1210
60     Finland    NaN            mcap_gdp_twfe        0.0796             NaN
61       Italy    NaN            mcap_gdp_twfe        0.1018             NaN
62       Japan    NaN            mcap_gdp_twfe        0.1544             NaN
63 Netherlands    NaN            mcap_gdp_twfe        0.1311             NaN

    beta_simple
0        0.1581
1        0.1451
2        0.1493
3        0.1170
4           NaN
..          ...
59          NaN
60          NaN
61          NaN
62          NaN
63          NaN

[64 rows x 6 columns]
```

```
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
```

Question 12 --> TWFE transforms with the regression of the two Proxies of Economic Growth and then also for the explantory varibels.

In [167…

```python
counts = df_twfe.groupby("country").size()
df_twfe = df_twfe[df_twfe["country"].isin(counts[counts >= 2].index)].copy()

# makig my yr dummies

year_dummies = pd.get_dummies(df_twfe["year"], prefix ="year", drop_first=False)

# doing my regressions first for teh two dependent varibels;

dummy_reg_gdp = df_twfe["ln_gdp_growth_5y_within"].notna()
dummy_y_gdp_reg = df_twfe.loc[dummy_reg_gdp, "ln_gdp_growth_5y_within"]
dummy_x_gdp_reg = year_dummies.loc[dummy_reg_gdp]

ln_gdp_time_fe = sm.OLS(dummy_y_gdp_reg, dummy_x_gdp_reg).fit()

# second one

dummy_reg_gdp_labprod = df_twfe["ln_labour_productivity_growth_5y_within"].notna()
dummy_y_gdp_labprod = df_twfe.loc[dummy_reg_gdp_labprod, "ln_labour_productivity_growth_5y_within"]
dummy_x_gdo_labprod = year_dummies.loc[dummy_reg_gdp_labprod]
ln_labprod_time_fe = sm.OLS(dummy_y_gdp_labprod, dummy_x_gdo_labprod).fit()

# now explantory varibles

dummy_reg_liquid_liabilities = df_twfe["liquid_liabilities_within"].notna()
dummy_y_reg_liq = df_twfe.loc[dummy_reg_liquid_liabilities, "liquid_liabilities_within"]
dummy_x_reg_liq = year_dummies.loc[dummy_reg_liquid_liabilities]
liquid_liability_time_fe = sm.OLS(dummy_y_reg_liq, dummy_x_reg_liq).fit()


dummy_reg_mcap = df_twfe["mcap_gdp_within"].notna()
dummy_y_reg_mcap = df_twfe.loc[dummy_reg_mcap, "mcap_gdp_within"]
dummy_x_reg_mcap = year_dummies.loc[dummy_reg_mcap]

mcap_time_fe = sm.OLS(dummy_y_reg_mcap, dummy_x_reg_mcap).fit()


# bank_loans_gdp_within

dummy_reg_bank = df_twfe["bank_loans_gdp_within"].notna()
dummy_y_reg_bank = df_twfe.loc[dummy_reg_bank, "bank_loans_gdp_within"]
dummy_x_reg_bank = year_dummies.loc[dummy_reg_bank]

bank_loans_time_fe = sm.OLS(dummy_y_reg_bank, dummy_x_reg_bank).fit()

print("Time FE – ln GDP growth (5y)")
print(ln_gdp_time_fe.summary())

print("\nTime FE – ln labour productivity growth (5y)")
print(ln_labprod_time_fe.summary())

print("\nTime FE – liquid liabilities (within)")
print(liquid_liability_time_fe.summary())

print("\nTime FE – market cap / GDP (within)")
print(mcap_time_fe.summary())

print("\nTime FE – bank loans / GDP (within)")
print(bank_loans_time_fe.summary())
```

```
Time FE – ln GDP growth (5y)
                                OLS Regression Results
==============================================================================
Dep. Variable:     ln_gdp_growth_5y_within   R-squared:                       0.300
Model:                                 OLS   Adj. R-squared:                  0.253
Method:                      Least Squares   F-statistic:                     6.466
Date:                     Sat, 27 Dec 2025   Prob (F-statistic):           1.16e-84
Time:                             17:14:14   Log-Likelihood:                1707.8
No. Observations:                     2144   AIC:                           -3148.
Df Residuals:                         2010   BIC:                           -2388.
Df Model:                              133
Covariance Type:                 nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882  -9.301e-19   7.04e-19     -1.322      0.186   -2.31e-18     4.5e-19
year_1883           0          0        nan        nan           0           0
year_1884  -1.114e-18   7.04e-19     -1.583      0.114   -2.49e-18    2.66e-19
year_1885  -7.576e-19   7.04e-19     -1.077      0.282   -2.14e-18    6.22e-19
year_1886   -1.16e-18   7.04e-19     -1.649      0.099   -2.54e-18     2.2e-19
year_1887     -0.0372      0.028     -1.322      0.186      -0.092       0.018
year_1888     -0.0558      0.028     -1.981      0.048      -0.111      -0.001
year_1889     -0.0446      0.028     -1.583      0.114      -0.100       0.011
year_1890     -0.0303      0.028     -1.077      0.282      -0.086       0.025
year_1891     -0.0464      0.028     -1.649      0.099      -0.102       0.009
year_1892     -0.0575      0.028     -2.040      0.041      -0.113      -0.002
year_1893     -0.0670      0.028     -2.378      0.017      -0.122      -0.012
year_1894     -0.0596      0.028     -2.117      0.034      -0.115      -0.004
year_1895     -0.0542      0.028     -1.922      0.055      -0.109       0.001
year_1896     -0.0451      0.028     -1.601      0.110      -0.100       0.010
year_1897     -0.0263      0.028     -0.933      0.351      -0.082       0.029
year_1898      0.0147      0.028      0.522      0.601      -0.041       0.070
year_1899      0.0071      0.028      0.251      0.802      -0.048       0.062
year_1900      0.0050      0.028      0.179      0.858      -0.050       0.060
year_1901      0.0068      0.028      0.243      0.808      -0.048       0.062
year_1902     -0.0175      0.028     -0.620      0.535      -0.073       0.038
year_1903     -0.0341      0.028     -1.212      0.226      -0.089       0.021
year_1904     -0.0392      0.028     -1.393      0.164      -0.094       0.016
year_1905     -0.0435      0.028     -1.544      0.123      -0.099       0.012
year_1906     -0.0103      0.028     -0.367      0.714      -0.066       0.045
year_1907      0.0061      0.028      0.218      0.827      -0.049       0.061
year_1908     -0.0272      0.028     -0.964      0.335      -0.082       0.028
year_1909     -0.0079      0.028     -0.282      0.778      -0.063       0.047
year_1910     -0.0099      0.028     -0.350      0.726      -0.065       0.045
year_1911     -0.0273      0.028     -0.969      0.333      -0.083       0.028
year_1912     -0.0236      0.028     -0.838      0.402      -0.079       0.032
year_1913      0.0143      0.028      0.506      0.613      -0.041       0.070
year_1914     -0.0588      0.028     -2.088      0.037      -0.114      -0.004
year_1915     -0.0733      0.028     -2.602      0.009      -0.129      -0.018
year_1916     -0.0504      0.028     -1.789      0.074      -0.106       0.005
year_1917     -0.1147      0.028     -4.073      0.000      -0.170      -0.059
year_1918     -0.1941      0.028     -6.891      0.000      -0.249      -0.139
year_1919     -0.1181      0.028     -4.191      0.000      -0.173      -0.063
year_1920     -0.0840      0.028     -2.982      0.003      -0.139      -0.029
year_1921     -0.1557      0.028     -5.526      0.000      -0.211      -0.100
year_1922     -0.0321      0.028     -1.140      0.254      -0.087       0.023
year_1923      0.0504      0.028      1.789      0.074      -0.005       0.106
year_1924      0.0530      0.028      1.883      0.060      -0.002       0.108
year_1925      0.0545      0.028      1.935      0.053      -0.001       0.110
year_1926      0.0928      0.028      3.293      0.001       0.038       0.148
year_1927      0.0554      0.028      1.966      0.049       0.000       0.111
year_1928      0.0492      0.028      1.746      0.081      -0.006       0.104
year_1929      0.0470      0.028      1.668      0.095      -0.008       0.102
year_1930     -0.0144      0.028     -0.512      0.609      -0.070       0.041
year_1931     -0.0786      0.028     -2.789      0.005      -0.134      -0.023
year_1932     -0.1422      0.028     -5.049      0.000      -0.197      -0.087
year_1933     -0.1490      0.028     -5.289      0.000      -0.204      -0.094
year_1934     -0.1453      0.028     -5.158      0.000      -0.201      -0.090
year_1935     -0.0895      0.028     -3.177      0.002      -0.145      -0.034
year_1936     -0.0315      0.028     -1.118      0.264      -0.087       0.024
year_1937      0.0379      0.028      1.344      0.179      -0.017       0.093
year_1938      0.0283      0.028      1.004      0.316      -0.027       0.084
year_1939      0.0365      0.028      1.297      0.195      -0.019       0.092
year_1940     -0.0121      0.028     -0.428      0.669      -0.067       0.043
year_1941     -0.0125      0.028     -0.445      0.656      -0.068       0.043
year_1942     -0.0559      0.028     -1.984      0.047      -0.111      -0.001
year_1943     -0.0508      0.028     -1.802      0.072      -0.106       0.004
year_1944     -0.1321      0.028     -4.688      0.000      -0.187      -0.077
year_1945     -0.1693      0.028     -6.009      0.000      -0.225      -0.114
year_1946     -0.1364      0.028     -4.843      0.000      -0.192      -0.081
year_1947     -0.0927      0.028     -3.292      0.001      -0.148      -0.037
year_1948     -0.0647      0.028     -2.297      0.022      -0.120      -0.009
year_1949      0.0088      0.028      0.313      0.755      -0.046       0.064
year_1950      0.1164      0.028      4.134      0.000       0.061       0.172
year_1951      0.1207      0.028      4.285      0.000       0.065       0.176
year_1952      0.1021      0.028      3.625      0.000       0.047       0.157
year_1953      0.0971      0.028      3.446      0.001       0.042       0.152
year_1954      0.0978      0.028      3.474      0.001       0.043       0.153
year_1955      0.0938      0.028      3.328      0.001       0.039       0.149
year_1956      0.0815      0.028      2.892      0.004       0.026       0.137
year_1957      0.0822      0.028      2.919      0.004       0.027       0.137
year_1958      0.0577      0.028      2.048      0.041       0.002       0.113
year_1959      0.0618      0.028      2.195      0.028       0.007       0.117
year_1960      0.0644      0.028      2.286      0.022       0.009       0.120
year_1961      0.0748      0.028      2.656      0.008       0.020       0.130
year_1962      0.0902      0.028      3.202      0.001       0.035       0.145
year_1963      0.1178      0.028      4.183      0.000       0.063       0.173
year_1964      0.1307      0.028      4.638      0.000       0.075       0.186
year_1965      0.1250      0.028      4.437      0.000       0.070       0.180
year_1966      0.1142      0.028      4.054      0.000       0.059       0.169
year_1967      0.1060      0.028      3.765      0.000       0.051       0.161
year_1968      0.1128      0.028      4.003      0.000       0.058       0.168
year_1969      0.1098      0.028      3.896      0.000       0.055       0.165
year_1970      0.1077      0.028      3.824      0.000       0.052       0.163
year_1971      0.1036      0.028      3.679      0.000       0.048       0.159
year_1972      0.1097      0.028      3.895      0.000       0.054       0.165
year_1973      0.1154      0.028      4.095      0.000       0.060       0.171
year_1974      0.0783      0.028      2.778      0.006       0.023       0.133
year_1975      0.0354      0.028      1.257      0.209      -0.020       0.091
year_1976      0.0405      0.028      1.439      0.150      -0.015       0.096
year_1977      0.0198      0.028      0.702      0.483      -0.035       0.075
year_1978     -0.0051      0.028     -0.182      0.856      -0.060       0.050
year_1979      0.0163      0.028      0.580      0.562      -0.039       0.072
year_1980      0.0361      0.028      1.282      0.200      -0.019       0.091
year_1981      0.0069      0.028      0.244      0.807      -0.048       0.062
year_1982     -0.0122      0.028     -0.432      0.666      -0.067       0.043
year_1983     -0.0219      0.028     -0.779      0.436      -0.077       0.033
year_1984     -0.0270      0.028     -0.959      0.338      -0.082       0.028
year_1985     -0.0117      0.028     -0.414      0.679      -0.067       0.044
year_1986      0.0101      0.028      0.360      0.719      -0.045       0.065
year_1987      0.0372      0.028      1.319      0.187      -0.018       0.092
year_1988      0.0548      0.028      1.945      0.052      -0.000       0.110
year_1989      0.0539      0.028      1.912      0.056      -0.001       0.109
year_1990      0.0368      0.028      1.306      0.192      -0.018       0.092
year_1991      0.0172      0.028      0.610      0.542      -0.038       0.072
year_1992      0.0001      0.028      0.005      0.996      -0.055       0.055
year_1993     -0.0279      0.028     -0.989      0.323      -0.083       0.027
year_1994     -0.0235      0.028     -0.834      0.405      -0.079       0.032
year_1995     -0.0080      0.028     -0.283      0.778      -0.063       0.047
year_1996      0.0121      0.028      0.431      0.667      -0.043       0.067
year_1997      0.0352      0.028      1.249      0.212      -0.020       0.090
year_1998      0.0643      0.028      2.281      0.023       0.009       0.120
year_1999      0.0659      0.028      2.340      0.019       0.011       0.121
year_2000      0.0773      0.028      2.743      0.006       0.022       0.133
year_2001      0.0695      0.028      2.467      0.014       0.014       0.125
year_2002      0.0483      0.028      1.715      0.087      -0.007       0.104
year_2003      0.0270      0.028      0.959      0.338      -0.028       0.082
year_2004      0.0216      0.028      0.769      0.442      -0.034       0.077
year_2005      0.0053      0.028      0.187      0.852      -0.050       0.061
year_2006      0.0157      0.028      0.558      0.577      -0.040       0.071
year_2007      0.0292      0.028      1.035      0.301      -0.026       0.084
year_2008      0.0189      0.028      0.670      0.503      -0.036       0.074
year_2009     -0.0486      0.028     -1.726      0.085      -0.104       0.007
year_2010     -0.0476      0.028     -1.690      0.091      -0.103       0.008
year_2011     -0.0649      0.028     -2.304      0.021      -0.120      -0.010
year_2012     -0.0970      0.028     -3.445      0.001      -0.152      -0.042
year_2013     -0.0996      0.028     -3.535      0.000      -0.155      -0.044
year_2014     -0.0520      0.028     -1.845      0.065      -0.107       0.003
year_2015     -0.0653      0.028     -2.319      0.020      -0.121      -0.010
year_2016     -0.0725      0.028     -2.574      0.010      -0.128      -0.017
year_2017     -0.0491      0.028     -1.744      0.081      -0.104       0.006
year_2018     -0.0239      0.028     -0.848      0.396      -0.079       0.031
year_2019     -0.0233      0.028     -0.827      0.408      -0.079       0.032
year_2020     -0.0909      0.028     -3.226      0.001      -0.146      -0.036
==============================================================================
Omnibus:                      667.485   Durbin-Watson:                   0.375
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            14650.498
Skew:                          -0.934   Prob(JB):                         0.00
Kurtosis:                      15.669   Cond. No.                     1.03e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.5e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE – ln labour productivity growth (5y)
                                       OLS Regression Results
==================================================================================================
Dep. Variable:     ln_labour_productivity_growth_5y_within   R-squared:                       0.322
Model:                                                 OLS   Adj. R-squared:                  0.277
Method:                                      Least Squares   F-statistic:                     7.173
Date:                                     Sat, 27 Dec 2025   Prob (F-statistic):           4.79e-91
Time:                                             17:14:14   Log-Likelihood:                1680.4
No. Observations:                                     2016   AIC:                           -3109.
Df Residuals:                                         1890   BIC:                           -2402.
Df Model:                                              125
Covariance Type:                                 nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882    -1.9e-18   6.78e-19     -2.803      0.005   -3.23e-18   -5.71e-19
year_1883           0          0        nan        nan           0           0
year_1884  -1.184e-18   6.78e-19     -1.746      0.081   -2.51e-18    1.46e-19
year_1885  -4.603e-19   6.78e-19     -0.679      0.497   -1.79e-18     8.7e-19
year_1886  -6.938e-19   6.78e-19     -1.023      0.306   -2.02e-18    6.36e-19
year_1887  -6.183e-19   6.78e-19     -0.912      0.362   -1.95e-18    7.12e-19
year_1888  -2.466e-19   6.78e-19     -0.364      0.716   -1.58e-18    1.08e-18
year_1889  -7.673e-19   6.78e-19     -1.132      0.258    -2.1e-18    5.63e-19
year_1890  -1.224e-18   6.78e-19     -1.806      0.071   -2.55e-18    1.06e-19
year_1891  -1.218e-18   6.78e-19     -1.796      0.073   -2.55e-18    1.12e-19
year_1892  -1.123e-18   6.78e-19     -1.656      0.098   -2.45e-18    2.07e-19
year_1893  -5.648e-19   6.78e-19     -0.833      0.405   -1.89e-18    7.65e-19
year_1894  -1.508e-19   6.78e-19     -0.222      0.824   -1.48e-18    1.18e-18
year_1895     -0.0761      0.027     -2.803      0.005      -0.129      -0.023
year_1896     -0.0691      0.027     -2.547      0.011      -0.122      -0.016
year_1897     -0.0474      0.027     -1.746      0.081      -0.101       0.006
year_1898     -0.0184      0.027     -0.679      0.497      -0.072       0.035
year_1899     -0.0278      0.027     -1.023      0.306      -0.081       0.025
year_1900     -0.0248      0.027     -0.912      0.362      -0.078       0.028
year_1901     -0.0099      0.027     -0.364      0.716      -0.063       0.043
year_1902     -0.0307      0.027     -1.132      0.258      -0.084       0.023
year_1903     -0.0490      0.027     -1.806      0.071      -0.102       0.004
year_1904     -0.0487      0.027     -1.796      0.073      -0.102       0.004
year_1905     -0.0449      0.027     -1.656      0.098      -0.098       0.008
year_1906     -0.0226      0.027     -0.833      0.405      -0.076       0.031
year_1907     -0.0060      0.027     -0.222      0.824      -0.059       0.047
year_1908     -0.0279      0.027     -1.028      0.304      -0.081       0.025
year_1909     -0.0134      0.027     -0.495      0.621      -0.067       0.040
year_1910     -0.0179      0.027     -0.658      0.510      -0.071       0.035
year_1911     -0.0213      0.027     -0.786      0.432      -0.075       0.032
year_1912     -0.0231      0.027     -0.852      0.394      -0.076       0.030
year_1913      0.0016      0.027      0.057      0.954      -0.052       0.055
year_1914     -0.0484      0.027     -1.783      0.075      -0.102       0.005
year_1915     -0.0649      0.027     -2.392      0.017      -0.118      -0.012
year_1916     -0.0413      0.027     -1.520      0.129      -0.095       0.012
year_1917     -0.0947      0.027     -3.489      0.000      -0.148      -0.041
year_1918     -0.1324      0.027     -4.879      0.000      -0.186      -0.079
year_1919     -0.0395      0.027     -1.453      0.146      -0.093       0.014
year_1920      0.0143      0.027      0.528      0.598      -0.039       0.068
year_1921     -0.0153      0.027     -0.564      0.573      -0.069       0.038
year_1922      0.0760      0.027      2.799      0.005       0.023       0.129
year_1923      0.1179      0.027      4.344      0.000       0.065       0.171
year_1924      0.0864      0.027      3.184      0.001       0.033       0.140
year_1925      0.0637      0.027      2.348      0.019       0.010       0.117
year_1926      0.0683      0.027      2.517      0.012       0.015       0.122
year_1927      0.0400      0.027      1.473      0.141      -0.013       0.093
year_1928      0.0346      0.027      1.276      0.202      -0.019       0.088
year_1929      0.0244      0.027      0.898      0.369      -0.029       0.078
year_1930     -0.0071      0.027     -0.263      0.792      -0.060       0.046
year_1931     -0.0417      0.027     -1.535      0.125      -0.095       0.012
year_1932     -0.0573      0.027     -2.112      0.035      -0.111      -0.004
year_1933     -0.0650      0.027     -2.395      0.017      -0.118      -0.012
year_1934     -0.0801      0.027     -2.950      0.003      -0.133      -0.027
year_1935     -0.0582      0.027     -2.144      0.032      -0.111      -0.005
year_1936     -0.0559      0.027     -2.059      0.040      -0.109      -0.003
year_1937     -0.0323      0.027     -1.189      0.235      -0.086       0.021
year_1938     -0.0265      0.027     -0.977      0.329      -0.080       0.027
year_1939     -0.0174      0.027     -0.642      0.521      -0.071       0.036
year_1940     -0.0760      0.027     -2.800      0.005      -0.129      -0.023
year_1941     -0.0512      0.027     -1.886      0.059      -0.104       0.002
year_1942     -0.0846      0.027     -3.116      0.002      -0.138      -0.031
year_1943     -0.0932      0.027     -3.432      0.001      -0.146      -0.040
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year_1944     -0.1488      0.027     -5.483      0.000      -0.202      -0.096
year_1945     -0.1919      0.027     -7.070      0.000      -0.245      -0.139
year_1946     -0.1046      0.027     -3.852      0.000      -0.158      -0.051
year_1947     -0.0382      0.027     -1.405      0.160      -0.091       0.015
year_1948     -0.0068      0.027     -0.250      0.802      -0.060       0.046
year_1949      0.0641      0.027      2.362      0.018       0.011       0.117
year_1950      0.1925      0.027      7.092      0.000       0.139       0.246
year_1951      0.1341      0.027      4.939      0.000       0.081       0.187
year_1952      0.0992      0.027      3.653      0.000       0.046       0.152
year_1953      0.0977      0.027      3.598      0.000       0.044       0.151
year_1954      0.1006      0.027      3.707      0.000       0.047       0.154
year_1955      0.1018      0.027      3.748      0.000       0.049       0.155
year_1956      0.0892      0.027      3.287      0.001       0.036       0.142
year_1957      0.0884      0.027      3.257      0.001       0.035       0.142
year_1958      0.0708      0.027      2.607      0.009       0.018       0.124
year_1959      0.0785      0.027      2.890      0.004       0.025       0.132
year_1960      0.0781      0.027      2.878      0.004       0.025       0.131
year_1961      0.0893      0.027      3.288      0.001       0.036       0.143
year_1962      0.1028      0.027      3.788      0.000       0.050       0.156
year_1963      0.1245      0.027      4.587      0.000       0.071       0.178
year_1964      0.1297      0.027      4.777      0.000       0.076       0.183
year_1965      0.1280      0.027      4.715      0.000       0.075       0.181
year_1966      0.1170      0.027      4.311      0.000       0.064       0.170
year_1967      0.1171      0.027      4.313      0.000       0.064       0.170
year_1968      0.1297      0.027      4.779      0.000       0.076       0.183
year_1969      0.1315      0.027      4.845      0.000       0.078       0.185
year_1970      0.1299      0.027      4.787      0.000       0.077       0.183
year_1971      0.1344      0.027      4.952      0.000       0.081       0.188
year_1972      0.1375      0.027      5.063      0.000       0.084       0.191
year_1973      0.1295      0.027      4.769      0.000       0.076       0.183
year_1974      0.0945      0.027      3.482      0.001       0.041       0.148
year_1975      0.0674      0.027      2.483      0.013       0.014       0.121
year_1976      0.0598      0.027      2.202      0.028       0.007       0.113
year_1977      0.0377      0.027      1.390      0.165      -0.015       0.091
year_1978      0.0201      0.027      0.739      0.460      -0.033       0.073
year_1979      0.0295      0.027      1.087      0.277      -0.024       0.083
year_1980      0.0306      0.027      1.127      0.260      -0.023       0.084
year_1981      0.0153      0.027      0.564      0.573      -0.038       0.069
year_1982      0.0024      0.027      0.089      0.929      -0.051       0.056
year_1983     -0.0053      0.027     -0.196      0.844      -0.059       0.048
year_1984     -0.0073      0.027     -0.269      0.788      -0.061       0.046
year_1985     -0.0003      0.027     -0.010      0.992      -0.054       0.053
year_1986     -0.0020      0.027     -0.075      0.941      -0.055       0.051
year_1987     -0.0023      0.027     -0.083      0.934      -0.055       0.051
year_1988     -0.0075      0.027     -0.276      0.783      -0.061       0.046
year_1989     -0.0117      0.027     -0.431      0.667      -0.065       0.042
year_1990     -0.0174      0.027     -0.641      0.522      -0.071       0.036
year_1991     -0.0181      0.027     -0.667      0.505      -0.071       0.035
year_1992     -0.0111      0.027     -0.408      0.683      -0.064       0.042
year_1993     -0.0121      0.027     -0.445      0.657      -0.065       0.041
year_1994     -0.0115      0.027     -0.425      0.671      -0.065       0.042
year_1995     -0.0133      0.027     -0.491      0.623      -0.067       0.040
year_1996     -0.0139      0.027     -0.514      0.608      -0.067       0.039
year_1997     -0.0168      0.027     -0.619      0.536      -0.070       0.036
year_1998     -0.0195      0.027     -0.720      0.472      -0.073       0.034
year_1999     -0.0283      0.027     -1.042      0.297      -0.082       0.025
year_2000     -0.0176      0.027     -0.649      0.517      -0.071       0.036
year_2001     -0.0185      0.027     -0.682      0.496      -0.072       0.035
year_2002     -0.0257      0.027     -0.946      0.344      -0.079       0.028
year_2003     -0.0273      0.027     -1.007      0.314      -0.081       0.026
year_2004     -0.0260      0.027     -0.957      0.339      -0.079       0.027
year_2005     -0.0386      0.027     -1.424      0.155      -0.092       0.015
year_2006     -0.0415      0.027     -1.527      0.127      -0.095       0.012
year_2007     -0.0475      0.027     -1.751      0.080      -0.101       0.006
year_2008     -0.0628      0.027     -2.312      0.021      -0.116      -0.010
year_2009     -0.0898      0.027     -3.306      0.001      -0.143      -0.037
year_2010     -0.0802      0.027     -2.953      0.003      -0.133      -0.027
year_2011     -0.0851      0.027     -3.135      0.002      -0.138      -0.032
year_2012     -0.0884      0.027     -3.257      0.001      -0.142      -0.035
year_2013     -0.0766      0.027     -2.823      0.005      -0.130      -0.023
year_2014     -0.0593      0.027     -2.184      0.029      -0.113      -0.006
year_2015     -0.0737      0.027     -2.716      0.007      -0.127      -0.020
year_2016     -0.0777      0.027     -2.862      0.004      -0.131      -0.024
year_2017     -0.0686      0.027     -2.528      0.012      -0.122      -0.015
year_2018     -0.0759      0.027     -2.797      0.005      -0.129      -0.023
year_2019     -0.0763      0.027     -2.810      0.005      -0.130      -0.023
year_2020     -0.0724      0.027     -2.667      0.008      -0.126      -0.019
==============================================================================
Omnibus:                      628.516   Durbin-Watson:                   0.432
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            11200.041
Skew:                          -0.996   Prob(JB):                         0.00
Kurtosis:                      14.374   Cond. No.                     1.03e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.5e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE — liquid liabilities (within)
                                OLS Regression Results
==============================================================================
Dep. Variable:     liquid_liabilities_within   R-squared:                  0.375
Model:                                   OLS   Adj. R-squared:             0.333
Method:                        Least Squares   F-statistic:                9.053
Date:                       Sat, 27 Dec 2025   Prob (F-statistic):     1.95e-132
Time:                               17:14:14   Log-Likelihood:            614.04
No. Observations:                       2224   AIC:                       -950.1
Df Residuals:                           2085   BIC:                       -156.8
Df Model:                                138
Covariance Type:                   nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882     -0.2593      0.047     -5.471      0.000      -0.352      -0.166
year_1883     -0.2499      0.047     -5.272      0.000      -0.343      -0.157
year_1884     -0.2346      0.047     -4.949      0.000      -0.328      -0.142
year_1885     -0.2225      0.047     -4.693      0.000      -0.315      -0.130
year_1886     -0.2125      0.047     -4.482      0.000      -0.305      -0.120
year_1887     -0.1996      0.047     -4.211      0.000      -0.293      -0.107
year_1888     -0.1947      0.047     -4.107      0.000      -0.288      -0.102
year_1889     -0.1992      0.047     -4.201      0.000      -0.292      -0.106
year_1890     -0.2061      0.047     -4.347      0.000      -0.299      -0.113
year_1891     -0.2124      0.047     -4.481      0.000      -0.305      -0.119
year_1892     -0.1979      0.047     -4.174      0.000      -0.291      -0.105
year_1893     -0.1837      0.047     -3.875      0.000      -0.277      -0.091
year_1894     -0.1702      0.047     -3.591      0.000      -0.263      -0.077
year_1895     -0.1588      0.047     -3.351      0.001      -0.252      -0.066
year_1896     -0.1540      0.047     -3.248      0.001      -0.247      -0.061
year_1897     -0.1496      0.047     -3.157      0.002      -0.243      -0.057
year_1898     -0.1525      0.047     -3.217      0.001      -0.245      -0.060
year_1899     -0.1408      0.047     -2.971      0.003      -0.234      -0.048
year_1900     -0.1409      0.047     -2.973      0.003      -0.234      -0.048
year_1901     -0.1266      0.047     -2.670      0.008      -0.220      -0.034
year_1902     -0.1152      0.047     -2.430      0.015      -0.208      -0.022
year_1903     -0.1167      0.047     -2.462      0.014      -0.210      -0.024
year_1904     -0.0998      0.047     -2.105      0.035      -0.193      -0.007
year_1905     -0.0864      0.047     -1.822      0.069      -0.179       0.006
year_1906     -0.0837      0.047     -1.765      0.078      -0.177       0.010
year_1907     -0.0959      0.047     -2.024      0.043      -0.189      -0.003
year_1908     -0.0651      0.047     -1.373      0.170      -0.158       0.027
year_1909     -0.0557      0.047     -1.176      0.240      -0.149       0.037
year_1910     -0.0517      0.047     -1.092      0.275      -0.145       0.041
year_1911     -0.0571      0.047     -1.205      0.228      -0.150       0.036
year_1912     -0.0682      0.047     -1.440      0.150      -0.161       0.025
year_1913     -0.0711      0.047     -1.499      0.134      -0.164       0.022
year_1914     -0.0183      0.047     -0.386      0.700      -0.111       0.075
year_1915     -0.0040      0.047     -0.085      0.932      -0.097       0.089
year_1916     -0.0323      0.047     -0.681      0.496      -0.125       0.061
year_1917      0.0049      0.047      0.104      0.917      -0.088       0.098
year_1918      0.0656      0.047      1.383      0.167      -0.027       0.158
year_1919      0.0616      0.047      1.300      0.194      -0.031       0.155
year_1920     -0.0968      0.047     -2.042      0.041      -0.190      -0.004
year_1921     -0.0499      0.047     -1.052      0.293      -0.143       0.043
year_1922     -0.0394      0.047     -0.831      0.406      -0.132       0.054
year_1923     -0.0609      0.047     -1.285      0.199      -0.154       0.032
year_1924     -0.0899      0.047     -1.897      0.058      -0.183       0.003
year_1925     -0.1025      0.047     -2.163      0.031      -0.195      -0.010
year_1926     -0.0850      0.047     -1.794      0.073      -0.178       0.008
year_1927     -0.0662      0.047     -1.396      0.163      -0.159       0.027
year_1928     -0.0660      0.047     -1.392      0.164      -0.159       0.027
year_1929     -0.0638      0.047     -1.347      0.178      -0.157       0.029
year_1930     -0.0205      0.047     -0.433      0.665      -0.113       0.072
year_1931      0.0354      0.047      0.747      0.455      -0.058       0.128
year_1932      0.0766      0.047      1.616      0.106      -0.016       0.170
year_1933      0.0776      0.047      1.637      0.102      -0.015       0.170
year_1934      0.0592      0.047      1.250      0.212      -0.034       0.152
year_1935      0.0491      0.047      1.036      0.300      -0.044       0.142
year_1936      0.0364      0.047      0.768      0.442      -0.057       0.129
year_1937      0.0155      0.047      0.327      0.744      -0.077       0.108
year_1938      0.0440      0.047      0.928      0.354      -0.049       0.137
year_1939      0.0338      0.047      0.714      0.475      -0.059       0.127
year_1940      0.0429      0.047      0.904      0.366      -0.050       0.136
year_1941      0.0726      0.047      1.532      0.126      -0.020       0.166
year_1942      0.0946      0.047      1.996      0.046       0.002       0.188
year_1943      0.1500      0.047      3.165      0.002       0.057       0.243
year_1944      0.2553      0.047      5.386      0.000       0.162       0.348
year_1945      0.2183      0.047      4.604      0.000       0.125       0.311
year_1946      0.0824      0.047      1.738      0.082      -0.011       0.175
year_1947      0.0121      0.047      0.255      0.799      -0.081       0.105
year_1948     -0.0260      0.047     -0.549      0.583      -0.119       0.067
year_1949     -0.0407      0.047     -0.858      0.391      -0.134       0.052
year_1950     -0.0675      0.047     -1.425      0.154      -0.160       0.025
year_1951     -0.1071      0.047     -2.258      0.024      -0.200      -0.014
year_1952     -0.1062      0.047     -2.240      0.025      -0.199      -0.013
year_1953     -0.0974      0.047     -2.055      0.040      -0.190      -0.005
year_1954     -0.0933      0.047     -1.967      0.049      -0.186      -0.000
year_1955     -0.0938      0.047     -1.979      0.048      -0.187      -0.001
year_1956     -0.1086      0.047     -2.291      0.022      -0.202      -0.016
year_1957     -0.1104      0.047     -2.329      0.020      -0.203      -0.017
year_1958     -0.0942      0.047     -1.987      0.047      -0.187      -0.001
year_1959     -0.0857      0.047     -1.808      0.071      -0.179       0.007
year_1960     -0.0837      0.047     -1.766      0.078      -0.177       0.009
year_1961     -0.0813      0.047     -1.715      0.087      -0.174       0.011
year_1962     -0.0748      0.047     -1.578      0.115      -0.168       0.018
year_1963     -0.0641      0.047     -1.353      0.176      -0.157       0.029
year_1964     -0.0639      0.047     -1.349      0.178      -0.157       0.029
year_1965     -0.0618      0.047     -1.304      0.192      -0.155       0.031
year_1966     -0.0602      0.047     -1.269      0.205      -0.153       0.032
year_1967     -0.0519      0.047     -1.095      0.273      -0.145       0.041
year_1968     -0.0402      0.047     -0.848      0.396      -0.133       0.053
year_1969     -0.0440      0.047     -0.928      0.354      -0.137       0.049
year_1970     -0.0555      0.047     -1.171      0.242      -0.148       0.037
year_1971     -0.0380      0.047     -0.801      0.423      -0.131       0.055
year_1972     -0.0160      0.047     -0.337      0.736      -0.109       0.077
year_1973     -0.0121      0.047     -0.255      0.799      -0.105       0.081
year_1974     -0.0331      0.047     -0.698      0.486      -0.126       0.060
year_1975     -0.0258      0.047     -0.543      0.587      -0.119       0.067
year_1976     -0.0237      0.047     -0.500      0.617      -0.117       0.069
year_1977     -0.0286      0.047     -0.603      0.546      -0.122       0.064
year_1978     -0.0037      0.047     -0.077      0.938      -0.097       0.089
year_1979      0.0115      0.047      0.243      0.808      -0.081       0.104
year_1980     -0.0090      0.047     -0.189      0.850      -0.102       0.084
year_1981     -0.0072      0.047     -0.151      0.880      -0.100       0.086
year_1982      0.0134      0.047      0.283      0.777      -0.080       0.106
year_1983      0.0168      0.047      0.353      0.724      -0.076       0.110
year_1984      0.0173      0.047      0.364      0.716      -0.076       0.110
year_1985      0.0237      0.047      0.499      0.618      -0.069       0.117
year_1986      0.0292      0.047      0.617      0.538      -0.064       0.122
year_1987      0.0377      0.047      0.795      0.427      -0.055       0.130
year_1988      0.0425      0.047      0.896      0.371      -0.051       0.135
year_1989      0.0467      0.047      0.984      0.325      -0.046       0.139
year_1990      0.0483      0.047      1.019      0.308      -0.045       0.141
year_1991      0.0504      0.047      1.063      0.288      -0.043       0.144
year_1992      0.0549      0.047      1.159      0.247      -0.038       0.148
year_1993      0.0674      0.047      1.422      0.155      -0.026       0.160
year_1994      0.0580      0.047      1.224      0.221      -0.035       0.151
year_1995      0.0587      0.047      1.239      0.215      -0.034       0.152
year_1996      0.0621      0.047      1.309      0.191      -0.031       0.155
year_1997      0.0584      0.047      1.231      0.218      -0.035       0.151
year_1998      0.0596      0.047      1.256      0.209      -0.033       0.152
year_1999      0.0704      0.047      1.486      0.137      -0.023       0.163
year_2000      0.0569      0.047      1.201      0.230      -0.036       0.150
year_2001      0.0762      0.047      1.606      0.108      -0.017       0.169
year_2002      0.0914      0.047      1.928      0.054      -0.002       0.184
year_2003      0.1107      0.047      2.336      0.020       0.018       0.203
year_2004      0.1167      0.047      2.463      0.014       0.024       0.209
year_2005      0.1378      0.047      2.908      0.004       0.045       0.230
year_2006      0.1549      0.047      3.267      0.001       0.062       0.248
year_2007      0.1864      0.047      3.933      0.000       0.093       0.279
year_2008      0.2170      0.047      4.579      0.000       0.124       0.310
year_2009      0.2579      0.047      5.441      0.000       0.165       0.351
year_2010      0.2527      0.047      5.331      0.000       0.160       0.345
year_2011      0.2581      0.047      5.444      0.000       0.165       0.351
year_2012      0.2832      0.047      5.974      0.000       0.190       0.376
year_2013      0.2941      0.047      6.204      0.000       0.201       0.387
year_2014      0.3054      0.047      6.442      0.000       0.212       0.398
year_2015      0.3298      0.047      6.958      0.000       0.237       0.423
year_2016      0.3545      0.047      7.479      0.000       0.262       0.447
year_2017      0.3639      0.047      7.676      0.000       0.271       0.457
year_2018      0.3644      0.047      7.686      0.000       0.271       0.457
year_2019      0.3782      0.047      7.979      0.000       0.285       0.471
year_2020      0.5318      0.047     11.219      0.000       0.439       0.625
==============================================================================
Omnibus:                      472.680   Durbin-Watson:                   0.125
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2197.817
Skew:                           0.940   Prob(JB):                         0.00
Kurtosis:                       7.492   Cond. No.                         1.00
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Time FE — market cap / GDP (within)
                            OLS Regression Results
==============================================================================
Dep. Variable:         mcap_gdp_within   R-squared:                       0.607
Model:                             OLS   Adj. R-squared:                  0.570
Method:                  Least Squares   F-statistic:                     16.67
Date:                 Sat, 27 Dec 2025   Prob (F-statistic):          1.03e-198
Time:                         17:14:14   Log-Likelihood:                 339.75
No. Observations:                 1488   AIC:                            -425.5
Df Residuals:                     1361   BIC:                             248.3
Df Model:                          126
Covariance Type:             nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882  -3.387e-18   1.68e-18     -2.020      0.044   -6.68e-18    -9.6e-20
year_1883  -4.404e+12   1.53e+13     -0.288      0.774   -3.44e+13    2.56e+13
year_1884  -3.106e-18   1.68e-18     -1.852      0.064   -6.39e-18    1.83e-19
year_1885  -1.312e+13   4.23e+13     -0.310      0.757   -9.61e+13    6.99e+13
year_1886  -3.292e-18   1.68e-18     -1.963      0.050   -6.58e-18    -2.9e-21
year_1887   2.126e+12    6.8e+12      0.313      0.754   -1.12e+13    1.55e+13
year_1888  -1.509e-18   1.68e-18     -0.900      0.368    -4.8e-18    1.78e-18
year_1889     -0.0924      0.071     -1.297      0.195      -0.232       0.047
year_1890     -0.1089      0.071     -1.530      0.126      -0.249       0.031
year_1891     -0.0999      0.071     -1.403      0.161      -0.240       0.040
year_1892     -0.0927      0.071     -1.302      0.193      -0.232       0.047
year_1893     -0.0832      0.071     -1.168      0.243      -0.223       0.058
year_1894     -0.0712      0.071     -1.000      0.318      -0.211       0.067
year_1895     -0.0529      0.071     -0.744      0.457      -0.193       0.086
year_1896     -0.0539      0.071     -0.757      0.449      -0.194       0.086
year_1897     -0.0474      0.071     -0.666      0.506      -0.187       0.092
year_1898     -0.1174      0.058     -2.020      0.044      -0.231      -0.003
year_1899  -1.091e+12   1.57e+13     -0.069      0.945   -3.19e+13    2.97e+13
year_1900     -0.0784      0.213     -0.369      0.712      -0.495       0.338
year_1901     -0.1077      0.058     -1.852      0.064      -0.222       0.006
year_1902     -0.0843      0.069     -1.219      0.223      -0.220       0.051
year_1903     -0.1138      0.058     -1.957      0.051      -0.228       0.001
year_1904     -0.0769      0.058     -1.318      0.188      -0.191       0.052
year_1905     -0.0537      0.058     -0.921      0.357      -0.168       0.077
year_1906     -0.0521      0.058     -0.897      0.370      -0.166       0.062
year_1907     -0.0939      0.058     -1.615      0.107      -0.208       0.020
year_1908     -0.0517      0.058     -0.889      0.374      -0.166       0.062
year_1909     -0.0417      0.058     -0.717      0.474      -0.156       0.072
year_1910     -0.0324      0.058     -0.557      0.578      -0.146       0.081
year_1911     -0.0334      0.058     -0.575      0.565      -0.147       0.078
year_1912     -0.0478      0.058     -0.822      0.411      -0.162       0.066
year_1913     -0.0697      0.058     -1.198      0.231      -0.184       0.046
year_1914     -0.0835      0.058     -1.437      0.151      -0.198       0.031
year_1915     -0.0933      0.058     -1.604      0.109      -0.207       0.020
year_1916     -0.1234      0.069     -1.798      0.072      -0.258       0.011
year_1917     -0.1343      0.058     -2.311      0.021      -0.248      -0.020
year_1918     -0.1590      0.058     -2.735      0.006      -0.273      -0.045
year_1919     -0.1685      0.058     -2.899      0.004      -0.283      -0.054
year_1920     -0.2460      0.058     -4.232      0.000      -0.360      -0.132
year_1921     -0.2183      0.058     -3.756      0.000      -0.332      -0.104
year_1922     -0.1769      0.058     -3.043      0.002      -0.291      -0.063
year_1923     -0.1790      0.058     -3.079      0.002      -0.293      -0.065
year_1924     -0.1735      0.058     -2.986      0.003      -0.288      -0.060
year_1925     -0.1624      0.058     -2.794      0.005      -0.276      -0.048
year_1926     -0.1239      0.058     -2.131      0.033      -0.238      -0.010
year_1927     -0.0673      0.058     -1.157      0.247      -0.181       0.047
year_1928     -0.0007      0.058     -0.012      0.990      -0.115       0.113
year_1929     -0.0194      0.058     -0.333      0.739      -0.133       0.095
year_1930     -0.0595      0.058     -1.024      0.306      -0.174       0.055
year_1931     -0.1256      0.058     -2.161      0.031      -0.240      -0.012
year_1932     -0.0846      0.058     -1.455      0.146      -0.199       0.029
year_1933     -0.0230      0.058     -0.395      0.693      -0.137       0.091
year_1934     -0.0337      0.058     -0.579      0.562      -0.148       0.081
year_1935      0.0037      0.058      0.064      0.949      -0.110       0.118
year_1936      0.0668      0.058      1.149      0.251      -0.047       0.181
year_1937     -0.0471      0.058     -0.810      0.418      -0.161       0.067
year_1938     -0.0519      0.058     -0.893      0.372      -0.166       0.062
year_1939     -0.0897      0.058     -1.543      0.123      -0.204       0.024
year_1940     -0.1428      0.058     -2.457      0.014      -0.257      -0.029
year_1941     -0.1637      0.058     -2.816      0.005      -0.278      -0.050
year_1942     -0.1714      0.058     -2.948      0.003      -0.285      -0.057
year_1943     -0.1706      0.058     -2.935      0.003      -0.285      -0.057
year_1944     -0.1584      0.058     -2.725      0.007      -0.272      -0.045
year_1945     -0.1167      0.058     -2.007      0.045      -0.231      -0.003
year_1946     -0.1138      0.058     -1.958      0.050      -0.228       0.000
year_1947     -0.1397      0.058     -2.404      0.016      -0.254      -0.026
year_1948     -0.2036      0.058     -3.503      0.000      -0.318      -0.090
year_1949     -0.2150      0.058     -3.699      0.000      -0.329      -0.101
year_1950     -0.2061      0.058     -3.545      0.000      -0.320      -0.092
year_1951     -0.2070      0.058     -3.562      0.000      -0.321      -0.093
year_1952     -0.2076      0.058     -3.571      0.000      -0.322      -0.094
year_1953     -0.1948      0.058     -3.351      0.001      -0.309      -0.081
year_1954     -0.1273      0.058     -2.190      0.029      -0.241      -0.013
year_1955     -0.1018      0.058     -1.752      0.080      -0.216       0.012
year_1956     -0.1008      0.058     -1.734      0.083      -0.215       0.013
year_1957     -0.1345      0.058     -2.314      0.021      -0.249      -0.020
year_1958     -0.0879      0.058     -1.511      0.131      -0.202       0.026
year_1959     -0.0326      0.058     -0.561      0.575      -0.147       0.081
year_1960     -0.0424      0.058     -0.729      0.466      -0.156       0.071
year_1961      0.0063      0.058      0.108      0.914      -0.108       0.120
year_1962     -0.0494      0.058     -0.850      0.395      -0.163       0.064
year_1963     -0.0296      0.058     -0.510      0.610      -0.144       0.084
year_1964     -0.0415      0.058     -0.714      0.475      -0.156       0.073
year_1965     -0.0577      0.058     -0.992      0.321      -0.172       0.056
year_1966     -0.1137      0.058     -1.957      0.051      -0.228       0.000
year_1967     -0.0688      0.058     -1.184      0.237      -0.183       0.046
year_1968     -0.0199      0.058     -0.342      0.732      -0.134       0.094
year_1969     -0.0326      0.058     -0.561      0.575      -0.147       0.081
year_1970     -0.0900      0.058     -1.548      0.122      -0.204       0.024
year_1971     -0.0871      0.058     -1.498      0.134      -0.201       0.027
year_1972     -0.0384      0.058     -0.660      0.509      -0.152       0.076
year_1973     -0.0751      0.058     -1.293      0.196      -0.189       0.039
year_1974     -0.2085      0.058     -3.587      0.000      -0.323      -0.094
year_1975     -0.1870      0.058     -3.216      0.001      -0.301      -0.073
year_1976     -0.2178      0.058     -3.747      0.000      -0.332      -0.104
year_1977     -0.2443      0.058     -4.203      0.000      -0.358      -0.130
year_1978     -0.2464      0.058     -4.239      0.000      -0.360      -0.132
year_1979     -0.2384      0.058     -4.102      0.000      -0.352      -0.124
year_1980     -0.2094      0.058     -3.602      0.000      -0.323      -0.095
year_1981     -0.2314      0.058     -3.982      0.000      -0.345      -0.117
year_1982     -0.2305      0.058     -3.966      0.000      -0.345      -0.116
year_1983     -0.1719      0.058     -2.958      0.003      -0.286      -0.058
year_1984     -0.1800      0.058     -3.096      0.002      -0.294      -0.066
year_1985     -0.1083      0.058     -1.862      0.063      -0.222       0.006
year_1986     -0.0282      0.058     -0.486      0.627      -0.142       0.086
year_1987     -0.0422      0.058     -0.726      0.468      -0.156       0.072
year_1988     -0.0280      0.058     -0.482      0.630      -0.142       0.086
year_1989      0.0405      0.058      0.697      0.486      -0.073       0.154
year_1990     -0.0681      0.058     -1.172      0.241      -0.182       0.046
year_1991     -0.0310      0.058     -0.533      0.594      -0.145       0.083
year_1992     -0.0322      0.058     -0.554      0.580      -0.146       0.082
year_1993      0.0992      0.058      1.707      0.088      -0.015       0.213
year_1994      0.0653      0.058      1.124      0.261      -0.049       0.179
year_1995      0.1271      0.058      2.187      0.029       0.013       0.241
year_1996      0.2383      0.058      4.099      0.000       0.124       0.352
year_1997      0.4340      0.058      7.466      0.000       0.320       0.548
year_1998      0.5024      0.058      8.642      0.000       0.388       0.616
year_1999      0.6879      0.058     11.835      0.000       0.574       0.802
year_2000      0.6889      0.058     11.851      0.000       0.575       0.803
year_2001      0.4963      0.058      8.537      0.000       0.382       0.610
year_2002      0.2506      0.058      4.312      0.000       0.137       0.365
year_2003      0.3757      0.058      6.463      0.000       0.262       0.490
year_2004      0.4389      0.058      7.550      0.000       0.325       0.553
year_2005      0.5846      0.058     10.057      0.000       0.471       0.699
year_2006      0.7156      0.058     12.311      0.000       0.602       0.830
year_2007      0.6849      0.058     11.782      0.000       0.571       0.799
year_2008      0.2069      0.058      3.560      0.000       0.093       0.321
year_2009      0.4241      0.058      7.297      0.000       0.310       0.538
year_2010      0.4866      0.058      8.372      0.000       0.373       0.601
year_2011      0.3345      0.058      5.755      0.000       0.221       0.449
year_2012      0.3973      0.058      6.834      0.000       0.283       0.511
year_2013      0.5462      0.058      9.397      0.000       0.432       0.660
year_2014      0.5658      0.058      9.733      0.000       0.452       0.680
year_2015      0.5766      0.058      9.920      0.000       0.463       0.691
year_2016      0.6045      0.058     10.399      0.000       0.490       0.719
year_2017           0          0        nan        nan           0           0
year_2018           0          0        nan        nan           0           0
year_2019           0          0        nan        nan           0           0
year_2020           0          0        nan        nan           0           0
==============================================================================
Omnibus:                      681.683   Durbin-Watson:                   0.160
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6313.782
Skew:                           1.904   Prob(JB):                         0.00
Kurtosis:                      12.345   Cond. No.                     1.05e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.1e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE — bank loans / GDP (within)
                              OLS Regression Results
==============================================================================
Dep. Variable:     bank_loans_gdp_within   R-squared:                       0.682
Model:                               OLS   Adj. R-squared:                  0.654
Method:                    Least Squares   F-statistic:                     23.81
Date:                   Sat, 27 Dec 2025   Prob (F-statistic):          7.71e-292
Time:                           17:14:14   Log-Likelihood:                 818.80
No. Observations:                   1668   AIC:                            -1360.
Df Residuals:                       1529   BIC:                            -606.3
Df Model:                            138
Covariance Type:               nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882     -0.2828      0.045     -6.333      0.000      -0.370      -0.195
year_1883     -0.2797      0.045     -6.264      0.000      -0.367      -0.192
year_1884     -0.2580      0.045     -5.778      0.000      -0.346      -0.170
year_1885     -0.2407      0.045     -5.390      0.000      -0.328      -0.153
year_1886     -0.2255      0.045     -5.049      0.000      -0.313      -0.138
year_1887     -0.2148      0.045     -4.810      0.000      -0.302      -0.127
year_1888     -0.2069      0.045     -4.634      0.000      -0.295      -0.119
year_1889     -0.2015      0.045     -4.511      0.000      -0.289      -0.114
year_1890     -0.2038      0.045     -4.564      0.000      -0.291      -0.116
year_1891     -0.1979      0.045     -4.432      0.000      -0.285      -0.110
year_1892     -0.1849      0.045     -4.140      0.000      -0.272      -0.097
year_1893     -0.1650      0.045     -3.695      0.000      -0.253      -0.077
year_1894     -0.1492      0.045     -3.341      0.001      -0.237      -0.061
year_1895     -0.1430      0.045     -3.202      0.001      -0.231      -0.055
year_1896     -0.1231      0.045     -2.756      0.006      -0.211      -0.035
year_1897     -0.1237      0.045     -2.769      0.006      -0.211      -0.036
year_1898     -0.1159      0.045     -2.596      0.010      -0.204      -0.028
year_1899     -0.0995      0.045     -2.228      0.026      -0.187      -0.012
year_1900     -0.0959      0.045     -2.147      0.032      -0.183      -0.008
year_1901     -0.0844      0.045     -1.891      0.059      -0.172       0.004
year_1902     -0.0722      0.045     -1.616      0.106      -0.160       0.016
year_1903     -0.0638      0.045     -1.428      0.153      -0.151       0.024
year_1904     -0.0258      0.045     -0.578      0.563      -0.113       0.062
year_1905     -0.0169      0.045     -0.378      0.706      -0.104       0.071
year_1906     -0.0051      0.045     -0.114      0.909      -0.093       0.083
year_1907      0.0038      0.045      0.086      0.931      -0.084       0.092
year_1908      0.0321      0.045      0.719      0.472      -0.055       0.119
year_1909      0.0376      0.045      0.842      0.400      -0.050       0.125
year_1910      0.0488      0.045      1.093      0.274      -0.039       0.136
year_1911      0.0497      0.045      1.112      0.266      -0.038       0.137
year_1912      0.0476      0.045      1.065      0.287      -0.040       0.135
year_1913      0.0429      0.045      0.961      0.336      -0.045       0.131
year_1914      0.0718      0.045      1.608      0.108      -0.016       0.160
year_1915      0.0168      0.045      0.377      0.706      -0.071       0.104
year_1916     -0.0672      0.045     -1.505      0.132      -0.155       0.020
year_1917     -0.0793      0.045     -1.775      0.076      -0.167       0.009
year_1918     -0.0835      0.045     -1.869      0.062      -0.171       0.004
year_1919     -0.0647      0.045     -1.449      0.148      -0.152       0.023
year_1920     -0.1321      0.045     -2.957      0.003      -0.220      -0.045
year_1921     -0.0478      0.045     -1.071      0.284      -0.135       0.040
year_1922     -0.0523      0.045     -1.171      0.242      -0.140       0.035
year_1923     -0.0710      0.045     -1.590      0.112      -0.159       0.017
year_1924     -0.1028      0.045     -2.302      0.021      -0.190      -0.015
year_1925     -0.1060      0.045     -2.374      0.018      -0.194      -0.018
year_1926     -0.0616      0.045     -1.379      0.168      -0.149       0.026
year_1927     -0.0349      0.045     -0.781      0.435      -0.122       0.053
year_1928     -0.0305      0.045     -0.683      0.495      -0.118       0.057
year_1929     -0.0198      0.045     -0.443      0.658      -0.107       0.068
year_1930      0.0190      0.045      0.426      0.670      -0.069       0.107
year_1931      0.0774      0.045      1.732      0.083      -0.010       0.165
year_1932      0.1000      0.045      2.239      0.025       0.012       0.188
year_1933      0.0699      0.045      1.565      0.118      -0.018       0.158
year_1934      0.0286      0.045      0.641      0.522      -0.059       0.116
year_1935      0.0045      0.045      0.101      0.919      -0.083       0.092
year_1936     -0.0237      0.045     -0.532      0.595      -0.111       0.064
year_1937     -0.0702      0.045     -1.573      0.116      -0.158       0.018
year_1938     -0.0761      0.045     -1.704      0.089      -0.164       0.012
year_1939     -0.0883      0.045     -1.977      0.048      -0.176      -0.000
year_1940     -0.1357      0.045     -3.038      0.002      -0.223      -0.048
year_1941     -0.1864      0.045     -4.174      0.000      -0.274      -0.099
year_1942     -0.2231      0.045     -4.995      0.000      -0.311      -0.135
year_1943     -0.2566      0.045     -5.746      0.000      -0.344      -0.169
year_1944     -0.2716      0.045     -6.082      0.000      -0.359      -0.184
year_1945     -0.2944      0.045     -6.592      0.000      -0.382      -0.207
year_1946     -0.3033      0.045     -6.791      0.000      -0.391      -0.216
year_1947     -0.3110      0.045     -6.965      0.000      -0.399      -0.223
year_1948     -0.3048      0.045     -6.826      0.000      -0.392      -0.217
year_1949     -0.2878      0.045     -6.445      0.000      -0.375      -0.200
year_1950     -0.2748      0.045     -6.153      0.000      -0.362      -0.187
year_1951     -0.2822      0.045     -6.320      0.000      -0.370      -0.195
year_1952     -0.2721      0.045     -6.093      0.000      -0.360      -0.184
year_1953     -0.2598      0.045     -5.817      0.000      -0.347      -0.172
year_1954     -0.2464      0.045     -5.518      0.000      -0.334      -0.158
year_1955     -0.2393      0.045     -5.359      0.000      -0.327      -0.152
year_1956     -0.2370      0.045     -5.307      0.000      -0.325      -0.149
year_1957     -0.2345      0.045     -5.251      0.000      -0.322      -0.147
year_1958     -0.2216      0.045     -4.962      0.000      -0.309      -0.134
year_1959     -0.2054      0.045     -4.601      0.000      -0.293      -0.118
year_1960     -0.1785      0.045     -3.997      0.000      -0.266      -0.091
year_1961     -0.1712      0.045     -3.834      0.000      -0.259      -0.084
year_1962     -0.1525      0.045     -3.415      0.001      -0.240      -0.065
year_1963     -0.1318      0.045     -2.950      0.003      -0.219      -0.044
year_1964     -0.1282      0.045     -2.871      0.004      -0.216      -0.041
year_1965     -0.1174      0.045     -2.628      0.009      -0.205      -0.030
year_1966     -0.1130      0.045     -2.531      0.011      -0.201      -0.025
year_1967     -0.1021      0.045     -2.287      0.022      -0.190      -0.014
year_1968     -0.0912      0.045     -2.043      0.041      -0.179      -0.004
year_1969     -0.0807      0.045     -1.806      0.071      -0.168       0.007
year_1970     -0.0904      0.045     -2.025      0.043      -0.178      -0.002
year_1971     -0.0750      0.045     -1.679      0.093      -0.163       0.013
year_1972     -0.0588      0.045     -1.317      0.188      -0.146       0.029
year_1973     -0.0525      0.045     -1.175      0.240      -0.140       0.035
year_1974     -0.0447      0.045     -1.000      0.317      -0.132       0.043
year_1975     -0.0416      0.045     -0.932      0.351      -0.129       0.046
year_1976     -0.0438      0.045     -0.981      0.327      -0.131       0.044
year_1977     -0.0353      0.045     -0.790      0.430      -0.123       0.052
year_1978     -0.0281      0.045     -0.629      0.529      -0.116       0.060
year_1979     -0.0200      0.045     -0.448      0.654      -0.108       0.068
year_1980     -0.0198      0.045     -0.444      0.657      -0.107       0.068
year_1981     -0.0120      0.045     -0.269      0.788      -0.100       0.076
year_1982     -0.0071      0.045     -0.158      0.875      -0.095       0.081
year_1983     -0.0019      0.045     -0.042      0.967      -0.089       0.085
year_1984      0.0202      0.045      0.451      0.652      -0.067       0.108
year_1985      0.0568      0.045      1.272      0.203      -0.031       0.145
year_1986      0.1037      0.045      2.323      0.020       0.016       0.191
year_1987      0.1421      0.045      3.183      0.001       0.055       0.230
year_1988      0.1776      0.045      3.977      0.000       0.090       0.265
year_1989      0.2099      0.045      4.701      0.000       0.122       0.298
year_1990      0.2251      0.045      5.040      0.000       0.137       0.313
year_1991      0.2140      0.045      4.793      0.000       0.126       0.302
year_1992      0.2167      0.045      4.852      0.000       0.129       0.304
year_1993      0.2064      0.045      4.621      0.000       0.119       0.294
year_1994      0.1743      0.045      3.903      0.000       0.087       0.262
year_1995      0.1691      0.045      3.787      0.000       0.082       0.257
year_1996      0.1674      0.045      3.748      0.000       0.080       0.255
year_1997      0.1728      0.045      3.870      0.000       0.085       0.261
year_1998      0.1931      0.045      4.323      0.000       0.105       0.281
year_1999      0.2093      0.045      4.688      0.000       0.122       0.297
year_2000      0.2096      0.045      4.693      0.000       0.122       0.297
year_2001      0.2318      0.045      5.191      0.000       0.144       0.319
year_2002      0.2529      0.045      5.663      0.000       0.165       0.340
year_2003      0.2743      0.045      6.143      0.000       0.187       0.362
year_2004      0.2877      0.045      6.442      0.000       0.200       0.375
year_2005      0.3245      0.045      7.267      0.000       0.237       0.412
year_2006      0.3564      0.045      7.982      0.000       0.269       0.444
year_2007      0.4041      0.045      9.050      0.000       0.317       0.492
year_2008      0.4376      0.045      9.800      0.000       0.350       0.525
year_2009      0.4828      0.045     10.812      0.000       0.395       0.570
year_2010      0.4506      0.045     10.091      0.000       0.363       0.538
year_2011      0.4381      0.045      9.810      0.000       0.350       0.526
year_2012      0.4381      0.045      9.810      0.000       0.350       0.526
year_2013      0.4398      0.045      9.849      0.000       0.352       0.527
year_2014      0.4522      0.045     10.125      0.000       0.365       0.540
year_2015      0.4656      0.045     10.427      0.000       0.378       0.553
year_2016      0.4767      0.045     10.674      0.000       0.389       0.564
year_2017      0.4719      0.045     10.568      0.000       0.384       0.560
year_2018      0.5182      0.045     11.605      0.000       0.431       0.606
year_2019      0.5240      0.045     11.735      0.000       0.436       0.612
year_2020      0.5910      0.045     13.235      0.000       0.503       0.679
==============================================================================
Omnibus:                       96.861   Durbin-Watson:                   0.085
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              135.646
Skew:                           0.509   Prob(JB):                     3.51e-30
Kurtosis:                       3.956   Cond. No.                         1.00
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```
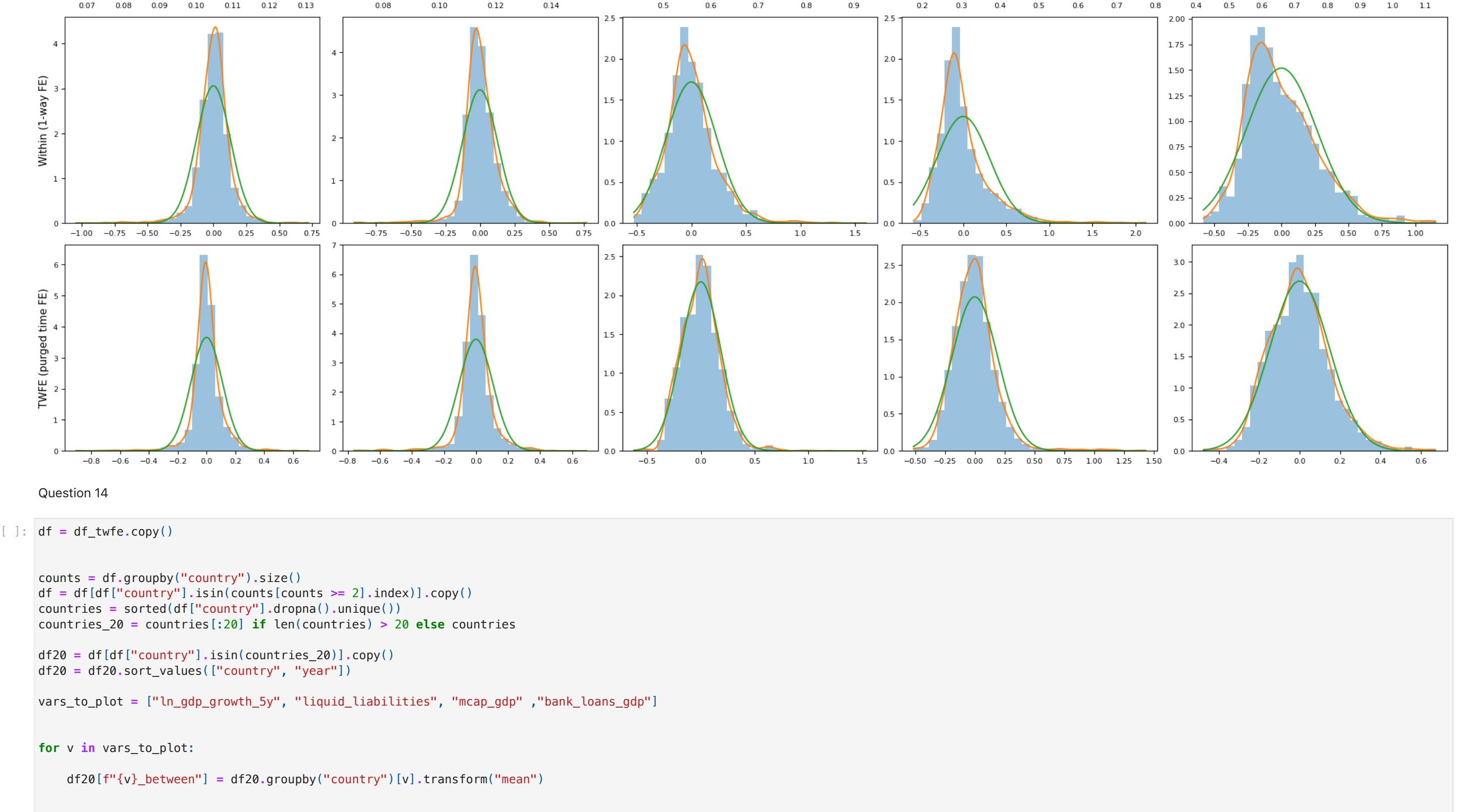

Question 13

In [ ]:

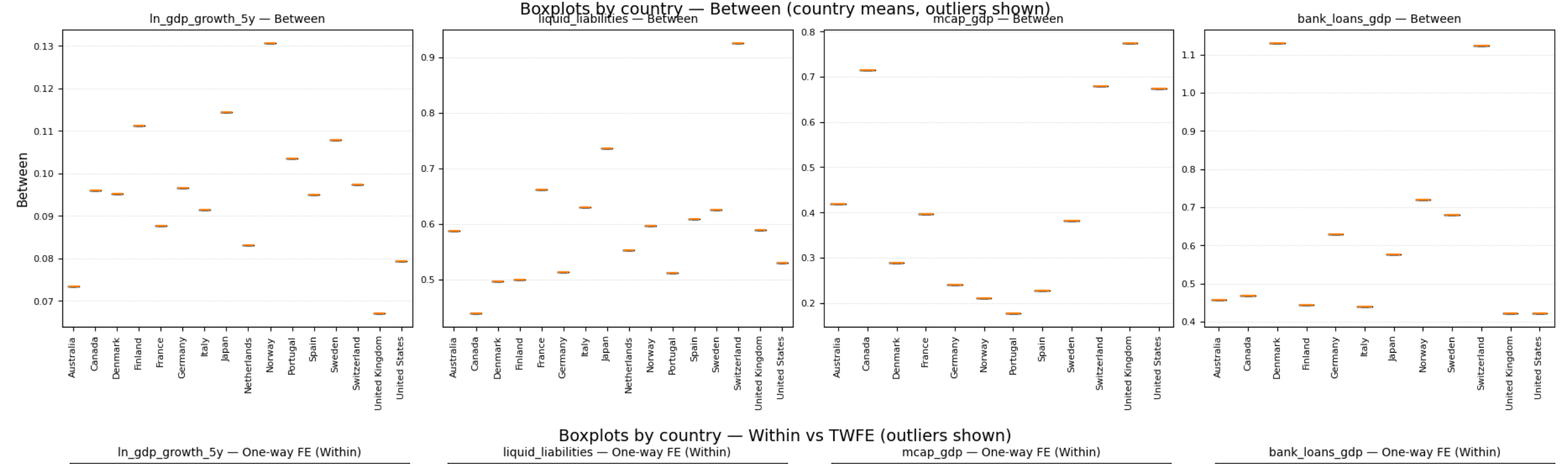
```
df_twfe["ln_gdp_growth_5y_twfe"] = np.nan
df_twfe.loc[dummy_reg_gdp, "ln_gdp_growth_5y_twfe"] = ln_gdp_time_fe.resid

df_twfe["ln_labour_productivity_growth_5y_twfe"] = np.nan
df_twfe.loc[dummy_reg_gdp_labprod, "ln_labour_productivity_growth_5y_twfe"] = ln_labprod_time_fe.resid

df_twfe["liquid_liabilities_twfe"] = np.nan
df_twfe.loc[dummy_reg_liquid_liabilities, "liquid_liabilities_twfe"] = liquid_liability_time_fe.resid

df_twfe["mcap_gdp_twfe"] = np.nan
df_twfe.loc[dummy_reg_mcap, "mcap_gdp_twfe"] = mcap_time_fe.resid

df_twfe["bank_loans_gdp_twfe"] = np.nan
df_twfe.loc[dummy_reg_bank, "bank_loans_gdp_twfe"] = bank_loans_time_fe.resid

df_twfe["ln_gdp_growth_5y_between"] = df_twfe.groupby("country")["ln_gdp_growth_5y"].transform("mean")
df_twfe["ln_labour_productivity_growth_5y_between"] = df_twfe.groupby("country")["ln_labour_productivity_growth_5y"].transform("mean")
df_twfe["liquid_liabilities_between"] = df_twfe.groupby("country")["liquid_liabilities"].transform("mean")
df_twfe["mcap_gdp_between"] = df_twfe.groupby("country")["mcap_gdp"].transform("mean")
df_twfe["bank_loans_gdp_between"] = df_twfe.groupby("country")["bank_loans_gdp"].transform("mean")

vars_info = [
    ("ln_gdp_growth_5y", "ln_gdp_growth_5y_between", "ln_gdp_growth_5y_within", "ln_gdp_growth_5y_twfe", "GDP growth (5y, log diff)"),
    ("ln_labour_productivity_growth_5y", "ln_labour_productivity_growth_5y_between", "ln_labour_productivity_growth_5y_within", "ln_labour_productivity_growth_5y_twfe", "Labour prod growth (5y)"),
    ("liquid_liabilities", "liquid_liabilities_between", "liquid_liabilities_within", "liquid_liabilities_twfe", "Liquid liabilities"),
    ("mcap_gdp", "mcap_gdp_between", "mcap_gdp_within", "mcap_gdp_twfe", "Market cap / GDP"),
    ("bank_loans_gdp", "bank_loans_gdp_between", "bank_loans_gdp_within", "bank_loans_gdp_twfe", "Bank loans / GDP"),
]

row_names = ["Between", "Within (1-way FE)", "TWFE (purged time FE)"]

nvars = len(vars_info)
fig, axes = plt.subplots(3, nvars, figsize=(4.2*nvars, 10.5), constrained_layout=True)

def plot_dist(ax, s, title=None):
    s = pd.Series(s).dropna().astype(float)

    if s.empty or s.var(ddof=1) == 0:
        ax.text(0.5, 0.5, "No variation / empty", ha="center", va="center")
        ax.set_axis_off()
        return


    ax.hist(s, bins=30, density=True, alpha=0.45)


    kde = gaussian_kde(s.values)
    x = np.linspace(s.min(), s.max(), 400)
    ax.plot(x, kde(x), linewidth=1.5)


    mu = s.mean()
    sigma = s.std(ddof=1)
    ax.plot(x, norm.pdf(x, loc=mu, scale=sigma), linewidth=1.5)

    if title is not None:
        ax.set_title(title, fontsize=10)

    ax.tick_params(axis="both", labelsize=8)

for j, (base, between_col, within_col, twfe_col, nice) in enumerate(vars_info):
    plot_dist(axes[0, j], df_twfe[between_col], title=nice)   # Between
    plot_dist(axes[1, j], df_twfe[within_col])                # Within
    plot_dist(axes[2, j], df_twfe[twfe_col])                  # TWFE

for i in range(3):
    axes[i, 0].set_ylabel(row_names[i], fontsize=11)

plt.suptitle("Distributions: Between vs Within vs TWFE (Hist + KDE + Normal fit)", fontsize=14)
plt.show()
```

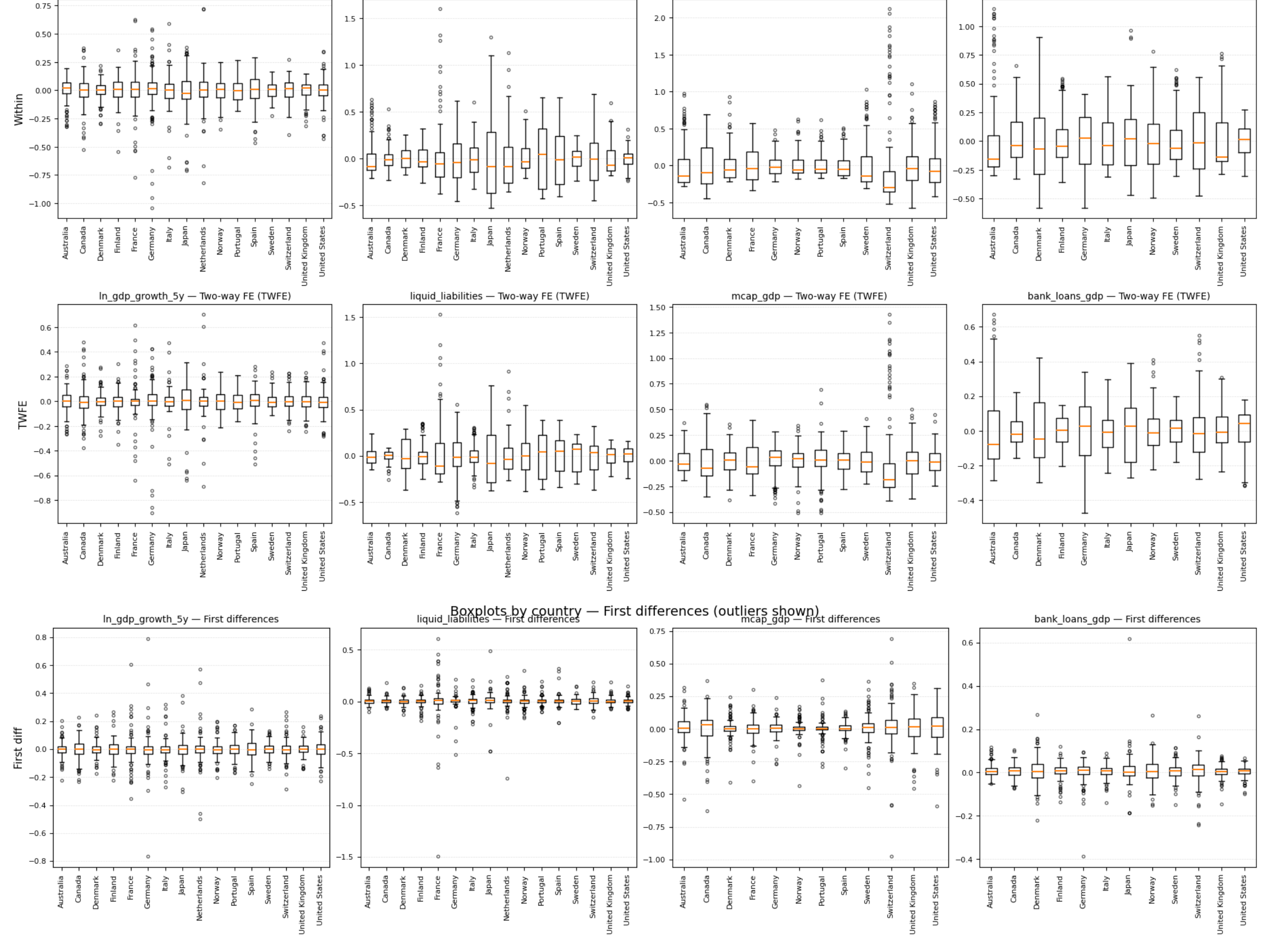


Question 14

In [ ]:

```
df = df_twfe.copy()


counts = df.groupby("country").size()
df = df[df["country"].isin(counts[counts >= 2].index)].copy()
countries = sorted(df["country"].dropna().unique())
countries_20 = countries[:20] if len(countries) > 20 else countries

df20 = df[df["country"].isin(countries_20)].copy()
df20 = df20.sort_values(["country", "year"])

vars_to_plot = ["ln_gdp_growth_5y", "liquid_liabilities", "mcap_gdp" ,"bank_loans_gdp"]


for v in vars_to_plot:

    df20[f"{v}_between"] = df20.groupby("country")[v].transform("mean")


    df20[f"{v}_within"] = df20[v] - df20.groupby("country")[v].transform("mean")


    df20[f"{v}_fd"] = df20.groupby("country")[v].diff()

# two-way FE: within then remove year effects
for v in vars_to_plot:
    w = df20[f"{v}_within"]
    df20[f"{v}_twfe"] = w - w.groupby(df20["year"]).transform("mean")


def boxplot_by_country(ax, df_in, value_col, title, country_list):
    data = []
    labels = []

    for c in country_list:
        s = df_in.loc[df_in["country"] == c, value_col].dropna()
        if len(s) > 0:
            data.append(s.to_numpy())
            labels.append(str(c))

    if len(data) == 0:
        ax.set_title(title, fontsize=10)
        ax.text(0.5, 0.5, "No data", ha="center", va="center", transform=ax.transAxes)
        ax.set_xticks([])
        return

    ax.boxplot(
        data,
        showfliers=True,
        flierprops=dict(marker="o", markersize=3, alpha=0.6),
        medianprops=dict(linewidth=1.5),
        whiskerprops=dict(linewidth=1.0),
        capprops=dict(linewidth=1.0),
        boxprops=dict(linewidth=1.0),
    )
    ax.set_title(title, fontsize=10)
    ax.set_xticklabels(labels, rotation=90, fontsize=8)
    ax.tick_params(axis="y", labelsize=8)
    ax.yaxis.grid(True, linestyle="--", linewidth=0.5, alpha=0.4)


def _axes_2d(axes, nrows, ncols):

    axes = np.array(axes)
    if nrows == 1 and ncols == 1:
        return axes.reshape(1, 1)
    if nrows == 1:
        return axes.reshape(1, -1)
    if ncols == 1:
        return axes.reshape(-1, 1)
    return axes

# between
cols = len(vars_to_plot)
fig1, axes1 = plt.subplots(1, cols, figsize=(4.6 * cols, 4.8), constrained_layout=True)
axes1 = _axes_2d(axes1, 1, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes1[0, j], df20, f"{v}_between", f"{v} — Between", countries_20)

axes1[0, 0].set_ylabel("Between", fontsize=11)
fig1.suptitle("Boxplots by country — Between (country means, outliers shown)", fontsize=14, y=1.02)

plt.show()

# within
fig2, axes2 = plt.subplots(2, cols, figsize=(4.6 * cols, 9.0), constrained_layout=True)
axes2 = _axes_2d(axes2, 2, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes2[0, j], df20, f"{v}_within", f"{v} — One-way FE (Within)", countries_20)
    boxplot_by_country(axes2[1, j], df20, f"{v}_twfe",   f"{v} — Two-way FE (TWFE)",     countries_20)

axes2[0, 0].set_ylabel("Within", fontsize=11)
axes2[1, 0].set_ylabel("TWFE", fontsize=11)
fig2.suptitle("Boxplots by country — Within vs TWFE (outliers shown)", fontsize=14, y=1.02)

plt.show()

#firsyt diff
fig3, axes3 = plt.subplots(1, cols, figsize=(4.6 * cols, 4.8), constrained_layout=True)
axes3 = _axes_2d(axes3, 1, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes3[0, j], df20, f"{v}_fd", f"{v} — First differences", countries_20)

axes3[0, 0].set_ylabel("First diff", fontsize=11)
fig3.suptitle("Boxplots by country — First differences (outliers shown)", fontsize=14, y=1.02)

plt.show()
```





Question 14 --> Dessy Stats

In [ ]:
```
def descriptive_table(df, col):
    """
    Return a Series with descriptive stats for df[col].
    Handles empty / all-NaN columns safely.
    """
    # select series and drop missing values
    s = df[col].dropna().astype(float)

    n = len(s)
    if n == 0:
        return pd.Series({
            "Mean": np.nan,
            "SE": np.nan,
            "Q1": np.nan,
            "Median": np.nan,
            "Q3": np.nan,
            "Std_MAX": np.nan,
            "Std_MIN": np.nan
        })

    mean = s.mean()
    se = s.std(ddof=1) / np.sqrt(n) if n > 1 else np.nan

    #
    q1 = s.quantile(0.25)
    median = s.median()
    q3 = s.quantile(0.75)

    std_max = (s.max() - mean) / se if se and not np.isclose(se, 0) else np.nan
    std_min = (s.min() - mean) / se if se and not np.isclose(se, 0) else np.nan

    return pd.Series({
        "Mean": mean,
        "SE": se,
        "Q1": q1,
        "Median": median,
        "Q3": q3,
        "Std_MAX": std_max,
        "Std_MIN": std_min
    })


tables = {}

for v in vars_to_plot:
    tables[v] = pd.DataFrame({
        "Between": descriptive_table(df20, f"{v}_between"),
        "Within": descriptive_table(df20, f"{v}_within"),
        "TWFE": descriptive_table(df20, f"{v}_twfe"),
        "FirstDiff": descriptive_table(df20, f"{v}_fd"),
    })

for v in ["ln_gdp_growth_5y", "liquid_liabilities", "mcap_gdp", "bank_loans_gdp"]:
    print(f"\n=== {v} ===")
    print(tables[v])
```

```
=== ln_gdp_growth_5y ===
          Between    Within      TWFE  FirstDiff
Mean       0.0956    0.0000    0.0000    -0.0004
SE         0.0003    0.0028    0.0024     0.0016
Q1         0.0866   -0.0574   -0.0415    -0.0289
Median     0.0956    0.0064   -0.0023    -0.0017
Q3         0.1047    0.0617    0.0418     0.0267
Std_MAX  106.5262  255.6753  299.3752   489.1175
Std_MIN  -86.7047 -370.2491 -384.5361  -473.1952

=== liquid_liabilities ===
          Between    Within      TWFE   FirstDiff
Mean       0.5938   -0.0000    0.0000      0.0057
SE         0.0024    0.0049    0.0039      0.0014
Q1         0.5121   -0.1393   -0.1271     -0.0107
Median     0.5888   -0.0253   -0.0005      0.0056
Q3         0.6261    0.1109    0.1041      0.0224
Std_MAX  140.3621  324.6846  393.7554    420.9124
Std_MIN  -65.6070 -106.8822 -159.7807 -1,046.0670

=== mcap_gdp ===
          Between   Within      TWFE  FirstDiff
Mean       0.4319  -0.0000   -0.0000     0.0061
SE         0.0052   0.0080    0.0050     0.0028
Q1         0.2362  -0.1747   -0.1155    -0.0237
Median     0.3893  -0.0742   -0.0116     0.0041
Q3         0.6758   0.0952    0.0836     0.0445
Std_MAX   66.2633 265.8348  286.0876   249.4248
Std_MIN  -49.2983 -72.4132 -102.6897  -356.5606

=== bank_loans_gdp ===
          Between   Within      TWFE  FirstDiff
Mean       0.6261   0.0000   -0.0000     0.0063
SE         0.0060   0.0064    0.0036     0.0011
Q1         0.4420  -0.1897   -0.1039    -0.0112
Median     0.5229  -0.0508   -0.0070     0.0072
Q3         0.6909   0.1562    0.0868     0.0249
Std_MAX   83.9762 178.6632  185.6218   541.8276
Std_MIN  -34.0340 -90.4951 -131.4073  -348.0149
```

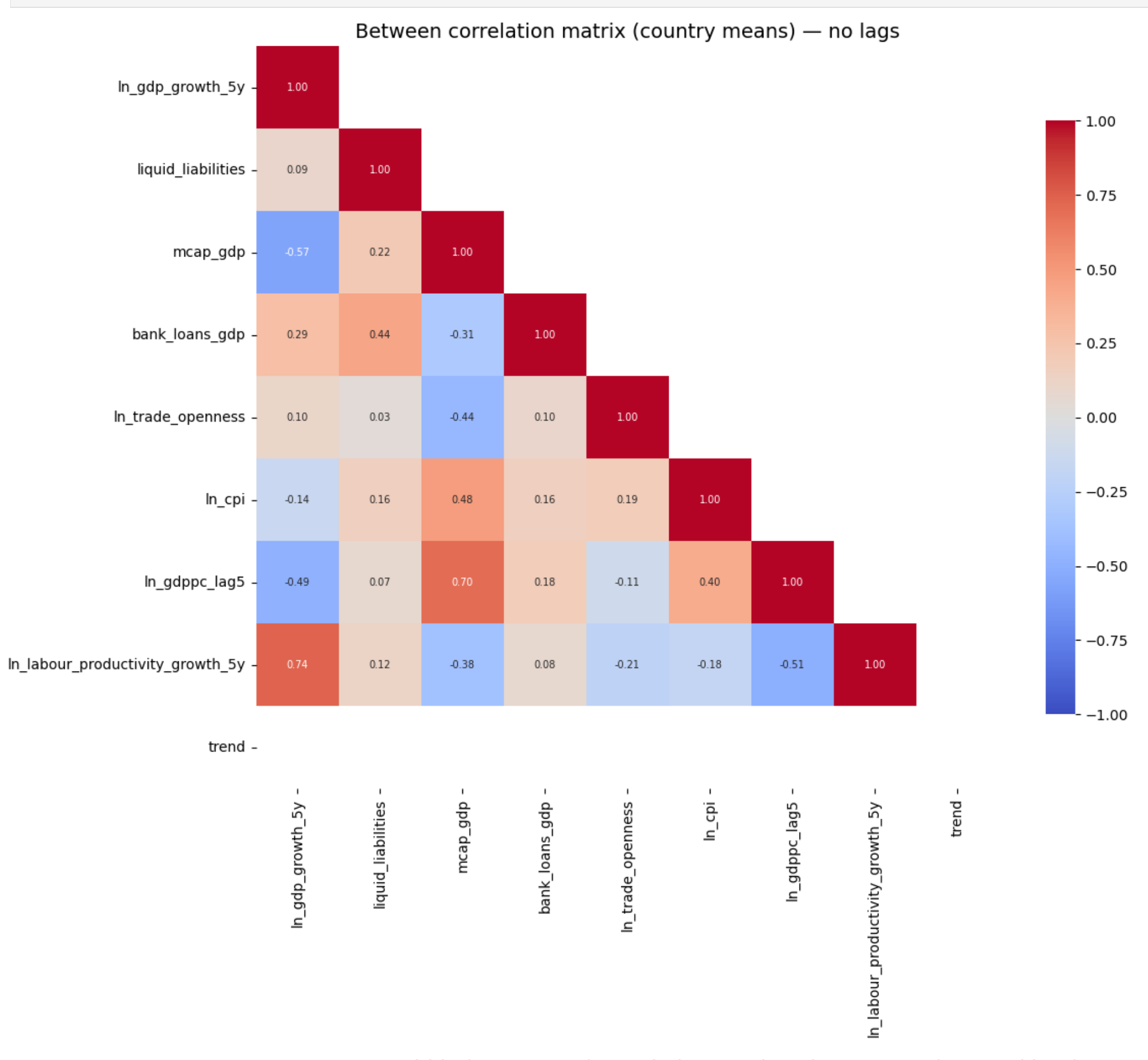
Question 16

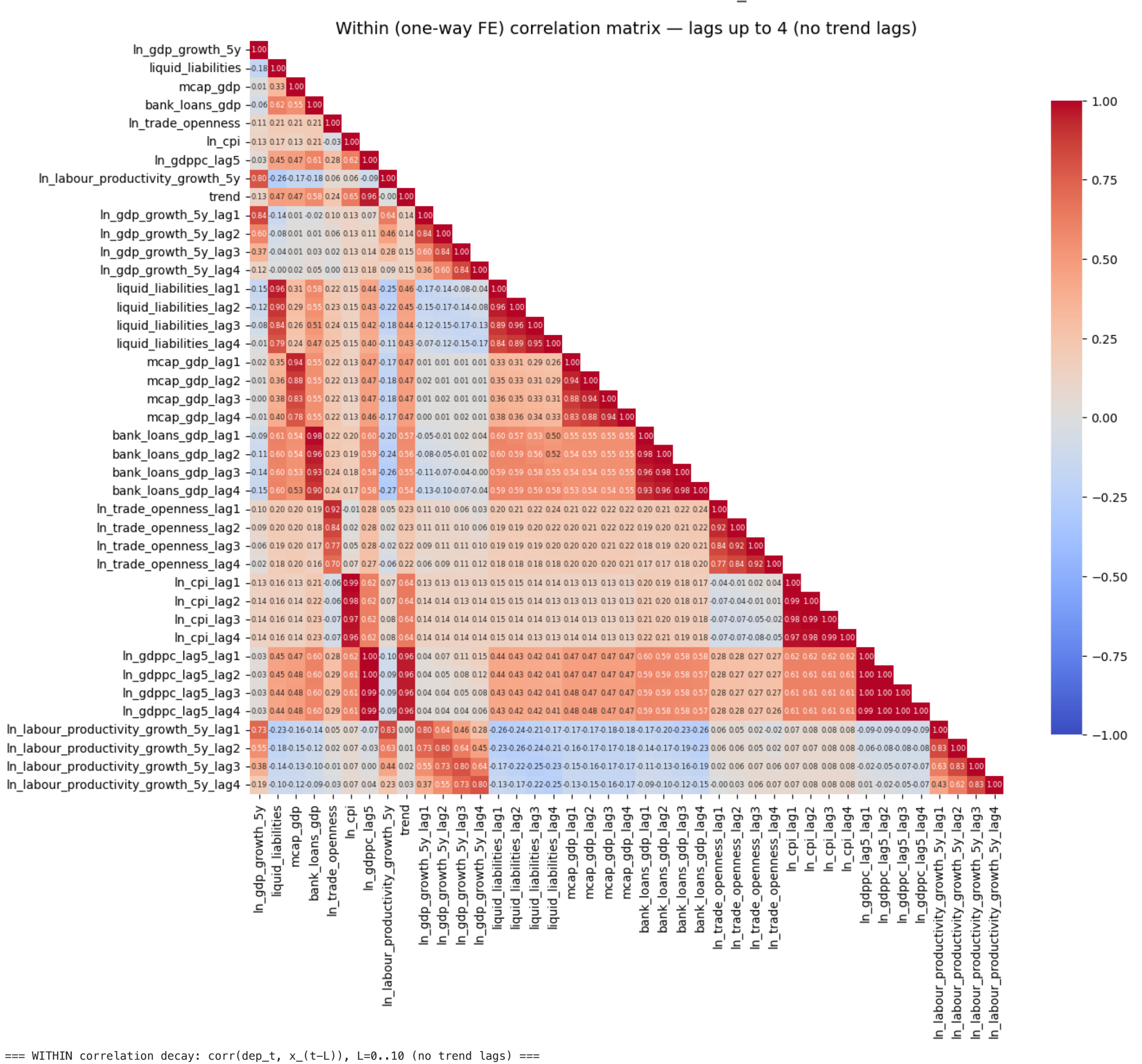
In [171…
```
df_Q16 = df_twfe.copy()
df_Q16["trend"] = df_Q16.groupby("country").cumcount() + 1
dep_var = "ln_gdp_growth_5y"
x_vars = [
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_trade_openness",
    "ln_cpi",
    "ln_gdppc_lag5",
    "ln_labour_productivity_growth_5y"
]
all_vars = [dep_var] + x_vars + ["trend"]


for v in [dep_var] + x_vars:
    df_Q16[f"{v}_lag1"] = df_Q16.groupby("country")[v].shift(1)

lag_vars = [f"{v}_lag1" for v in [dep_var] + x_vars]


df_between = (
    df_Q16
   .groupby("country")[all_vars + lag_vars].mean())

corr_between = df_between.corr()

corr_dep_between = (corr_between.loc[dep_var].sort_values())


df_within = df_Q16.copy()

# demean by country
for v in all_vars + lag_vars:
    df_within[f"{v}_within"] = (
        df_within[v] - df_within.groupby("country")[v].transform("mean")
    )

within_vars = [f"{v}_within" for v in all_vars + lag_vars]

corr_within = df_within[within_vars].corr()

# Correlations with dependent variable
corr_dep_within = (
    corr_within.loc[f"{dep_var}_within"]
    .sort_values()
)


corr_between, corr_within, corr_dep_between, corr_dep_within
```

Out[171…
```
(                                       ln_gdp_growth_5y  liquid_liabilities  \
 ln_gdp_growth_5y                                 1.0000              0.0942
 liquid_liabilities                               0.0942              1.0000
 mcap_gdp                                        -0.5680              0.2152
 bank_loans_gdp                                   0.2851              0.4389
 ln_trade_openness                                0.1020              0.0317
 ln_cpi                                          -0.1434              0.1583
 ln_gdppc_lag5                                   -0.4902              0.0733
 ln_labour_productivity_growth_5y                 0.7401              0.1193
 trend                                               NaN                 NaN
 ln_gdp_growth_5y_lag1                            0.9999              0.0973
 liquid_liabilities_lag1                          0.0968              0.9999
 mcap_gdp_lag1                                   -0.5680              0.2152
 bank_loans_gdp_lag1                              0.2834              0.4377
 ln_trade_openness_lag1                           0.1020              0.0317
 ln_cpi_lag1                                     -0.1431              0.1587
 ln_gdppc_lag5_lag1                              -0.4937              0.0723
 ln_labour_productivity_growth_5y_lag1            0.7398              0.1210

                                       mcap_gdp  bank_loans_gdp  \
 ln_gdp_growth_5y                       -0.5680          0.2851
 liquid_liabilities                      0.2152          0.4389
 mcap_gdp                                1.0000         -0.3147
 bank_loans_gdp                         -0.3147          1.0000
 ln_trade_openness                      -0.4438          0.0965
 ln_cpi                                  0.4840          0.1569
 ln_gdppc_lag5                           0.7031          0.1813
 ln_labour_productivity_growth_5y       -0.3787          0.0751
 trend                                      NaN             NaN
 ln_gdp_growth_5y_lag1                  -0.5655          0.2800
 liquid_liabilities_lag1                 0.2134          0.4458
 mcap_gdp_lag1                           1.0000         -0.3147
 bank_loans_gdp_lag1                    -0.3131          1.0000
 ln_trade_openness_lag1                 -0.4437          0.0966
 ln_cpi_lag1                             0.4842          0.1572
 ln_gdppc_lag5_lag1                      0.7050          0.1789
 ln_labour_productivity_growth_5y_lag1  -0.3849          0.0734

                                       ln_trade_openness  ln_cpi  \
 ln_gdp_growth_5y                                 0.1020 -0.1434
 liquid_liabilities                               0.0317  0.1583
 mcap_gdp                                        -0.4438  0.4840
 bank_loans_gdp                                   0.0965  0.1569
 ln_trade_openness                                1.0000  0.1861
 ln_cpi                                           0.1861  1.0000
 ln_gdppc_lag5                                   -0.1071  0.4035
 ln_labour_productivity_growth_5y                -0.2136 -0.1751
 trend                                               NaN     NaN
 ln_gdp_growth_5y_lag1                            0.0993 -0.1442
 liquid_liabilities_lag1                          0.0387  0.1627
 mcap_gdp_lag1                                   -0.4438  0.4840
 bank_loans_gdp_lag1                              0.0969  0.1547
 ln_trade_openness_lag1                           1.0000  0.1862
 ln_cpi_lag1                                      0.1858  1.0000
 ln_gdppc_lag5_lag1                              -0.1079  0.4033
 ln_labour_productivity_growth_5y_lag1           -0.2103 -0.1758

                                       ln_gdppc_lag5  \
 ln_gdp_growth_5y                            -0.4902
 liquid_liabilities                           0.0733
 mcap_gdp                                     0.7031
 bank_loans_gdp                               0.1813
 ln_trade_openness                           -0.1071
 ln_cpi                                       0.4035
 ln_gdppc_lag5                                1.0000
 ln_labour_productivity_growth_5y            -0.5062
 trend                                           NaN
 ln_gdp_growth_5y_lag1                       -0.4931
 liquid_liabilities_lag1                      0.0800
 mcap_gdp_lag1                                0.7031
 bank_loans_gdp_lag1                          0.1830
 ln_trade_openness_lag1                      -0.1068
 ln_cpi_lag1                                  0.4036
 ln_gdppc_lag5_lag1                           1.0000
 ln_labour_productivity_growth_5y_lag1       -0.5097

                                       ln_labour_productivity_growth_5y  \
 ln_gdp_growth_5y                                                0.7401
 liquid_liabilities                                              0.1193
 mcap_gdp                                                       -0.3787
 bank_loans_gdp                                                  0.0751
 ln_trade_openness                                              -0.2136
 ln_cpi                                                         -0.1751
 ln_gdppc_lag5                                                  -0.5062
 ln_labour_productivity_growth_5y                                1.0000
 trend                                                              NaN
 ln_gdp_growth_5y_lag1                                           0.7405
 liquid_liabilities_lag1                                         0.1190
 mcap_gdp_lag1                                                  -0.3787
 bank_loans_gdp_lag1                                             0.0748
 ln_trade_openness_lag1                                         -0.2136
 ln_cpi_lag1                                                    -0.1747
 ln_gdppc_lag5_lag1                                             -0.5076
 ln_labour_productivity_growth_5y_lag1                           0.9999

                                       trend  ln_gdp_growth_5y_lag1  \
 ln_gdp_growth_5y                        NaN                 0.9999
 liquid_liabilities                      NaN                 0.0973
 mcap_gdp                                NaN                -0.5655
 bank_loans_gdp                          NaN                 0.2800
 ln_trade_openness                       NaN                 0.0993
 ln_cpi                                  NaN                -0.1442
 ln_gdppc_lag5                           NaN                -0.4931
 ln_labour_productivity_growth_5y        NaN                 0.7405
 trend                                   NaN                    NaN
 ln_gdp_growth_5y_lag1                   NaN                 1.0000
 liquid_liabilities_lag1                 NaN                 0.0999
 mcap_gdp_lag1                           NaN                -0.5655
 bank_loans_gdp_lag1                     NaN                 0.2783
 ln_trade_openness_lag1                  NaN                 0.0992
 ln_cpi_lag1                             NaN                -0.1440
 ln_gdppc_lag5_lag1                      NaN                -0.4966
 ln_labour_productivity_growth_5y_lag1   NaN                 0.7403

                                       liquid_liabilities_lag1  mcap_gdp_lag1  \
 ln_gdp_growth_5y                                       0.0968        -0.5680
 liquid_liabilities                                     0.9999         0.2152
 mcap_gdp                                               0.2134         1.0000
 bank_loans_gdp                                         0.4458        -0.3147
 ln_trade_openness                                      0.0387        -0.4438
 ln_cpi                                                 0.1627         0.4840
 ln_gdppc_lag5                                          0.0800         0.7031
 ln_labour_productivity_growth_5y                       0.1190        -0.3787
 trend                                                     NaN            NaN
 ln_gdp_growth_5y_lag1                                  0.0999        -0.5655
 liquid_liabilities_lag1                                1.0000         0.2134
 mcap_gdp_lag1                                          0.2134         1.0000
 bank_loans_gdp_lag1                                    0.4447        -0.3131
 ln_trade_openness_lag1                                 0.0387        -0.4437
 ln_cpi_lag1                                            0.1631         0.4842
 ln_gdppc_lag5_lag1                                     0.0789         0.7050
 ln_labour_productivity_growth_5y_lag1                  0.1206        -0.3849

                                       bank_loans_gdp_lag1  \
 ln_gdp_growth_5y                                   0.2834
 liquid_liabilities                                 0.4377
 mcap_gdp                                          -0.3131
 bank_loans_gdp                                     1.0000
 ln_trade_openness                                  0.0969
 ln_cpi                                             0.1547
 ln_gdppc_lag5                                      0.1830
 ln_labour_productivity_growth_5y                   0.0748
 trend                                                 NaN
 ln_gdp_growth_5y_lag1                              0.2783
 liquid_liabilities_lag1                            0.4447
 mcap_gdp_lag1                                     -0.3131
 bank_loans_gdp_lag1                                1.0000
 ln_trade_openness_lag1                             0.0969
 ln_cpi_lag1                                        0.1550
 ln_gdppc_lag5_lag1                                 0.1807
 ln_labour_productivity_growth_5y_lag1              0.0732

                                       ln_trade_openness_lag1  ln_cpi_lag1  \
 ln_gdp_growth_5y                                      0.1020      -0.1431
 liquid_liabilities                                    0.0317       0.1587
 mcap_gdp                                             -0.4437       0.4842
 bank_loans_gdp                                        0.0966       0.1572
 ln_trade_openness                                     1.0000       0.1858
 ln_cpi                                                0.1862       1.0000
 ln_gdppc_lag5                                        -0.1068       0.4036
 ln_labour_productivity_growth_5y                     -0.2136      -0.1747
 trend                                                    NaN          NaN
 ln_gdp_growth_5y_lag1                                 0.0992      -0.1440
 liquid_liabilities_lag1                               0.0387       0.1631
 mcap_gdp_lag1                                        -0.4437       0.4842
 bank_loans_gdp_lag1                                   0.0969       0.1550
 ln_trade_openness_lag1                                1.0000       0.1859
 ln_cpi_lag1                                           0.1859       1.0000
 ln_gdppc_lag5_lag1                                   -0.1076       0.4034
 ln_labour_productivity_growth_5y_lag1                -0.2103      -0.1754

                                       ln_gdppc_lag5_lag1  \
 ln_gdp_growth_5y                                 -0.4937
 liquid_liabilities                                0.0723
 mcap_gdp                                          0.7050
 bank_loans_gdp                                    0.1789
 ln_trade_openness                                -0.1079
 ln_cpi                                            0.4033
 ln_gdppc_lag5                                     1.0000
 ln_labour_productivity_growth_5y                 -0.5076
 trend                                                NaN
 ln_gdp_growth_5y_lag1                            -0.4966
 liquid_liabilities_lag1                           0.0789
 mcap_gdp_lag1                                     0.7050
 bank_loans_gdp_lag1                               0.1807
 ln_trade_openness_lag1                           -0.1076
 ln_cpi_lag1                                       0.4034
 ln_gdppc_lag5_lag1                                1.0000
 ln_labour_productivity_growth_5y_lag1            -0.5111

                                       ln_labour_productivity_growth_5y_lag1
 ln_gdp_growth_5y                                                     0.7398
 liquid_liabilities                                                   0.1210
 mcap_gdp                                                            -0.3849
 bank_loans_gdp                                                       0.0734
 ln_trade_openness                                                   -0.2103
 ln_cpi                                                              -0.1758
 ln_gdppc_lag5                                                       -0.5097
 ln_labour_productivity_growth_5y                                     0.9999
 trend                                                                   NaN
 ln_gdp_growth_5y_lag1                                                0.7403
 liquid_liabilities_lag1                                              0.1206
 mcap_gdp_lag1                                                       -0.3849
 bank_loans_gdp_lag1                                                  0.0732
 ln_trade_openness_lag1                                              -0.2103
 ln_cpi_lag1                                                         -0.1754
 ln_gdppc_lag5_lag1                                                  -0.5111
 ln_labour_productivity_growth_5y_lag1                                1.0000  ,
                                              ln_gdp_growth_5y_within  \
 ln_gdp_growth_5y_within                                       1.0000
 liquid_liabilities_within                                    -0.1762
 mcap_gdp_within                                               0.0108
 bank_loans_gdp_within                                        -0.0636
 ln_trade_openness_within                                      0.1072
 ln_cpi_within                                                 0.1283
 ln_gdppc_lag5_within                                          0.0332
 ln_labour_productivity_growth_5y_within                       0.7965
 trend_within                                                  0.1263
 ln_gdp_growth_5y_lag1_within                                  0.8367
 liquid_liabilities_lag1_within                               -0.1541
 mcap_gdp_lag1_within                                          0.0226
 bank_loans_gdp_lag1_within                                   -0.0865
 ln_trade_openness_lag1_within                                 0.1024
 ln_cpi_lag1_within                                            0.1349
 ln_gdppc_lag5_lag1_within                                     0.0278
 ln_labour_productivity_growth_5y_lag1_within                  0.7261

                                              liquid_liabilities_within  \
 ln_gdp_growth_5y_within                                        -0.1762
 liquid_liabilities_within                                       1.0000
 mcap_gdp_within                                                 0.3312
 bank_loans_gdp_within                                           0.6160
 ln_trade_openness_within                                        0.2073
 ln_cpi_within                                                   0.1650
 ln_gdppc_lag5_within                                            0.4506
 ln_labour_productivity_growth_5y_within                        -0.2637
 trend_within                                                    0.4698
 ln_gdp_growth_5y_lag1_within                                   -0.1367
 liquid_liabilities_lag1_within                                  0.9572
 mcap_gdp_lag1_within                                            0.3451
 bank_loans_gdp_lag1_within                                      0.6119
 ln_trade_openness_lag1_within                                   0.2010
 ln_cpi_lag1_within                                              0.1620
 ln_gdppc_lag5_lag1_within                                       0.4496
 ln_labour_productivity_growth_5y_lag1_within                   -0.2326

                                              mcap_gdp_within  \
 ln_gdp_growth_5y_within                               0.0108
 liquid_liabilities_within                             0.3312
 mcap_gdp_within                                       1.0000
 bank_loans_gdp_within                                 0.5476
 ln_trade_openness_within                              0.2097
 ln_cpi_within                                         0.1328
 ln_gdppc_lag5_within                                  0.4716
 ln_labour_productivity_growth_5y_within              -0.1708
 trend_within                                          0.4714
 ln_gdp_growth_5y_lag1_within                          0.0076
 liquid_liabilities_lag1_within                        0.3085
 mcap_gdp_lag1_within                                  0.9408
 bank_loans_gdp_lag1_within                            0.5435
 ln_trade_openness_lag1_within                         0.2033
 ln_cpi_lag1_within                                    0.1341
 ln_gdppc_lag5_lag1_within                             0.4740
 ln_labour_productivity_growth_5y_lag1_within         -0.1629

                                              bank_loans_gdp_within  \
 ln_gdp_growth_5y_within                                    -0.0636
 liquid_liabilities_within                                   0.6160
 mcap_gdp_within                                             0.5476
 bank_loans_gdp_within                                       1.0000
 ln_trade_openness_within                                    0.2056
 ln_cpi_within                                               0.2094
 ln_gdppc_lag5_within                                        0.6057
 ln_labour_productivity_growth_5y_within                    -0.1776
 trend_within                                                0.5772
 ln_gdp_growth_5y_lag1_within                               -0.0187
 liquid_liabilities_lag1_within                              0.5819
 mcap_gdp_lag1_within                                        0.5492
 bank_loans_gdp_lag1_within                                  0.9845
 ln_trade_openness_lag1_within                               0.1910
 ln_cpi_lag1_within                                          0.2136
 ln_gdppc_lag5_lag1_within                                   0.6040
 ln_labour_productivity_growth_5y_lag1_within               -0.1434

                                              ln_trade_openness_within  \
 ln_gdp_growth_5y_within                                        0.1072
 liquid_liabilities_within                                      0.2073
 mcap_gdp_within                                                0.2097
 bank_loans_gdp_within                                          0.2056
 ln_trade_openness_within                                       1.0000
 ln_cpi_within                                                 -0.0325
 ln_gdppc_lag5_within                                           0.2848
 ln_labour_productivity_growth_5y_within                        0.0575
 trend_within                                                   0.2404
 ln_gdp_growth_5y_lag1_within                                   0.0967
 liquid_liabilities_lag1_within                                 0.2162
 mcap_gdp_lag1_within                                           0.2179
 bank_loans_gdp_lag1_within                                     0.2170
 ln_trade_openness_lag1_within                                  0.9235
 ln_cpi_lag1_within                                            -0.0607
 ln_gdppc_lag5_lag1_within                                      0.2846
 ln_labour_productivity_growth_5y_lag1_within                   0.0521

                                              ln_cpi_within  \
 ln_gdp_growth_5y_within                             0.1283
 liquid_liabilities_within                           0.1650
 mcap_gdp_within                                     0.1328
 bank_loans_gdp_within                               0.2094
 ln_trade_openness_within                           -0.0325
 ln_cpi_within                                       1.0000
 ln_gdppc_lag5_within                                0.6181
 ln_labour_productivity_growth_5y_within             0.0648
 trend_within                                        0.6451
 ln_gdp_growth_5y_lag1_within                        0.1316
 liquid_liabilities_lag1_within                      0.1549
 mcap_gdp_lag1_within                                0.1332
 bank_loans_gdp_lag1_within                          0.1970
 ln_trade_openness_lag1_within                      -0.0052
 ln_cpi_lag1_within                                  0.9933
 ln_gdppc_lag5_lag1_within                           0.6163
 ln_labour_productivity_growth_5y_lag1_within        0.0666

                                              ln_gdppc_lag5_within  \
 ln_gdp_growth_5y_within                                    0.0332
 liquid_liabilities_within                                  0.4506
 mcap_gdp_within                                            0.4716
 bank_loans_gdp_within                                      0.6057
 ln_trade_openness_within                                   0.2848
 ln_cpi_within                                              0.6181
 ln_gdppc_lag5_within                                       1.0000
 ln_labour_productivity_growth_5y_within                   -0.0943
 trend_within                                               0.9584
 ln_gdp_growth_5y_lag1_within                               0.0701
 liquid_liabilities_lag1_within                             0.4371
 mcap_gdp_lag1_within                                       0.4693
 bank_loans_gdp_lag1_within                                 0.5979
 ln_trade_openness_lag1_within                              0.2807
 ln_cpi_lag1_within                                         0.6178
 ln_gdppc_lag5_lag1_within                                  0.9983
 ln_labour_productivity_growth_5y_lag1_within              -0.0676

                                              ln_labour_productivity_growth_5y_within  \
 ln_gdp_growth_5y_within                                                       0.7965
 liquid_liabilities_within                                                    -0.2637
 mcap_gdp_within                                                              -0.1708
 bank_loans_gdp_within                                                        -0.1776
 ln_trade_openness_within                                                      0.0575
 ln_cpi_within                                                                 0.0648
 ln_gdppc_lag5_within                                                         -0.0943
 ln_labour_productivity_growth_5y_within                                       1.0000
 trend_within                                                                 -0.0038
 ln_gdp_growth_5y_lag1_within                                                  0.6368
 liquid_liabilities_lag1_within                                               -0.2455
 mcap_gdp_lag1_within                                                         -0.1740
 bank_loans_gdp_lag1_within                                                   -0.2045
 ln_trade_openness_lag1_within                                                 0.0469
 ln_cpi_lag1_within                                                            0.0711
 ln_gdppc_lag5_lag1_within                                                    -0.0961
 ln_labour_productivity_growth_5y_lag1_within                                  0.8297

                                              trend_within  \
 ln_gdp_growth_5y_within                            0.1263
 liquid_liabilities_within                          0.4698
 mcap_gdp_within                                    0.4714
 bank_loans_gdp_within                              0.5772
 ln_trade_openness_within                           0.2404
 ln_cpi_within                                      0.6451
 ln_gdppc_lag5_within                               0.9584
 ln_labour_productivity_growth_5y_within           -0.0038
 trend_within                                       1.0000
 ln_gdp_growth_5y_lag1_within                       0.1371
 liquid_liabilities_lag1_within                     0.4571
 mcap_gdp_lag1_within                               0.4714
 bank_loans_gdp_lag1_within                         0.5674
 ln_trade_openness_lag1_within                      0.2347
 ln_cpi_lag1_within                                 0.6435
 ln_gdppc_lag5_lag1_within                          0.9577
 ln_labour_productivity_growth_5y_lag1_within       0.0041

                                              ln_gdp_growth_5y_lag1_within  \
 ln_gdp_growth_5y_within                                            0.8367
 liquid_liabilities_within                                         -0.1367
 mcap_gdp_within                                                    0.0076
 bank_loans_gdp_within                                             -0.0187
 ln_trade_openness_within                                           0.0967
 ln_cpi_within                                                      0.1316
 ln_gdppc_lag5_within                                               0.0701
 ln_labour_productivity_growth_5y_within                            0.6368
 trend_within                                                       0.1371
 ln_gdp_growth_5y_lag1_within                                       1.0000
 liquid_liabilities_lag1_within                                    -0.1681
 mcap_gdp_lag1_within                                               0.0108
 bank_loans_gdp_lag1_within                                        -0.0523
 ln_trade_openness_lag1_within                                      0.1113
 ln_cpi_lag1_within                                                 0.1336
 ln_gdppc_lag5_lag1_within                                          0.0426
 ln_labour_productivity_growth_5y_lag1_within                       0.7962

                                              liquid_liabilities_lag1_within  \
 ln_gdp_growth_5y_within                                             -0.1541
 liquid_liabilities_within                                            0.9572
 mcap_gdp_within                                                      0.3085
 bank_loans_gdp_within                                                0.5819
 ln_trade_openness_within                                             0.2162
 ln_cpi_within                                                        0.1549
 ln_gdppc_lag5_within                                                 0.4371
 ln_labour_productivity_growth_5y_within                             -0.2455
 trend_within                                                         0.4571
 ln_gdp_growth_5y_lag1_within                                        -0.1681
 liquid_liabilities_lag1_within                                       1.0000
 mcap_gdp_lag1_within                                                 0.3312
 bank_loans_gdp_lag1_within                                           0.6032
 ln_trade_openness_lag1_within                                        0.2001
 ln_cpi_lag1_within                                                   0.1543
 ln_gdppc_lag5_lag1_within                                            0.4374
 ln_labour_productivity_growth_5y_lag1_within                        -0.2599

                                              mcap_gdp_lag1_within  \
 ln_gdp_growth_5y_within                                    0.0226
 liquid_liabilities_within                                  0.3451
 mcap_gdp_within                                            0.9408
 bank_loans_gdp_within                                      0.5492
 ln_trade_openness_within                                   0.2179
 ln_cpi_within                                              0.1332
 ln_gdppc_lag5_within                                       0.4693
 ln_labour_productivity_growth_5y_within                   -0.1740
 trend_within                                               0.4714
 ln_gdp_growth_5y_lag1_within                               0.0108
 liquid_liabilities_lag1_within                             0.3312
 mcap_gdp_lag1_within                                       1.0000
 bank_loans_gdp_lag1_within                                 0.5478
 ln_trade_openness_lag1_within                              0.2097
 ln_cpi_lag1_within                                         0.1328
 ln_gdppc_lag5_lag1_within                                  0.4717
 ln_labour_productivity_growth_5y_lag1_within              -0.1708

                                              bank_loans_gdp_lag1_within  \
 ln_gdp_growth_5y_within                                         -0.0865
 liquid_liabilities_within                                        0.6119
 mcap_gdp_within                                                  0.5435
 bank_loans_gdp_within                                            0.9845
 ln_trade_openness_within                                         0.2170
 ln_cpi_within                                                    0.1970
 ln_gdppc_lag5_within                                             0.5979
 ln_labour_productivity_growth_5y_within                         -0.2045
 trend_within                                                     0.5674
 ln_gdp_growth_5y_lag1_within                                    -0.0523
 liquid_liabilities_lag1_within                                   0.6032
 mcap_gdp_lag1_within                                             0.5478
 bank_loans_gdp_lag1_within                                       1.0000
 ln_trade_openness_lag1_within                                    0.2023
 ln_cpi_lag1_within                                               0.2027
 ln_gdppc_lag5_lag1_within                                        0.5965
 ln_labour_productivity_growth_5y_lag1_within                    -0.1715

                                              ln_trade_openness_lag1_within  \
 ln_gdp_growth_5y_within                                             0.1024
 liquid_liabilities_within                                           0.2010
 mcap_gdp_within                                                     0.2033
 bank_loans_gdp_within                                               0.1910
 ln_trade_openness_within                                            0.9235
 ln_cpi_within                                                      -0.0052
 ln_gdppc_lag5_within                                                0.2807
 ln_labour_productivity_growth_5y_within                             0.0469
 trend_within                                                        0.2347
 ln_gdp_growth_5y_lag1_within                                        0.1113
 liquid_liabilities_lag1_within                                      0.2001
 mcap_gdp_lag1_within                                                0.2097
 bank_loans_gdp_lag1_within                                          0.2023
 ln_trade_openness_lag1_within                                       1.0000
 ln_cpi_lag1_within                                                 -0.0375
 ln_gdppc_lag5_lag1_within                                           0.2796
 ln_labour_productivity_growth_5y_lag1_within                        0.0610

                                              ln_cpi_lag1_within  \
 ln_gdp_growth_5y_within                                  0.1349
 liquid_liabilities_within                                0.1620
 mcap_gdp_within                                          0.1341
 bank_loans_gdp_within                                    0.2136
 ln_trade_openness_within                                -0.0607
 ln_cpi_within                                            0.9933
 ln_gdppc_lag5_within                                     0.6178
 ln_labour_productivity_growth_5y_within                  0.0711
 trend_within                                             0.6435
 ln_gdp_growth_5y_lag1_within                             0.1336
 liquid_liabilities_lag1_within                           0.1543
 mcap_gdp_lag1_within                                     0.1328
 bank_loans_gdp_lag1_within                               0.2027
 ln_trade_openness_lag1_within                           -0.0375
 ln_cpi_lag1_within                                       1.0000
 ln_gdppc_lag5_lag1_within                                0.6160
 ln_labour_productivity_growth_5y_lag1_within             0.0690

                                              ln_gdppc_lag5_lag1_within  \
 ln_gdp_growth_5y_within                                         0.0278
 liquid_liabilities_within                                       0.4496
 mcap_gdp_within                                                 0.4740
 bank_loans_gdp_within                                           0.6040
 ln_trade_openness_within                                        0.2846
 ln_cpi_within                                                   0.6163
 ln_gdppc_lag5_within                                            0.9983
 ln_labour_productivity_growth_5y_within                        -0.0961
 trend_within                                                    0.9577
 ln_gdp_growth_5y_lag1_within                                    0.0426
 liquid_liabilities_lag1_within                                  0.4374
 mcap_gdp_lag1_within                                            0.4717
 bank_loans_gdp_lag1_within                                      0.5965
 ln_trade_openness_lag1_within                                   0.2796
 ln_cpi_lag1_within                                              0.6160
 ln_gdppc_lag5_lag1_within                                       1.0000
 ln_labour_productivity_growth_5y_lag1_within                   -0.0876

                                              ln_labour_productivity_growth_5y_lag1_within
 ln_gdp_growth_5y_within                                                            0.7261
 liquid_liabilities_within                                                         -0.2326
 mcap_gdp_within                                                                   -0.1629
 bank_loans_gdp_within                                                             -0.1434
 ln_trade_openness_within                                                           0.0521
 ln_cpi_within                                                                      0.0666
 ln_gdppc_lag5_within                                                              -0.0676
 ln_labour_productivity_growth_5y_within                                            0.8297
 trend_within                                                                       0.0041
 ln_gdp_growth_5y_lag1_within                                                       0.7962
 liquid_liabilities_lag1_within                                                    -0.2599
 mcap_gdp_lag1_within                                                              -0.1708
 bank_loans_gdp_lag1_within                                                        -0.1715
 ln_trade_openness_lag1_within                                                      0.0610
 ln_cpi_lag1_within                                                                 0.0690
 ln_gdppc_lag5_lag1_within                                                         -0.0876
 ln_labour_productivity_growth_5y_lag1_within                                       1.0000  ,
 mcap_gdp                               -0.5680
 mcap_gdp_lag1                          -0.5680
 ln_gdppc_lag5_lag1                     -0.4937
 ln_gdppc_lag5                          -0.4902
 ln_cpi                                 -0.1434
 ln_cpi_lag1                            -0.1431
 liquid_liabilities                      0.0942
 liquid_liabilities_lag1                 0.0968
 ln_trade_openness_lag1                  0.1020
 ln_trade_openness                       0.1020
 bank_loans_gdp_lag1                     0.2834
 bank_loans_gdp                          0.2851
 ln_labour_productivity_growth_5y_lag1   0.7398
 ln_labour_productivity_growth_5y        0.7401
 ln_gdp_growth_5y_lag1                   0.9999
 ln_gdp_growth_5y                        1.0000
 trend                                      NaN
 Name: ln_gdp_growth_5y, dtype: float64,
 liquid_liabilities_within                      -0.1762
 liquid_liabilities_lag1_within                 -0.1541
 bank_loans_gdp_lag1_within                     -0.0865
 bank_loans_gdp_within                          -0.0636
 mcap_gdp_within                                 0.0108
 mcap_gdp_lag1_within                            0.0226
 ln_gdppc_lag5_lag1_within                       0.0278
 ln_gdppc_lag5_within                            0.0332
 ln_trade_openness_lag1_within                   0.1024
 ln_trade_openness_within                        0.1072
 trend_within                                    0.1263
 ln_cpi_within                                   0.1283
 ln_cpi_lag1_within                              0.1349
 ln_labour_productivity_growth_5y_lag1_within    0.7261
 ln_labour_productivity_growth_5y_within         0.7965
 ln_gdp_growth_5y_lag1_within                    0.8367
 ln_gdp_growth_5y_within                         1.0000
 Name: ln_gdp_growth_5y_within, dtype: float64)
```

In [172…
```
df_Q16 = df_twfe.copy()


df_Q16["trend"] = df_Q16.groupby("country").cumcount() + 1

dep_var = "ln_gdp_growth_5y"
x_vars = [
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_trade_openness",
    "ln_cpi",
    "ln_gdppc_lag5",
    "ln_labour_productivity_growth_5y"
]
all_vars = [dep_var] + x_vars + ["trend"]

K_HEAT  = 4
K_DECAY = 10


for v in [dep_var] + x_vars:
    for L in range(1, K_DECAY + 1):
        df_Q16[f"{v}_lag{L}"] = df_Q16.groupby("country")[v].shift(L)


df_between = df_Q16.groupby("country")[all_vars].mean()
corr_between = df_between.corr()


lag_vars_heat = [f"{v}_lag{L}" for v in ([dep_var] + x_vars) for L in range(1, K_HEAT + 1)]
within_cols_heat = ([dep_var] + x_vars) + ["trend"] + lag_vars_heat

df_within_heat = df_Q16.copy()
for c in within_cols_heat:
    df_within_heat[c] = df_within_heat[c] - df_within_heat.groupby("country")[c].transform("mean")

corr_within = df_within_heat[within_cols_heat].corr()


mask_between = np.triu(np.ones_like(corr_between, dtype=bool), k=1)
mask_within  = np.triu(np.ones_like(corr_within,  dtype=bool), k=1)

plt.figure(figsize=(12, 10))
sns.heatmap(
    corr_between,
    mask=mask_between,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    square=True,
    annot=True, fmt=".2f",
    annot_kws={"size": 7},
    cbar_kws={"shrink": 0.8}
)
plt.title("Between correlation matrix (country means) — no lags", fontsize=14)
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```

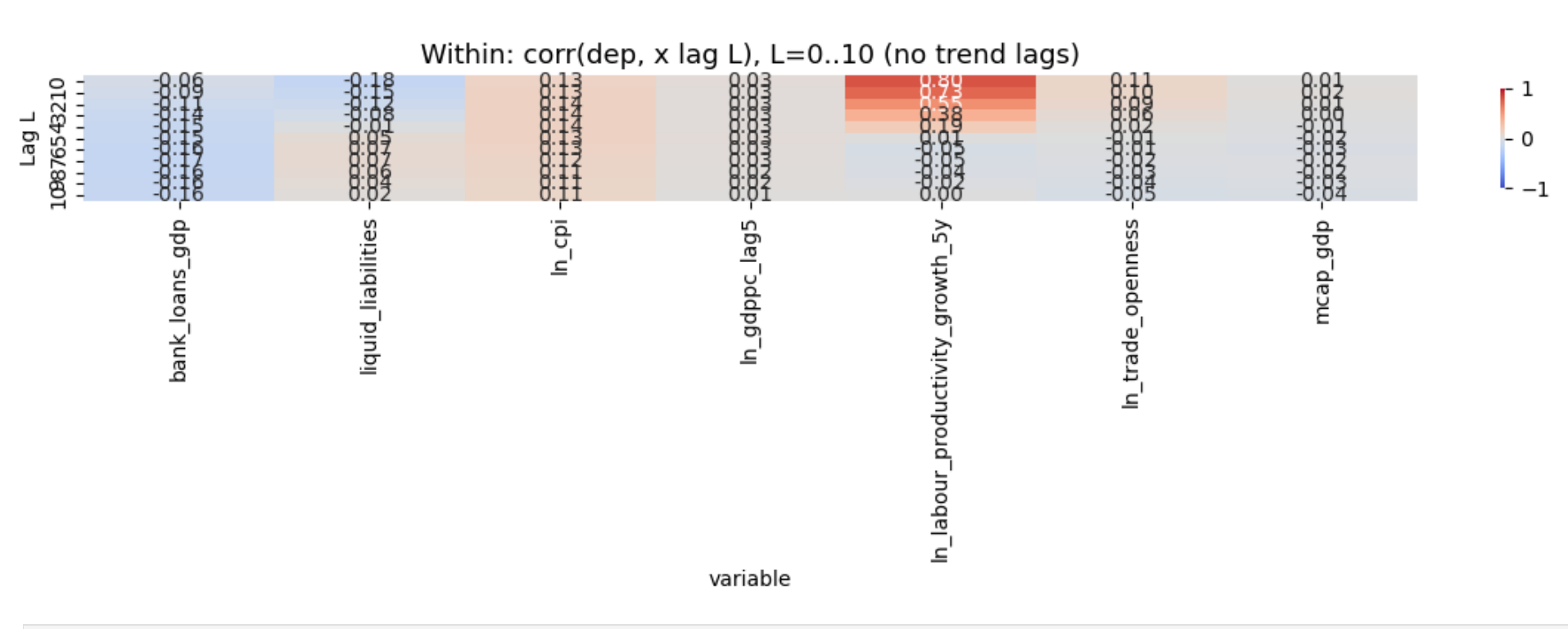
```python
plt.show()

plt.figure(figsize=(14, 12))
sns.heatmap(
    corr_within,
    mask=mask_within,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    square=True,
    annot=True, fmt=".2f",
    annot_kws={"size": 6},
    cbar_kws={"shrink": 0.8}
)
plt.title(f"Within (one-way FE) correlation matrix — lags up to {K_HEAT} (no trend lags)", fontsize=14)
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()


lag_vars_decay = [f"{v}_lag{L}" for v in ([dep_var] + x_vars) for L in range(1, K_DECAY + 1)]
within_cols_decay = ([dep_var] + x_vars) + ["trend"] + lag_vars_decay

df_within_decay = df_Q16.copy()
for c in within_cols_decay:
    df_within_decay[c] = df_within_decay[c] - df_within_decay.groupby("country")[c].transform("mean")

corr_within_decay = df_within_decay[within_cols_decay].corr()

def corr_decay_table(corr_mat, dep_var, x_vars, max_lag=10):
    rows = []
    for x in x_vars:
        rows.append({"variable": x, "lag": 0, "corr": corr_mat.loc[dep_var, x]})
        for L in range(1, max_lag + 1):
            col = f"{x}_lag{L}"
            if col in corr_mat.columns:
                rows.append({"variable": x, "lag": L, "corr": corr_mat.loc[dep_var, col]})
    return pd.DataFrame(rows).pivot(index="lag", columns="variable", values="corr")

decay_within = corr_decay_table(corr_within_decay, dep_var, x_vars, max_lag=K_DECAY)

print(f"\n=== WITHIN correlation decay: corr(dep_t, x_(t-L)), L=0..{K_DECAY} (no trend lags) ===")
display(decay_within.round(3))

plt.figure(figsize=(12, 4))
sns.heatmap(
    decay_within,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    annot=True, fmt=".2f",
    cbar_kws={"shrink": 0.8}
)
plt.title(f"Within: corr(dep, x lag L), L=0..{K_DECAY} (no trend lags)", fontsize=13)
plt.ylabel("Lag L")
plt.tight_layout()
plt.show()
```



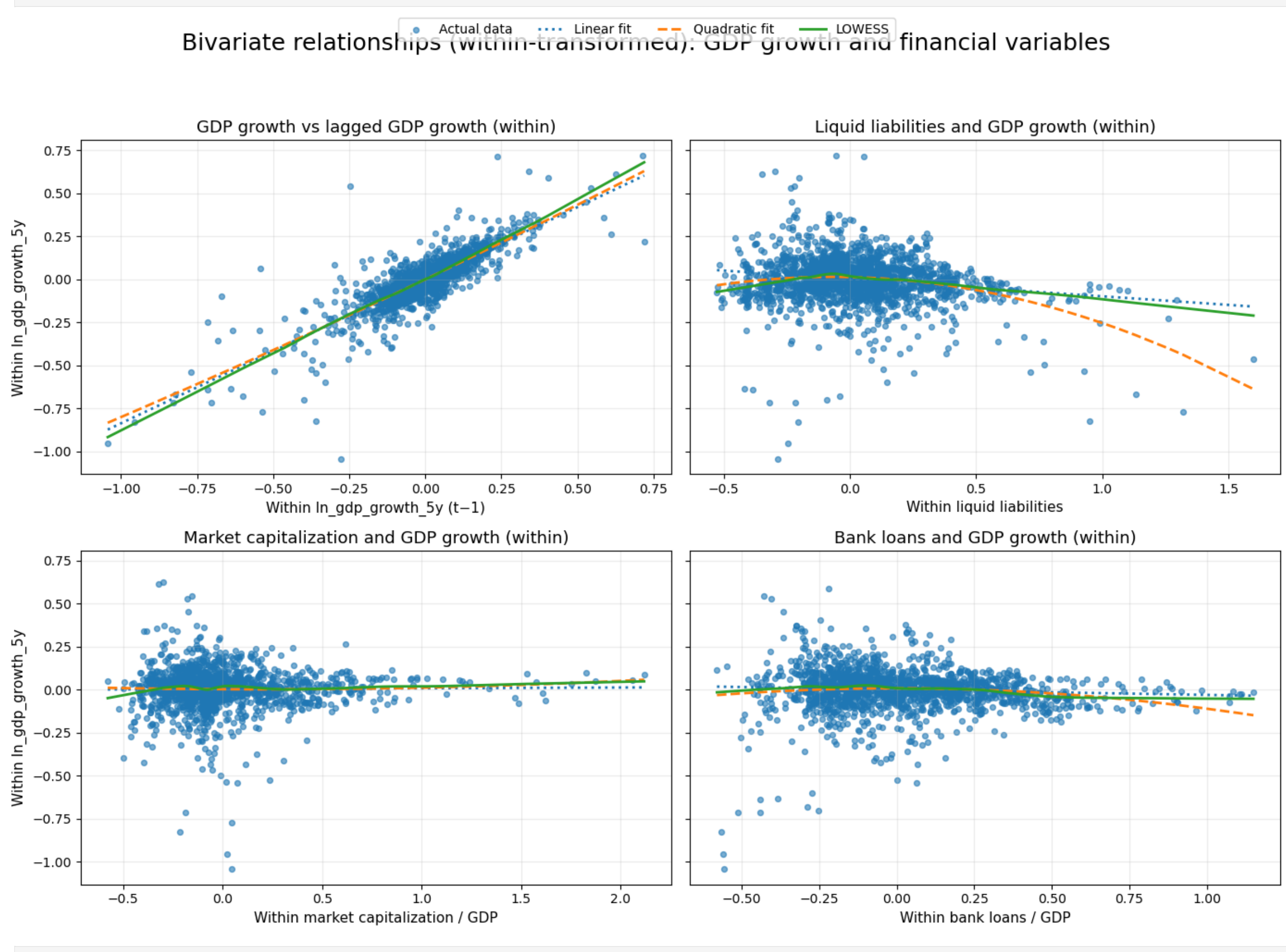


```
=== WITHIN correlation decay: corr(dep_t, x_(t-L)), L=0..10 (no trend lags) ===
```

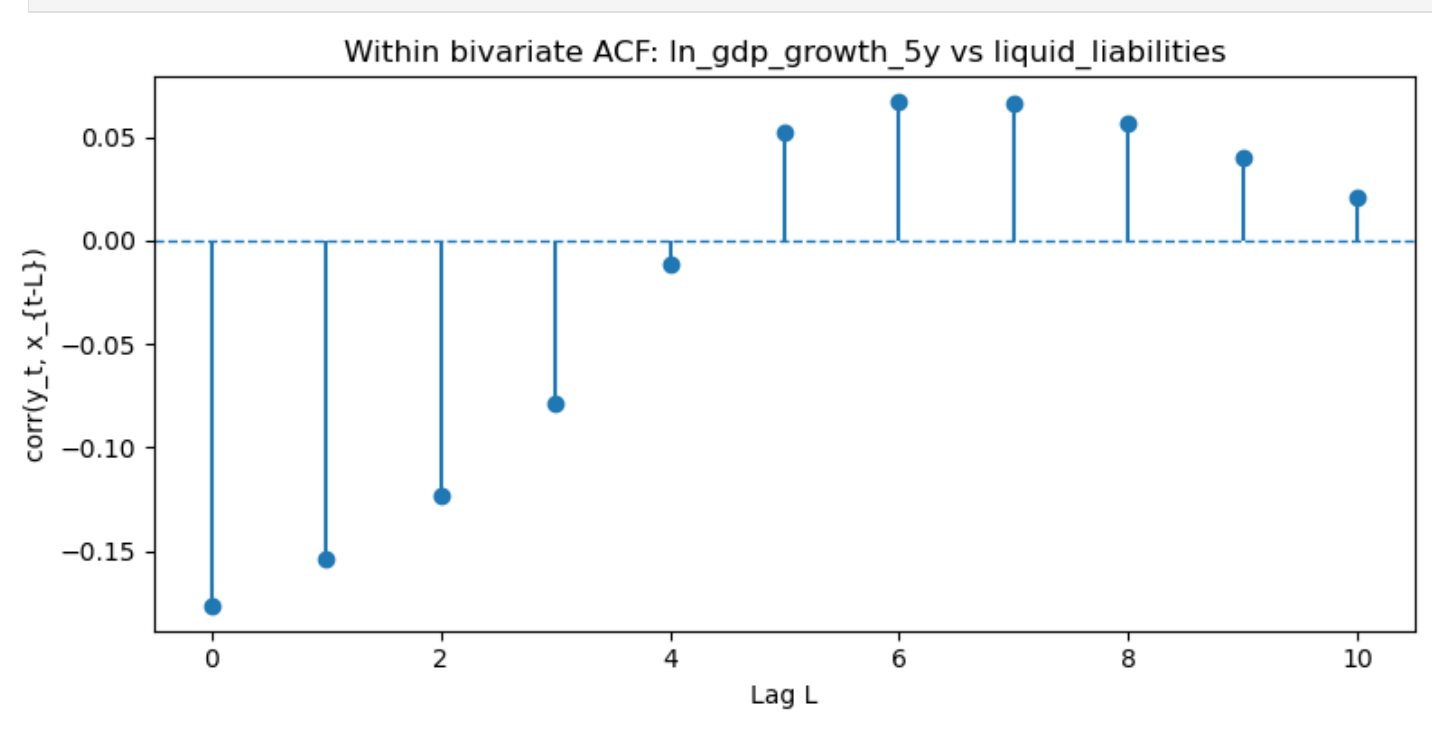
variable lag	bank_loans_gdp	liquid_liabilities	ln_cpi	ln_gdppc_lag5	ln_labour_productivity_growth_5y	ln_trade_openness	mcap_gdp
0	-0.0640	-0.1760	0.1280	0.0330	0.7960	0.1070	0.0110
1	-0.0860	-0.1540	0.1350	0.0280	0.7260	0.1020	0.0230
2	-0.1140	-0.1230	0.1370	0.0270	0.5490	0.0860	0.0130
3	-0.1390	-0.0790	0.1370	0.0280	0.3750	0.0550	0.0010
4	-0.1540	-0.0110	0.1370	0.0310	0.1920	0.0160	-0.0120
5	-0.1550	0.0520	0.1350	0.0320	0.0060	-0.0080	-0.0170
6	-0.1640	0.0670	0.1260	0.0300	-0.0500	-0.0150	-0.0310
7	-0.1680	0.0660	0.1170	0.0260	-0.0460	-0.0210	-0.0220
8	-0.1650	0.0570	0.1100	0.0230	-0.0380	-0.0280	-0.0190
9	-0.1600	0.0400	0.1070	0.0180	-0.0220	-0.0400	-0.0290
10	-0.1600	0.0210	0.1060	0.0140	0.0020	-0.0470	-0.0430



In [ ]:
```python
df_plot = df_twfe.copy()
df_plot = df_plot.sort_values(["country", "year"])

y = "ln_gdp_growth_5y"
x_list = ["liquid_liabilities", "mcap_gdp", "bank_loans_gdp"]

# lag of dependent variable
df_plot[f"{y}_lag1"] = df_plot.groupby("country")[y].shift(1)

def within_transform(df, var, group="country"):
    return df[var] - df.groupby(group)[var].transform("mean")

# dependent
df_plot[f"{y}_within"] = within_transform(df_plot, y)
df_plot[f"{y}_lag1_within"] = within_transform(df_plot, f"{y}_lag1")

# explanatory variables
for x in x_list:
    df_plot[f"{x}_within"] = within_transform(df_plot, x)
# ----------------------------
# Helper to plot one panel
# ----------------------------
def bivar_within_panel(ax, df, xcol, ycol, title, xlabel, ylabel, lowess_frac=0.35):
    d = df[[xcol, ycol]].dropna()
    X = d[xcol].to_numpy()
    Y = d[ycol].to_numpy()

    order = np.argsort(X)
    Xs, Ys = X[order], Y[order]

    # Scatter
    ax.scatter(X, Y, s=18, alpha=0.6, label="Actual data")

    # Linear fit
    b1 = np.polyfit(X, Y, 1)
    ax.plot(Xs, np.polyval(b1, Xs), linestyle=":", linewidth=2, label="Linear fit")

    # Quadratic fit
    b2 = np.polyfit(X, Y, 2)
    ax.plot(Xs, np.polyval(b2, Xs), linestyle="--", linewidth=2, label="Quadratic fit")

    # LOWESS
    lo = lowess(Y, X, frac=lowess_frac, return_sorted=True)
    ax.plot(lo[:, 0], lo[:, 1], linewidth=2, label="LOWESS")

    ax.set_title(title, fontsize=13)
    ax.set_xlabel(xlabel, fontsize=11)
    ax.set_ylabel(ylabel, fontsize=11)
    ax.grid(alpha=0.25)

fig, axes = plt.subplots(2, 2, figsize=(14, 10), sharey=True)


bivar_within_panel(
    axes[0, 0],
    df_plot,
    xcol=f"{y}_lag1_within",
    ycol=f"{y}_within",
    title="GDP growth vs lagged GDP growth (within)",
    xlabel="Within ln_gdp_growth_5y (t-1)",
    ylabel="Within ln_gdp_growth_5y"
)


bivar_within_panel(
    axes[0, 1],
    df_plot,
    xcol="liquid_liabilities_within",
    ycol=f"{y}_within",
    title="Liquid liabilities and GDP growth (within)",
    xlabel="Within liquid liabilities",
    ylabel=""
)

bivar_within_panel(
    axes[1, 0],
    df_plot,
    xcol="mcap_gdp_within",
    ycol=f"{y}_within",
    title="Market capitalization and GDP growth (within)",
    xlabel="Within market capitalization / GDP",
    ylabel="Within ln_gdp_growth_5y"
)


bivar_within_panel(
    axes[1, 1],
    df_plot,
    xcol="bank_loans_gdp_within",
    ycol=f"{y}_within",
    title="Bank loans and GDP growth (within)",
    xlabel="Within bank loans / GDP",
    ylabel=""
)
handles, labels = axes[0, 0].get_legend_handles_labels()
fig.legend(handles, labels, loc="upper center", ncol=4, frameon=True)

plt.suptitle(
    "Bivariate relationships (within-transformed): GDP growth and financial variables",
    fontsize=18,
    y=0.98
)

plt.tight_layout(rect=[0, 0, 1, 0.94])
plt.show()
```



In [174…
```python
def bivariate_acf_with_n(df, dep_var, x, max_lag=10):
    rows = []
    for L in range(0, max_lag + 1):
        if L == 0:
            y = df[dep_var]
            xL = df[x]
        else:
            y = df[dep_var]
            xL = df[f"{x}_lag{L}"]

        valid = y.notna() & xL.notna()
        rows.append({
            "lag": L,
            "corr": np.corrcoef(y[valid], xL[valid])[0, 1],
            "N": valid.sum()
        })
    return pd.DataFrame(rows)

x = "liquid_liabilities"

acf_liq = bivariate_acf_with_n(
    df_within_decay,
    dep_var=dep_var,
    x=x,
    max_lag=K_DECAY
)

plt.figure(figsize=(8, 4))
plt.stem(acf_liq["lag"], acf_liq["corr"], basefmt=" ")
plt.axhline(0, linestyle="--", linewidth=1)
plt.xlabel("Lag L")
plt.ylabel("corr(y_t, x_{t-L})")
plt.title(f"Within bivariate ACF: {dep_var} vs {x}")
plt.tight_layout()
plt.show()

acf_liq
```

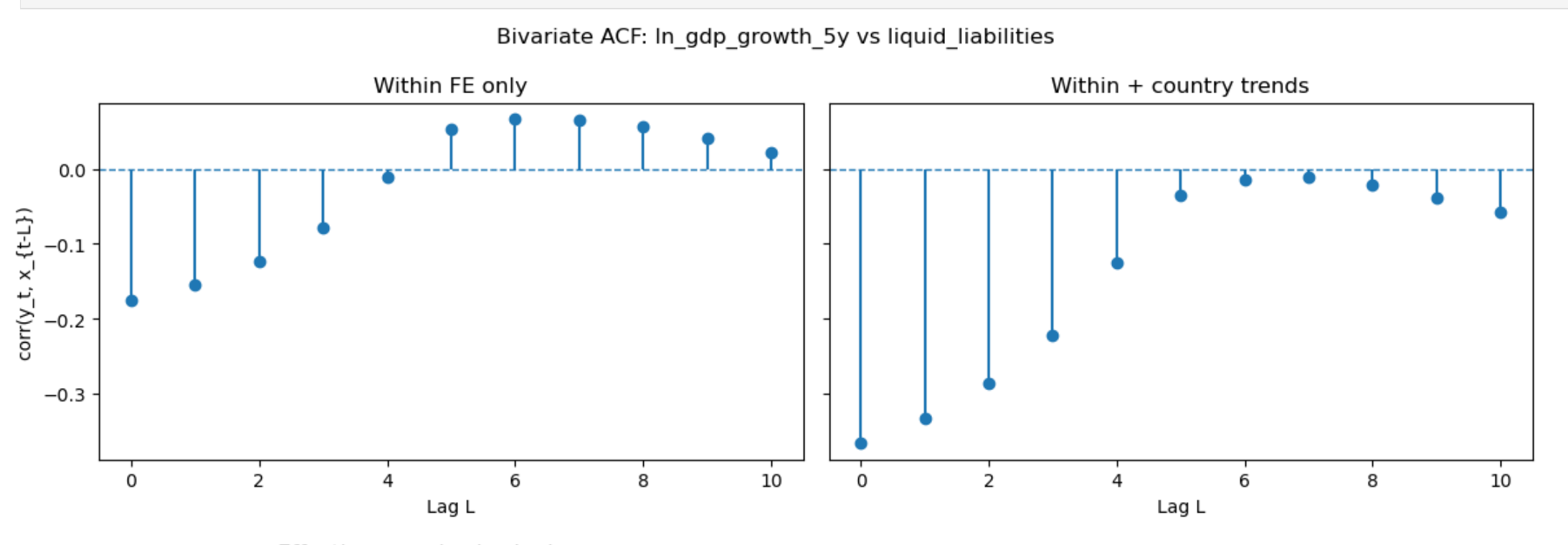


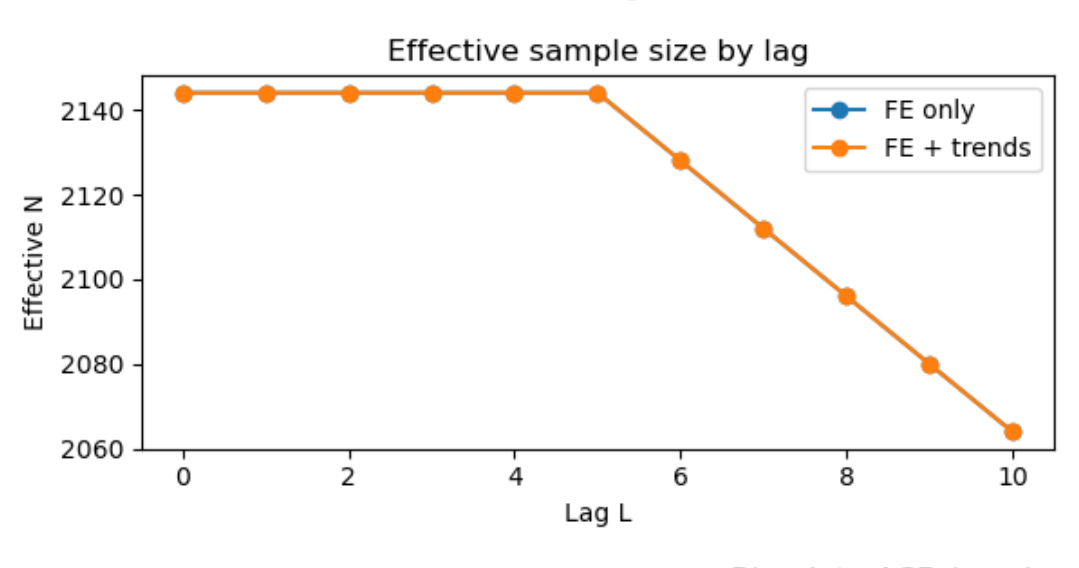
Out[174…

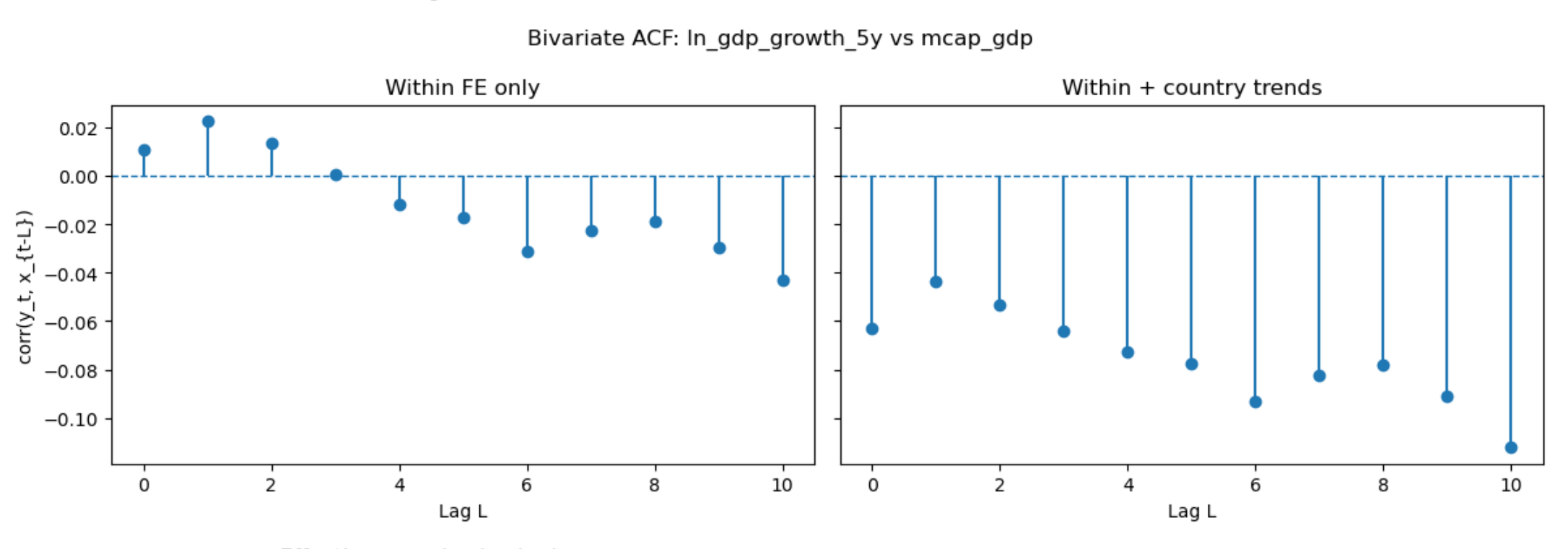
	lag	corr	N
0	0	-0.1762	2144
1	1	-0.1541	2144
2	2	-0.1228	2144
3	3	-0.0787	2144
4	4	-0.0113	2144
5	5	0.0524	2144
6	6	0.0669	2128
7	7	0.0659	2112
8	8	0.0566	2096
9	9	0.0401	2080
10	10	0.0212	2064

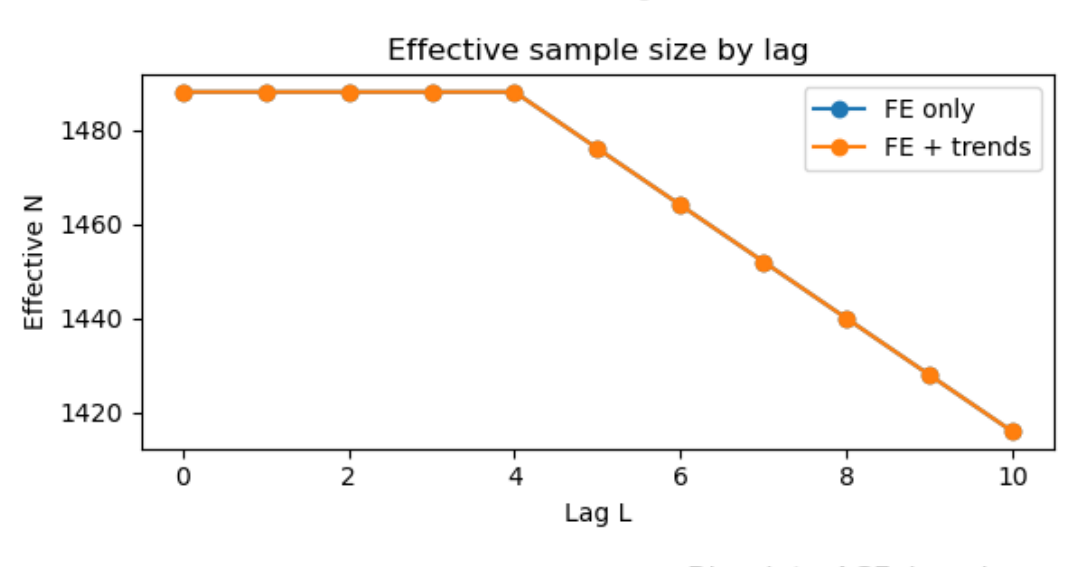
In [175…
```python
def detrend_within_safe(df, var, trend_col="trend", group_col="country", min_n=3):
    out = pd.Series(np.nan, index=df.index, dtype=float)
    for c, g in df.groupby(group_col):
        valid = g[var].notna() & g[trend_col].notna()
        gv = g.loc[valid, [var, trend_col]]
        if gv.shape[0] < min_n:
            continue
        y = gv[var].astype(float)
        X = add_constant(gv[trend_col].astype(float))
        res = OLS(y, X).fit().resid
        out.loc[gv.index] = res
    return out
df_trended = df_Q16.copy()

df_trended[dep_var] = detrend_within_safe(df_Q16, dep_var)

for x in x_vars:
    df_trended[x] = detrend_within_safe(df_Q16, x)
    for L in range(1, K_DECAY + 1):
        df_trended[f"{x}_lag{L}"] = df_trended.groupby("country")[x].shift(L)
```

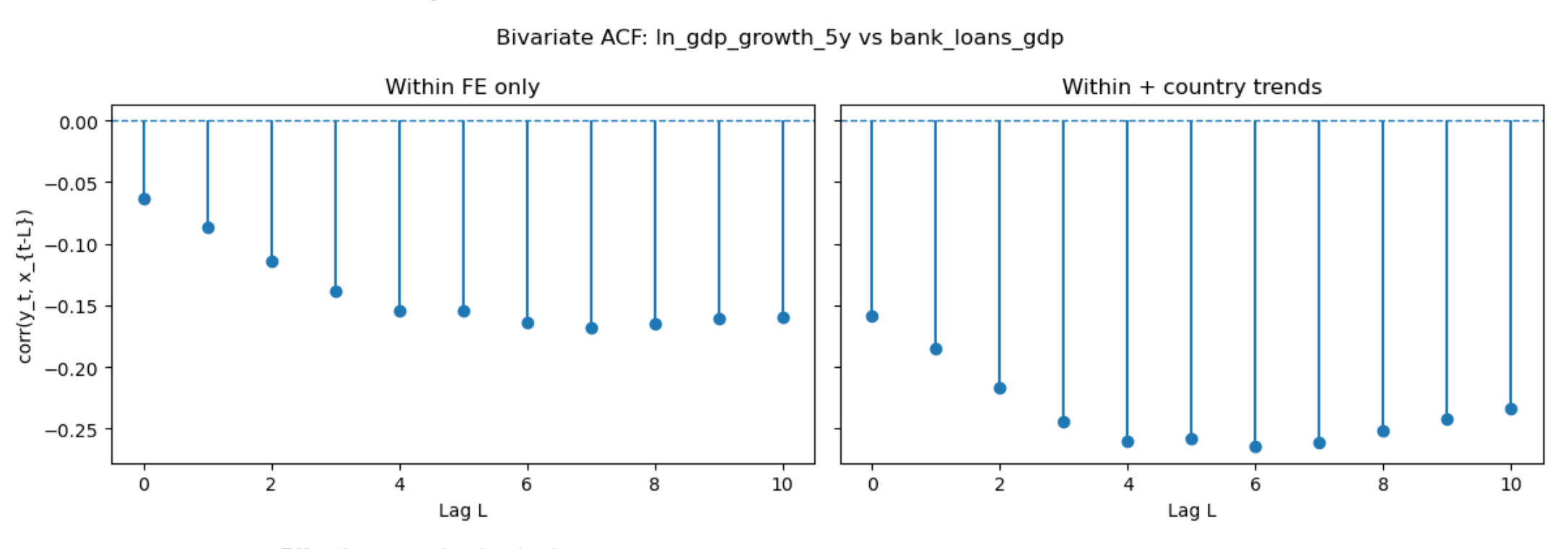
In [177…
```python
def bivariate_acf(df, dep_var, x, max_lag=10):
    rows = []
    for L in range(0, max_lag + 1):
        if L == 0:
            y = df[dep_var]
            xL = df[x]
        else:
            y = df[dep_var]
            xL = df[f"{x}_lag{L}"]

        valid = y.notna() & xL.notna()
        rows.append({
            "lag": L,
            "corr": np.corrcoef(y[valid], xL[valid])[0, 1] if valid.sum() > 2 else np.nan,
            "N": valid.sum()
        })
    return pd.DataFrame(rows)

acf_raw = bivariate_acf(df_within_decay, dep_var, "liquid_liabilities", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"liquid_liabilities_lag{L}"] = (
        df_trended.groupby("country")["liquid_liabilities"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "liquid_liabilities", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs liquid_liabilities")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()



acf_raw = bivariate_acf(df_within_decay, dep_var, "mcap_gdp", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"mcap_gdp_lag{L}"] = (
        df_trended.groupby("country")["mcap_gdp"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "mcap_gdp", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs mcap_gdp")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()


acf_raw = bivariate_acf(df_within_decay, dep_var, "bank_loans_gdp", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"bank_loans_gdp_lag{L}"] = (
        df_trended.groupby("country")["bank_loans_gdp"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "bank_loans_gdp", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs bank_loans_gdp")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()
```

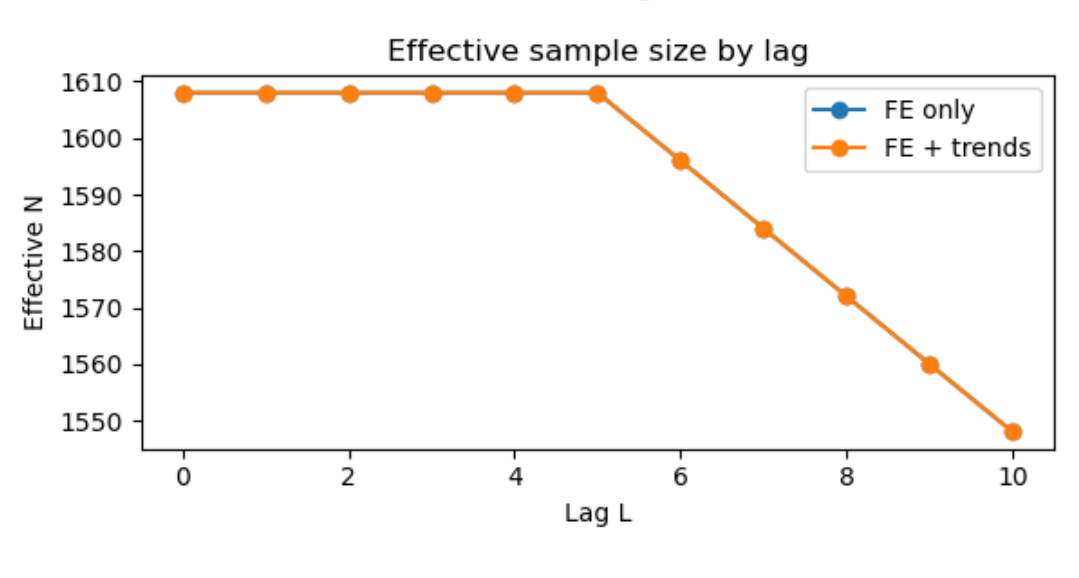












Question 18 --> Bivarite correaltions of all varibls withut time trend

In [ ]:
```python
individual_columns = "country"
time_columns       = "year"
dep_var            = "ln_gdp_growth_5y"

all_vars = [
    dep_var,
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_cpi",
    "trade_openness"
]

LOW_Threshold  = 0.10
HIGH_Threshold = 0.80

def ensure_sorted(df, id_col, time_col):
    return df.sort_values([id_col, time_col]).reset_index(drop=True)

def twoway_within_transform(df, cols, id_col, time_col):
    out = pd.DataFrame(index=df.index)
    overall = df[cols].mean(skipna=True)
    mean_i = df.groupby(id_col)[cols].transform("mean")
    mean_t = df.groupby(time_col)[cols].transform("mean")
    out[cols] = df[cols] - mean_i - mean_t + overall
    return out

def first_difference(df, cols, id_col, time_col):
    df = ensure_sorted(df, id_col, time_col)
    d = df.groupby(id_col)[cols].diff()
    d.columns = [f"d_{c}" for c in cols]
    return d

def lag_within_id(df, cols, id_col, L=1):
    lagged = df.groupby(id_col)[cols].shift(L)
    lagged.columns = [f"{c}_lag{L}" for c in cols]
    return lagged

def corr_matrix(df, cols, method="pearson", min_periods=3):
    return df[cols].corr(method=method, min_periods=min_periods)

def flag_low_corr_with_y(corr, y_name, low_thr=0.1):
    s = corr.loc[:, y_name].drop(y_name, errors="ignore")
    return s[abs(s) < low_thr].sort_values(key=lambda x: abs(x))

def flag_high_corr_among_x(corr, y_name, high_thr=0.8):
    xs = [c for c in corr.columns if c != y_name]
    C = corr.loc[xs, xs].copy()
    pairs = []
    for i, a in enumerate(xs):
        for b in xs[i+1:]:
            val = C.loc[a, b]
            if pd.notna(val) and abs(val) > high_thr:
                pairs.append((a, b, val))
    return pd.DataFrame(pairs, columns=["var1", "var2", "corr"]).sort_values(
        by="corr", key=lambda x: abs(x), ascending=False
    )

def check_boundary_missing_dots(df, id_col, cols_to_check):
    boundary = df[id_col].ne(df[id_col].shift(1))
    boundary_rows = df.loc[boundary, [id_col] + cols_to_check].copy()
    miss = boundary_rows[cols_to_check].isna()
    miss.columns = [f"{c}_isna_at_boundary" for c in miss.columns]
    return pd.concat([boundary_rows[[id_col]], miss], axis=1)

id_col, time_col = individual_columns, time_columns

required = [id_col, time_col] + all_vars
missing = [c for c in required if c not in df.columns]
if missing:
    raise KeyError(
        "Missing columns in df:\n"
        + "\n".join(missing)
        + "\n\nTip: print(df.columns) and fix typos (e.g., liquid_liabilities)."
    )

df0 = ensure_sorted(df.copy(), id_col, time_col)
df_tw = twoway_within_transform(df0, all_vars, id_col, time_col)
df_tw[id_col] = df0[id_col].values
df_tw[time_col] = df0[time_col].values
df_d = first_difference(df0, all_vars, id_col, time_col)
df_d[id_col] = df0[id_col].values
df_d[time_col] = df0[time_col].values
```

```python
d_cols = [f"d_{c}" for c in all_vars]
df_dlag = lag_within_id(df_d, d_cols, id_col, L=1)
df_fd = pd.concat([df_d, df_dlag], axis=1)
corr_tw = corr_matrix(df_tw, all_vars)
fd_cols_all = d_cols + list(df_dlag.columns)
corr_fd = corr_matrix(df_fd, fd_cols_all)
low_tw  = flag_low_corr_with_y(corr_tw, dep_var, low_thr=LOW_Threshold)
high_tw = flag_high_corr_among_x(corr_tw, dep_var, high_thr=HIGH_Threshold)

d_dep = f"d_{dep_var}"
low_fd  = flag_low_corr_with_y(corr_fd, d_dep, low_thr=LOW_Threshold)
high_fd = flag_high_corr_among_x(corr_fd, d_dep, high_thr=HIGH_Threshold)

print("\n=== TWO-WAY-WITHIN correlation matrix (rounded) ===")
print(corr_tw.round(3))

print("\n=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===")
print(low_tw)

print("\n=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===")
print(high_tw)

print("\n=== FIRST-DIFF correlation matrix (rounded) ===")
print(corr_fd.round(3))

print("\n=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===")
print(low_fd)

print("\n=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===")
print(high_fd)
cols_show = [id_col, time_col] + d_cols + list(df_dlag.columns)
print("\n=== First 30 observations: first differences + lag(first differences) ===")
print(df_fd[cols_show].head(30))
boundary_check = check_boundary_missing_dots(df_fd, id_col, d_cols + list(df_dlag.columns))
print("\n=== Boundary missing-dot check (should be True everywhere) ===")
print(boundary_check.head(30))
all_missing = boundary_check.drop(columns=[id_col]).all().all()
print("\nAll Δ and lag(Δ) missing at boundaries:", all_missing)
```

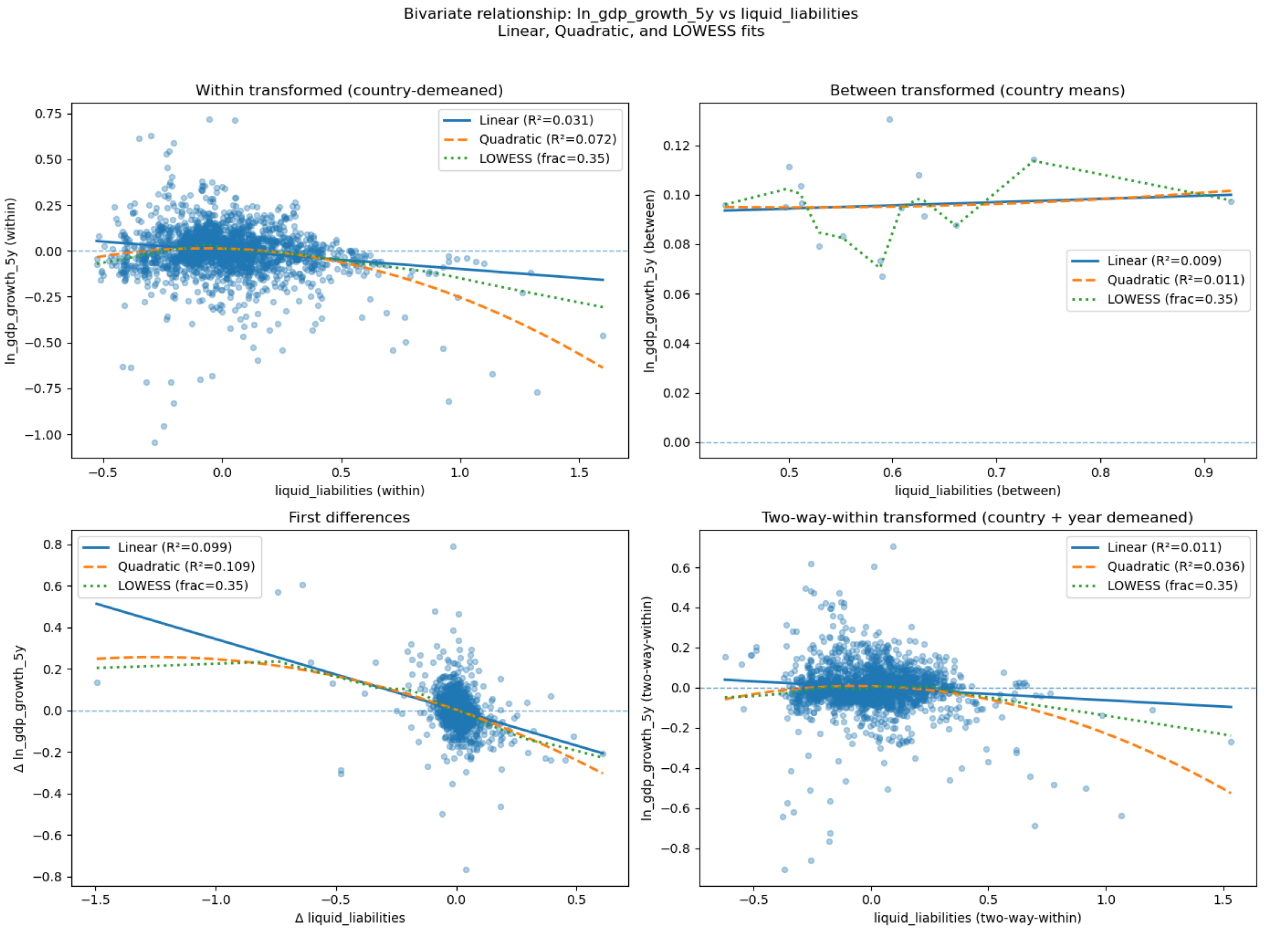
```
=== TWO-WAY-WITHIN correlation matrix (rounded) ===
                     ln_gdp_growth_5y  liquid_liabilities  mcap_gdp  \
ln_gdp_growth_5y               1.0000             -0.1070   -0.0130
liquid_liabilities            -0.1070              1.0000    0.1970
mcap_gdp                      -0.0130              0.1970    1.0000
bank_loans_gdp                 0.0100              0.4330    0.0520
ln_cpi                         0.0320             -0.1840   -0.2020
trade_openness                 0.0740              0.0710    0.0490

                     bank_loans_gdp  ln_cpi  trade_openness
ln_gdp_growth_5y             0.0100  0.0320          0.0740
liquid_liabilities           0.4330 -0.1840          0.0710
mcap_gdp                     0.0520 -0.2020          0.0490
bank_loans_gdp               1.0000 -0.1810         -0.0120
ln_cpi                      -0.1810  1.0000         -0.0080
trade_openness              -0.0120 -0.0080          1.0000

=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===
bank_loans_gdp    0.0103
mcap_gdp         -0.0132
ln_cpi            0.0318
trade_openness    0.0741
Name: ln_gdp_growth_5y, dtype: float64

=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== FIRST-DIFF correlation matrix (rounded) ===
                          d_ln_gdp_growth_5y  d_liquid_liabilities  \
d_ln_gdp_growth_5y                    1.0000               -0.3150
d_liquid_liabilities                 -0.3150                1.0000
d_mcap_gdp                           -0.0310                0.0790
d_bank_loans_gdp                     -0.1100                0.5430
d_ln_cpi                             -0.0670               -0.0610
d_trade_openness                      0.0360               -0.0380
d_ln_gdp_growth_5y_lag1               0.2330               -0.1370
d_liquid_liabilities_lag1            -0.0620                0.2230
d_mcap_gdp_lag1                       0.1260               -0.0410
d_bank_loans_gdp_lag1                 0.0280                0.1290
d_ln_cpi_lag1                         0.0660               -0.0460
d_trade_openness_lag1                 0.0090               -0.0020

                          d_mcap_gdp  d_bank_loans_gdp  d_ln_cpi  \
d_ln_gdp_growth_5y           -0.0310           -0.1100   -0.0670
d_liquid_liabilities          0.0790            0.5430   -0.0610
d_mcap_gdp                    1.0000            0.0520   -0.0390
d_bank_loans_gdp              0.0520            1.0000   -0.1140
d_ln_cpi                     -0.0390           -0.1140    1.0000
d_trade_openness              0.0140            0.0170   -0.0750
d_ln_gdp_growth_5y_lag1      -0.0450            0.0770    0.0060
d_liquid_liabilities_lag1     0.0130            0.0970   -0.0250
d_mcap_gdp_lag1               0.0360           -0.0530    0.0120
d_bank_loans_gdp_lag1         0.0130            0.2900   -0.0860
d_ln_cpi_lag1                -0.0350           -0.0800    0.2280
d_trade_openness_lag1         0.0080           -0.0150    0.0530

                          d_trade_openness  d_ln_gdp_growth_5y_lag1  \
d_ln_gdp_growth_5y                  0.0360                   0.2330
d_liquid_liabilities               -0.0380                  -0.1370
d_mcap_gdp                          0.0140                  -0.0450
d_bank_loans_gdp                    0.0170                   0.0770
d_ln_cpi                           -0.0750                   0.0060
d_trade_openness                    1.0000                   0.0510
d_ln_gdp_growth_5y_lag1             0.0510                   1.0000
d_liquid_liabilities_lag1          -0.0170                  -0.3050
d_mcap_gdp_lag1                     0.1130                  -0.0310
d_bank_loans_gdp_lag1               0.0050                  -0.1030
d_ln_cpi_lag1                      -0.0310                  -0.0680
d_trade_openness_lag1              -0.3240                   0.0380

                          d_liquid_liabilities_lag1  d_mcap_gdp_lag1  \
d_ln_gdp_growth_5y                          -0.0620           0.1260
d_liquid_liabilities                         0.2230          -0.0410
d_mcap_gdp                                   0.0130           0.0360
d_bank_loans_gdp                             0.0970          -0.0530
d_ln_cpi                                    -0.0250           0.0120
d_trade_openness                            -0.0170           0.1130
d_ln_gdp_growth_5y_lag1                     -0.3050          -0.0310
d_liquid_liabilities_lag1                    1.0000           0.0790
d_mcap_gdp_lag1                              0.0790           1.0000
d_bank_loans_gdp_lag1                        0.5360           0.0520
d_ln_cpi_lag1                               -0.0600          -0.0390
d_trade_openness_lag1                       -0.0450           0.0140

                          d_bank_loans_gdp_lag1  d_ln_cpi_lag1  \
d_ln_gdp_growth_5y                       0.0280         0.0660
d_liquid_liabilities                     0.1290        -0.0460
d_mcap_gdp                               0.0130        -0.0350
d_bank_loans_gdp                         0.2900        -0.0800
d_ln_cpi                                -0.0860         0.2280
d_trade_openness                         0.0050        -0.0310
d_ln_gdp_growth_5y_lag1                 -0.1030        -0.0680
d_liquid_liabilities_lag1                0.5360        -0.0600
d_mcap_gdp_lag1                          0.0520        -0.0390
d_bank_loans_gdp_lag1                    1.0000        -0.1150
d_ln_cpi_lag1                           -0.1150         1.0000
d_trade_openness_lag1                    0.0130        -0.0770

                          d_trade_openness_lag1
d_ln_gdp_growth_5y                       0.0090
d_liquid_liabilities                    -0.0020
d_mcap_gdp                               0.0080
d_bank_loans_gdp                        -0.0150
d_ln_cpi                                 0.0530
d_trade_openness                        -0.3240
d_ln_gdp_growth_5y_lag1                  0.0380
d_liquid_liabilities_lag1               -0.0450
d_mcap_gdp_lag1                          0.0140
d_bank_loans_gdp_lag1                    0.0130
d_ln_cpi_lag1                           -0.0770
d_trade_openness_lag1                    1.0000

=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===
d_trade_openness_lag1        0.0088
d_bank_loans_gdp_lag1        0.0277
d_mcap_gdp                  -0.0315
d_trade_openness             0.0362
d_liquid_liabilities_lag1   -0.0618
d_ln_cpi_lag1                0.0662
d_ln_cpi                    -0.0670
Name: d_ln_gdp_growth_5y, dtype: float64

=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== First 30 observations: first differences + lag(first differences) ===
      country  year  d_ln_gdp_growth_5y  d_liquid_liabilities  d_mcap_gdp  \
0   Australia  1882                 NaN                   NaN         NaN
1   Australia  1883                 NaN               -0.0122         NaN
2   Australia  1884                 NaN                0.0210         NaN
3   Australia  1885                 NaN                0.0148         NaN
4   Australia  1886                 NaN               -0.0028         NaN
5   Australia  1887                 NaN                0.0113         NaN
6   Australia  1888             -0.1246                0.0119         NaN
7   Australia  1889              0.0915               -0.0099         NaN
8   Australia  1890             -0.0899                0.0032         NaN
9   Australia  1891              0.0670                0.0139         NaN
10  Australia  1892             -0.2230                0.0436         NaN
11  Australia  1893             -0.0466                0.0198         NaN
12  Australia  1894             -0.0370                0.0347         NaN
13  Australia  1895             -0.0142                0.0366         NaN
14  Australia  1896              0.0096               -0.0116         NaN
15  Australia  1897              0.0806               -0.0323         NaN
16  Australia  1898              0.2037               -0.0346         NaN
17  Australia  1899             -0.0288               -0.0096         NaN
18  Australia  1900              0.1211                0.0056         NaN
19  Australia  1901             -0.0997                0.0036     -0.0216
20  Australia  1902              0.0706               -0.0166     -0.0157
21  Australia  1903             -0.0650                0.0162      0.0148
22  Australia  1904              0.0655               -0.0196     -0.0131
23  Australia  1905             -0.0472                0.0290      0.0390
24  Australia  1906              0.0957                0.0011      0.0164
25  Australia  1907              0.0280               -0.0242     -0.0598
26  Australia  1908             -0.0475                0.0188      0.0129
27  Australia  1909              0.0072               -0.0130      0.0130
28  Australia  1910              0.0480                0.0012     -0.0063
29  Australia  1911             -0.0718                0.0081     -0.0163

    d_bank_loans_gdp  d_ln_cpi  d_trade_openness  d_ln_gdp_growth_5y_lag1  \
0                NaN       NaN               NaN                      NaN
1             0.0015    0.1038           -0.0089                      NaN
2             0.0089    0.0140           -0.0185                      NaN
3             0.0292    0.0000            0.0011                      NaN
4             0.0055    0.0000           -0.0423                      NaN
5            -0.0015   -0.0282           -0.0119                      NaN
6             0.0382   -0.0438            0.0261                      NaN
7             0.0034    0.0000           -0.0116                  -0.1246
8             0.0075    0.0000           -0.0107                   0.0915
9             0.0194   -0.0150            0.0388                  -0.0899
10            0.0388   -0.0153           -0.0180                   0.0670
11            0.0177   -0.0313            0.0131                  -0.2230
12            0.0111   -0.0656            0.0060                  -0.0466
13           -0.0176   -0.0702            0.0404                  -0.0370
14           -0.0260    0.0000            0.0051                  -0.0142
15           -0.0427    0.0000           -0.0016                   0.0096
16           -0.0534   -0.0183            0.0294                   0.0806
17           -0.0422    0.0541           -0.0184                   0.2037
18           -0.0064    0.0174            0.0068                  -0.0288
19           -0.0057    0.0171           -0.0122                   0.1211
20           -0.0215    0.0656           -0.0375                  -0.0997
21            0.0165    0.0157            0.0124                   0.0706
22           -0.0198   -0.0480            0.0098                  -0.0650
23           -0.0045    0.0000            0.0022                   0.0655
24           -0.0180    0.0163            0.0274                  -0.0472
25           -0.0149    0.0000           -0.0063                   0.0957
26            0.0200    0.0317           -0.0204                   0.0280
27           -0.0259    0.0155           -0.0127                  -0.0475
28           -0.0176    0.0153            0.0152                   0.0072
29            0.0042    0.0730           -0.0027                   0.0480

    d_liquid_liabilities_lag1  d_mcap_gdp_lag1  d_bank_loans_gdp_lag1  \
0                         NaN              NaN                    NaN
1                         NaN              NaN                    NaN
2                     -0.0122              NaN                 0.0015
3                      0.0210              NaN                 0.0089
4                      0.0148              NaN                 0.0292
5                     -0.0028              NaN                 0.0055
6                      0.0113              NaN                -0.0015
7                      0.0119              NaN                 0.0382
8                     -0.0099              NaN                 0.0034
9                      0.0032              NaN                 0.0075
10                     0.0139              NaN                 0.0194
11                     0.0436              NaN                 0.0388
12                     0.0198              NaN                 0.0177
13                     0.0347              NaN                 0.0111
14                     0.0366              NaN                -0.0176
15                    -0.0116              NaN                -0.0260
16                    -0.0323              NaN                -0.0427
17                    -0.0346              NaN                -0.0534
18                    -0.0096              NaN                -0.0422
19                     0.0056              NaN                -0.0064
20                     0.0036          -0.0216                -0.0057
21                    -0.0166          -0.0157                -0.0215
22                     0.0162           0.0148                 0.0165
23                    -0.0196          -0.0131                -0.0198
24                     0.0290           0.0390                -0.0045
25                     0.0011           0.0164                -0.0180
26                    -0.0242          -0.0598                -0.0149
27                     0.0188           0.0129                 0.0200
28                    -0.0130           0.0130                -0.0259
29                     0.0012          -0.0063                -0.0176

    d_ln_cpi_lag1  d_trade_openness_lag1
0             NaN                    NaN
1             NaN                    NaN
2          0.1038                -0.0089
3          0.0140                -0.0185
4          0.0000                 0.0011
5          0.0000                -0.0423
6         -0.0282                -0.0119
7         -0.0438                 0.0261
8          0.0000                -0.0116
9          0.0000                -0.0107
10        -0.0150                 0.0388
11        -0.0153                -0.0180
12        -0.0313                 0.0131
13        -0.0656                 0.0060
14        -0.0702                 0.0404
15         0.0000                 0.0051
16         0.0000                -0.0016
17        -0.0183                 0.0294
18         0.0541                -0.0184
19         0.0174                 0.0068
20         0.0171                -0.0122
21         0.0656                -0.0375
22         0.0157                 0.0124
23        -0.0480                 0.0098
24         0.0000                 0.0022
25         0.0163                 0.0274
26         0.0000                -0.0063
27         0.0317                -0.0204
28         0.0155                -0.0127
29         0.0153                 0.0152

=== Boundary missing-dot check (should be True everywhere) ===
             country  d_ln_gdp_growth_5y_isna_at_boundary  \
0          Australia                                 True
139           Canada                                 True
278          Denmark                                 True
417          Finland                                 True
556           France                                 True
695          Germany                                 True
834            Italy                                 True
973            Japan                                 True
1112     Netherlands                                 True
1251          Norway                                 True
1390        Portugal                                 True
1529           Spain                                 True
1668          Sweden                                 True
1807     Switzerland                                 True
1946  United Kingdom                                 True
2085   United States                                 True

      d_liquid_liabilities_isna_at_boundary  d_mcap_gdp_isna_at_boundary  \
0                                      True                         True
139                                    True                         True
278                                    True                         True
417                                    True                         True
556                                    True                         True
695                                    True                         True
834                                    True                         True
973                                    True                         True
1112                                   True                         True
1251                                   True                         True
1390                                   True                         True
1529                                   True                         True
1668                                   True                         True
1807                                   True                         True
1946                                   True                         True
2085                                   True                         True

      d_bank_loans_gdp_isna_at_boundary  d_ln_cpi_isna_at_boundary  \
0                                  True                       True
139                                True                       True
278                                True                       True
417                                True                       True
556                                True                       True
695                                True                       True
834                                True                       True
973                                True                       True
1112                               True                       True
1251                               True                       True
1390                               True                       True
1529                               True                       True
1668                               True                       True
1807                               True                       True
1946                               True                       True
2085                               True                       True

      d_trade_openness_isna_at_boundary  \
0                                  True
139                                True
278                                True
417                                True
556                                True
695                                True
834                                True
973                                True
1112                               True
1251                               True
1390                               True
1529                               True
1668                               True
1807                               True
1946                               True
2085                               True

      d_ln_gdp_growth_5y_lag1_isna_at_boundary  \
0                                         True
139                                       True
278                                       True
417                                       True
556                                       True
695                                       True
834                                       True
973                                       True
1112                                      True
1251                                      True
1390                                      True
1529                                      True
1668                                      True
1807                                      True
1946                                      True
2085                                      True

      d_liquid_liabilities_lag1_isna_at_boundary  \
0                                           True
139                                         True
278                                         True
417                                         True
556                                         True
695                                         True
834                                         True
973                                         True
1112                                        True
1251                                        True
1390                                        True
1529                                        True
1668                                        True
1807                                        True
1946                                        True
2085                                        True

      d_mcap_gdp_lag1_isna_at_boundary  \
0                                 True
139                               True
278                               True
417                               True
556                               True
695                               True
834                               True
973                               True
1112                              True
1251                              True
1390                              True
1529                              True
1668                              True
1807                              True
1946                              True
2085                              True

      d_bank_loans_gdp_lag1_isna_at_boundary  d_ln_cpi_lag1_isna_at_boundary  \
0                                       True                            True
139                                     True                            True
278                                     True                            True
417                                     True                            True
556                                     True                            True
695                                     True                            True
834                                     True                            True
973                                     True                            True
1112                                    True                            True
1251                                    True                            True
1390                                    True                            True
1529                                    True                            True
1668                                    True                            True
1807                                    True                            True
1946                                    True                            True
2085                                    True                            True

      d_trade_openness_lag1_isna_at_boundary
0                                       True
139                                     True
278                                     True
417                                     True
556                                     True
695                                     True
834                                     True
973                                     True
1112                                    True
1251                                    True
1390                                    True
1529                                    True
1668                                    True
1807                                    True
1946                                    True
2085                                    True

All Δ and lag(Δ) missing at boundaries: True
```

```
In [ ]: z
```

```
=== TWO-WAY-WITHIN correlation matrix (rounded) ===
                     ln_gdp_growth_5y  liquid_liabilities  mcap_gdp  \
ln_gdp_growth_5y               1.0000             -0.1070   -0.0130
liquid_liabilities            -0.1070              1.0000    0.1970
mcap_gdp                      -0.0130              0.1970    1.0000
bank_loans_gdp                 0.0100              0.4330    0.0520
ln_cpi                         0.0320             -0.1840   -0.2020
trade_openness                 0.0740              0.0710    0.0490

                     bank_loans_gdp  ln_cpi  trade_openness
ln_gdp_growth_5y             0.0100  0.0320          0.0740
liquid_liabilities           0.4330 -0.1840          0.0710
mcap_gdp                     0.0520 -0.2020          0.0490
bank_loans_gdp               1.0000 -0.1810         -0.0120
ln_cpi                      -0.1810  1.0000         -0.0080
trade_openness              -0.0120 -0.0080          1.0000

=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===
bank_loans_gdp    0.0103
mcap_gdp         -0.0132
ln_cpi            0.0318
trade_openness    0.0741
Name: ln_gdp_growth_5y, dtype: float64

=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== FIRST-DIFF correlation matrix (rounded) ===
                          d_ln_gdp_growth_5y  d_liquid_liabilities  \
d_ln_gdp_growth_5y                    1.0000               -0.3150
d_liquid_liabilities                 -0.3150                1.0000
d_mcap_gdp                           -0.0310                0.0790
d_bank_loans_gdp                     -0.1100                0.5430
d_ln_cpi                             -0.0670               -0.0610
d_trade_openness                      0.0360               -0.0380
d_ln_gdp_growth_5y_lag1               0.2330               -0.1370
d_liquid_liabilities_lag1            -0.0620                0.2230
d_mcap_gdp_lag1                       0.1260               -0.0410
d_bank_loans_gdp_lag1                 0.0280                0.1290
d_ln_cpi_lag1                         0.0660               -0.0460
d_trade_openness_lag1                 0.0090               -0.0020

                          d_mcap_gdp  d_bank_loans_gdp  d_ln_cpi  \
d_ln_gdp_growth_5y           -0.0310           -0.1100   -0.0670
d_liquid_liabilities          0.0790            0.5430   -0.0610
d_mcap_gdp                    1.0000            0.0520   -0.0390
d_bank_loans_gdp              0.0520            1.0000   -0.1140
d_ln_cpi                     -0.0390           -0.1140    1.0000
d_trade_openness              0.0140            0.0170   -0.0750
d_ln_gdp_growth_5y_lag1      -0.0450            0.0770    0.0060
d_liquid_liabilities_lag1     0.0130            0.0970   -0.0250
d_mcap_gdp_lag1               0.0360           -0.0530    0.0120
d_bank_loans_gdp_lag1         0.0130            0.2900   -0.0860
d_ln_cpi_lag1                -0.0350           -0.0800    0.2280
d_trade_openness_lag1         0.0080           -0.0150    0.0530

                          d_trade_openness  d_ln_gdp_growth_5y_lag1  \
d_ln_gdp_growth_5y                  0.0360                   0.2330
d_liquid_liabilities               -0.0380                  -0.1370
d_mcap_gdp                          0.0140                  -0.0450
d_bank_loans_gdp                    0.0170                   0.0770
d_ln_cpi                           -0.0750                   0.0060
d_trade_openness                    1.0000                   0.0510
d_ln_gdp_growth_5y_lag1             0.0510                   1.0000
d_liquid_liabilities_lag1          -0.0170                  -0.3050
d_mcap_gdp_lag1                     0.1130                  -0.0310
d_bank_loans_gdp_lag1               0.0050                  -0.1030
d_ln_cpi_lag1                      -0.0310                  -0.0680
d_trade_openness_lag1              -0.3240                   0.0380

                          d_liquid_liabilities_lag1  d_mcap_gdp_lag1  \
d_ln_gdp_growth_5y                          -0.0620           0.1260
d_liquid_liabilities                         0.2230          -0.0410
d_mcap_gdp                                   0.0130           0.0360
d_bank_loans_gdp                             0.0970          -0.0530
d_ln_cpi                                    -0.0250           0.0120
d_trade_openness                            -0.0170           0.1130
d_ln_gdp_growth_5y_lag1                     -0.3050          -0.0310
d_liquid_liabilities_lag1                    1.0000           0.0790
d_mcap_gdp_lag1                              0.0790           1.0000
d_bank_loans_gdp_lag1                        0.5360           0.0520
d_ln_cpi_lag1                               -0.0600          -0.0390
d_trade_openness_lag1                       -0.0450           0.0140

                          d_bank_loans_gdp_lag1  d_ln_cpi_lag1  \
d_ln_gdp_growth_5y                       0.0280         0.0660
d_liquid_liabilities                     0.1290        -0.0460
d_mcap_gdp                               0.0130        -0.0350
d_bank_loans_gdp                         0.2900        -0.0800
d_ln_cpi                                -0.0860         0.2280
d_trade_openness                         0.0050        -0.0310
d_ln_gdp_growth_5y_lag1                 -0.1030        -0.0680
d_liquid_liabilities_lag1                0.5360        -0.0600
d_mcap_gdp_lag1                          0.0520        -0.0390
d_bank_loans_gdp_lag1                    1.0000        -0.1150
d_ln_cpi_lag1                           -0.1150         1.0000
d_trade_openness_lag1                    0.0130        -0.0770

                          d_trade_openness_lag1
d_ln_gdp_growth_5y                       0.0090
d_liquid_liabilities                    -0.0020
d_mcap_gdp                               0.0080
d_bank_loans_gdp                        -0.0150
d_ln_cpi                                 0.0530
d_trade_openness                        -0.3240
d_ln_gdp_growth_5y_lag1                  0.0380
d_liquid_liabilities_lag1               -0.0450
d_mcap_gdp_lag1                          0.0140
d_bank_loans_gdp_lag1                    0.0130
d_ln_cpi_lag1                           -0.0770
d_trade_openness_lag1                    1.0000

=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===
d_trade_openness_lag1        0.0088
d_bank_loans_gdp_lag1        0.0277
d_mcap_gdp                  -0.0315
d_trade_openness             0.0362
d_liquid_liabilities_lag1   -0.0618
d_ln_cpi_lag1                0.0662
d_ln_cpi                    -0.0670
Name: d_ln_gdp_growth_5y, dtype: float64

=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== First 30 observations: first differences + lag(first differences) ===
      country  year  d_ln_gdp_growth_5y  d_liquid_liabilities  d_mcap_gdp  \
0   Australia  1882                 NaN                   NaN         NaN
1   Australia  1883                 NaN               -0.0122         NaN
2   Australia  1884                 NaN                0.0210         NaN
3   Australia  1885                 NaN                0.0148         NaN
4   Australia  1886                 NaN               -0.0028         NaN
5   Australia  1887                 NaN                0.0113         NaN
6   Australia  1888             -0.1246                0.0119         NaN
7   Australia  1889              0.0915               -0.0099         NaN
8   Australia  1890             -0.0899                0.0032         NaN
9   Australia  1891              0.0670                0.0139         NaN
10  Australia  1892             -0.2230                0.0436         NaN
11  Australia  1893             -0.0466                0.0198         NaN
12  Australia  1894             -0.0370                0.0347         NaN
13  Australia  1895             -0.0142                0.0366         NaN
14  Australia  1896              0.0096               -0.0116         NaN
15  Australia  1897              0.0806               -0.0323         NaN
16  Australia  1898              0.2037               -0.0346         NaN
17  Australia  1899             -0.0288               -0.0096         NaN
18  Australia  1900              0.1211                0.0056         NaN
19  Australia  1901             -0.0997                0.0036     -0.0216
20  Australia  1902              0.0706               -0.0166     -0.0157
21  Australia  1903             -0.0650                0.0162      0.0148
22  Australia  1904              0.0655               -0.0196     -0.0131
23  Australia  1905             -0.0472                0.0290      0.0390
24  Australia  1906              0.0957                0.0011      0.0164
25  Australia  1907              0.0280               -0.0242     -0.0598
26  Australia  1908             -0.0475                0.0188      0.0129
27  Australia  1909              0.0072               -0.0130      0.0130
28  Australia  1910              0.0480                0.0012     -0.0063
29  Australia  1911             -0.0718                0.0081     -0.0163

    d_bank_loans_gdp  d_ln_cpi  d_trade_openness  d_ln_gdp_growth_5y_lag1  \
0                NaN       NaN               NaN                      NaN
1             0.0015    0.1038           -0.0089                      NaN
2             0.0089    0.0140           -0.0185                      NaN
3             0.0292    0.0000            0.0011                      NaN
4             0.0055    0.0000           -0.0423                      NaN
5            -0.0015   -0.0282           -0.0119                      NaN
6             0.0382   -0.0438            0.0261                      NaN
7             0.0034    0.0000           -0.0116                  -0.1246
8             0.0075    0.0000           -0.0107                   0.0915
9             0.0194   -0.0150            0.0388                  -0.0899
10            0.0388   -0.0153           -0.0180                   0.0670
11            0.0177   -0.0313            0.0131                  -0.2230
12            0.0111   -0.0656            0.0060                  -0.0466
13           -0.0176   -0.0702            0.0404                  -0.0370
14           -0.0260    0.0000            0.0051                  -0.0142
15           -0.0427    0.0000           -0.0016                   0.0096
16           -0.0534   -0.0183            0.0294                   0.0806
17           -0.0422    0.0541           -0.0184                   0.2037
18           -0.0064    0.0174            0.0068                  -0.0288
19           -0.0057    0.0171           -0.0122                   0.1211
20           -0.0215    0.0656           -0.0375                  -0.0997
21            0.0165    0.0157            0.0124                   0.0706
22           -0.0198   -0.0480            0.0098                  -0.0650
23           -0.0045    0.0000            0.0022                   0.0655
24           -0.0180    0.0163            0.0274                  -0.0472
25           -0.0149    0.0000           -0.0063                   0.0957
26            0.0200    0.0317           -0.0204                   0.0280
27           -0.0259    0.0155           -0.0127                  -0.0475
28           -0.0176    0.0153            0.0152                   0.0072
29            0.0042    0.0730           -0.0027                   0.0480

    d_liquid_liabilities_lag1  d_mcap_gdp_lag1  d_bank_loans_gdp_lag1  \
0                         NaN              NaN                    NaN
1                         NaN              NaN                    NaN
2                     -0.0122              NaN                 0.0015
3                      0.0210              NaN                 0.0089
4                      0.0148              NaN                 0.0292
5                     -0.0028              NaN                 0.0055
6                      0.0113              NaN                -0.0015
7                      0.0119              NaN                 0.0382
8                     -0.0099              NaN                 0.0034
9                      0.0032              NaN                 0.0075
10                     0.0139              NaN                 0.0194
11                     0.0436              NaN                 0.0388
12                     0.0198              NaN                 0.0177
13                     0.0347              NaN                 0.0111
14                     0.0366              NaN                -0.0176
15                    -0.0116              NaN                -0.0260
16                    -0.0323              NaN                -0.0427
17                    -0.0346              NaN                -0.0534
18                    -0.0096              NaN                -0.0422
19                     0.0056              NaN                -0.0064
20                     0.0036          -0.0216                -0.0057
21                    -0.0166          -0.0157                -0.0215
22                     0.0162           0.0148                 0.0165
23                    -0.0196          -0.0131                -0.0198
24                     0.0290           0.0390                -0.0045
25                     0.0011           0.0164                -0.0180
26                    -0.0242          -0.0598                -0.0149
27                     0.0188           0.0129                 0.0200
28                    -0.0130           0.0130                -0.0259
29                     0.0012          -0.0063                -0.0176

    d_ln_cpi_lag1  d_trade_openness_lag1
0             NaN                    NaN
1             NaN                    NaN
2          0.1038                -0.0089
3          0.0140                -0.0185
4          0.0000                 0.0011
5          0.0000                -0.0423
6         -0.0282                -0.0119
7         -0.0438                 0.0261
8          0.0000                -0.0116
9          0.0000                -0.0107
10        -0.0150                 0.0388
11        -0.0153                -0.0180
12        -0.0313                 0.0131
13        -0.0656                 0.0060
14        -0.0702                 0.0404
15         0.0000                 0.0051
16         0.0000                -0.0016
17        -0.0183                 0.0294
18         0.0541                -0.0184
19         0.0174                 0.0068
20         0.0171                -0.0122
21         0.0656                -0.0375
22         0.0157                 0.0124
23        -0.0480                 0.0098
24         0.0000                 0.0022
25         0.0163                 0.0274
26         0.0000                -0.0063
27         0.0317                -0.0204
28         0.0155                -0.0127
29         0.0153                 0.0152

=== Boundary missing-dot check (should be True everywhere) ===
             country  d_ln_gdp_growth_5y_isna_at_boundary  \
0          Australia                                 True
139           Canada                                 True
278          Denmark                                 True
417          Finland                                 True
556           France                                 True
695          Germany                                 True
834            Italy                                 True
973            Japan                                 True
1112     Netherlands                                 True
1251          Norway                                 True
1390        Portugal                                 True
1529           Spain                                 True
1668          Sweden                                 True
1807     Switzerland                                 True
1946  United Kingdom                                 True
2085   United States                                 True

      d_liquid_liabilities_isna_at_boundary  d_mcap_gdp_isna_at_boundary  \
0                                      True                         True
139                                    True                         True
278                                    True                         True
417                                    True                         True
556                                    True                         True
695                                    True                         True
834                                    True                         True
973                                    True                         True
1112                                   True                         True
1251                                   True                         True
1390                                   True                         True
1529                                   True                         True
1668                                   True                         True
1807                                   True                         True
1946                                   True                         True
2085                                   True                         True

      d_bank_loans_gdp_isna_at_boundary  d_ln_cpi_isna_at_boundary  \
0                                  True                       True
139                                True                       True
278                                True                       True
417                                True                       True
556                                True                       True
695                                True                       True
834                                True                       True
973                                True                       True
1112                               True                       True
1251                               True                       True
1390                               True                       True
1529                               True                       True
1668                               True                       True
1807                               True                       True
1946                               True                       True
2085                               True                       True

      d_trade_openness_isna_at_boundary  \
0                                  True
139                                True
278                                True
417                                True
556                                True
695                                True
834                                True
973                                True
1112                               True
1251                               True
1390                               True
1529                               True
1668                               True
1807                               True
1946                               True
2085                               True

      d_ln_gdp_growth_5y_lag1_isna_at_boundary  \
0                                         True
139                                       True
278                                       True
417                                       True
556                                       True
695                                       True
834                                       True
973                                       True
1112                                      True
1251                                      True
1390                                      True
1529                                      True
1668                                      True
1807                                      True
1946                                      True
2085                                      True

      d_liquid_liabilities_lag1_isna_at_boundary  \
0                                           True
139                                         True
278                                         True
417                                         True
556                                         True
695                                         True
834                                         True
973                                         True
1112                                        True
1251                                        True
1390                                        True
1529                                        True
1668                                        True
1807                                        True
1946                                        True
2085                                        True

      d_mcap_gdp_lag1_isna_at_boundary  \
0                                 True
139                               True
278                               True
417                               True
556                               True
695                               True
834                               True
973                               True
1112                              True
1251                              True
1390                              True
1529                              True
1668                              True
1807                              True
1946                              True
```

```
      d_bank_loans_gdp_lag1_isna_at_boundary  d_ln_cpi_lag1_isna_at_boundary  \
0                                       True                            True
139                                     True                            True
278                                     True                            True
417                                     True                            True
556                                     True                            True
695                                     True                            True
834                                     True                            True
973                                     True                            True
1112                                    True                            True
1251                                    True                            True
1390                                    True                            True
1529                                    True                            True
1668                                    True                            True
1807                                    True                            True
1946                                    True                            True
2085                                    True                            True

      d_trade_openness_lag1_isna_at_boundary
0                                       True
139                                     True
278                                     True
417                                     True
556                                     True
695                                     True
834                                     True
973                                     True
1112                                    True
1251                                    True
1390                                    True
1529                                    True
1668                                    True
1807                                    True
1946                                    True
2085                                    True

All Δ and lag(Δ) missing at boundaries: True
```

In [180…

```python
id_col   = "country"
time_col = "year"

y_col = "ln_gdp_growth_5y"
x_col = "liquid_liabilities"

LOWESS_FRAC = 0.35
LOWESS_IT   = 1

# ----------------------------

def ensure_sorted(df, id_col, time_col):
    return df.sort_values([id_col, time_col]).reset_index(drop=True)

def within_transform(df, cols, id_col):
    return df[cols] - df.groupby(id_col)[cols].transform("mean")

def between_transform(df, cols, id_col):
    return df.groupby(id_col)[cols].mean().reset_index()

def twoway_within_transform(df, cols, id_col, time_col):
    overall = df[cols].mean(skipna=True)
    mean_i = df.groupby(id_col)[cols].transform("mean")
    mean_t = df.groupby(time_col)[cols].transform("mean")
    return df[cols] - mean_i - mean_t + overall

def first_difference(df, cols, id_col, time_col):
    df = ensure_sorted(df, id_col, time_col)
    d = df.groupby(id_col)[cols].diff()
    d.columns = [f"d_{c}" for c in cols]
    return d



def _fit_linear(x, y):
    X = sm.add_constant(x)
    return sm.OLS(y, X, missing="drop").fit()

def _fit_quadratic(x, y):
    X = np.column_stack([x, x**2])
    X = sm.add_constant(X)
    return sm.OLS(y, X, missing="drop").fit()

def _predict_on_grid(x, model_type="linear", params=None, xgrid=None):
    if xgrid is None:
        xgrid = np.linspace(np.nanmin(x), np.nanmax(x), 200)

    params = np.asarray(params, dtype=float)

    if model_type == "linear":
        b0, b1 = params
        yhat = b0 + b1 * xgrid

    elif model_type == "quadratic":
        b0, b1, b2 = params
        yhat = b0 + b1 * xgrid + b2 * xgrid**2

    else:
        raise ValueError("model_type must be 'linear' or 'quadratic'")

    return xgrid, yhat

def plot_bivariate_with_fits(ax, x, y, title, xlabel, ylabel):
    mask = np.isfinite(x) & np.isfinite(y)
    x = np.asarray(x[mask], dtype=float)
    y = np.asarray(y[mask], dtype=float)

    ax.scatter(x, y, alpha=0.35, s=18)

    if x.size < 10:
        ax.set_title(f"{title}\n(too few obs for smooth fits)")
        ax.set_xlabel(xlabel)
        ax.set_ylabel(ylabel)
        return


    lin = _fit_linear(x, y)
    xg, y_lin = _predict_on_grid(x, "linear", lin.params)   # <-- FIXED

    # Quad
    quad = _fit_quadratic(x, y)
    xg2, y_quad = _predict_on_grid(x, "quadratic", quad.params, xgrid=xg)  # <-- FIXED

    # LOWESS
    lo = lowess(y, x, frac=LOWESS_FRAC, it=LOWESS_IT, return_sorted=True)
    x_lo, y_lo = lo[:, 0], lo[:, 1]

    ax.plot(xg, y_lin, linewidth=2, label=f"Linear (R²={lin.rsquared:.3f})")
    ax.plot(xg2, y_quad, linewidth=2, linestyle="--", label=f"Quadratic (R²={quad.rsquared:.3f})")
    ax.plot(x_lo, y_lo, linewidth=2, linestyle=":", label=f"LOWESS (frac={LOWESS_FRAC})")

    ax.axhline(0, linewidth=1, linestyle="--", alpha=0.6)
    ax.set_title(title)
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    ax.legend()


df0 = ensure_sorted(df.copy(), id_col, time_col)
needed = [id_col, time_col, y_col, x_col]
missing = [c for c in needed if c not in df0.columns]
if missing:
    raise KeyError(f"Missing columns in df: {missing}")

w = within_transform(df0, [y_col, x_col], id_col)
df_within = pd.concat([df0[[id_col, time_col]], w], axis=1)
df_between = between_transform(df0, [y_col, x_col], id_col)
d = first_difference(df0, [y_col, x_col], id_col, time_col)
df_fd = pd.concat([df0[[id_col, time_col]], d], axis=1)
tw = twoway_within_transform(df0, [y_col, x_col], id_col, time_col)
df_twoway = pd.concat([df0[[id_col, time_col]], tw], axis=1)
fig, axes = plt.subplots(2, 2, figsize=(14, 10), sharey=False)

plot_bivariate_with_fits(
    axes[0, 0], df_within[x_col], df_within[y_col],
    title="Within transformed (country-demeaned)",
    xlabel="liquid_liabilities (within)",
    ylabel="ln_gdp_growth_5y (within)"
)

plot_bivariate_with_fits(
    axes[0, 1], df_between[x_col], df_between[y_col],
    title="Between transformed (country means)",
    xlabel="liquid_liabilities (between)",
    ylabel="ln_gdp_growth_5y (between)"
)

plot_bivariate_with_fits(
    axes[1, 0], df_fd[f"d_{x_col}"], df_fd[f"d_{y_col}"],
    title="First differences",
    xlabel="Δ liquid_liabilities",
    ylabel="Δ ln_gdp_growth_5y"
)

plot_bivariate_with_fits(
    axes[1, 1], df_twoway[x_col], df_twoway[y_col],
    title="Two-way-within transformed (country + year demeaned)",
    xlabel="liquid_liabilities (two-way-within)",
    ylabel="ln_gdp_growth_5y (two-way-within)"
)

plt.suptitle(
    "Bivariate relationship: ln_gdp_growth_5y vs liquid_liabilities\n"
    "Linear, Quadratic, and LOWESS fits",
    y=1.02
)
plt.tight_layout()
plt.show()
```